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# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
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    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from flask_wtf.csrf import CSRFProtect, CSRFError
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
csrf = CSRFProtect()
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # CSRF 防护：WTF_CSRF_ENABLED 只对 FlaskForm 生效，页面上那些手写 POST
    表单
    # （账号增删改、项目 / 图斑删除等）以前是没校验的；同时模板里的 csrf_token()
    # 也需要它才会存在——少了这一行，保存过一次 AHP 权重后 /ahp/ 直接 500。
    csrf.init_app(app)
    # 模板里展示时间统一走 localdt：库里存 UTC，展示成本地时间
    from app.utils.timefmt import to_local
    app.jinja_env.filters["localdt"] = to_local
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
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app.register_blueprint(constraint.bp)
app.register_blueprint(ahp.bp)
app.register_blueprint(indicator.bp)
app.register_blueprint(evaluate.bp)
app.register_blueprint(report.bp)
app.register_blueprint(search.bp)
# 全局登录保护：除登录页与静态资源外，均需登录
from flask import request, redirect, url_for,
send_from_directory
from flask_login import current_user, login_user
PUBLIC_ENDPOINTS = {"auth.login", "auth.register",
"static", "favicon"}
# 浏览器默认会请求根路径下的 /favicon.ico。
# 页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
# 回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    # 截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"auth.register", "static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
# 侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
# 不使用模块编号，避免出现与实际流程不一致的代号。
# 每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
# 与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
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        ("适宜性定量评价", "/evaluate/", "target"),
        ("报告与结果输出", "/report/", "file"),
    ]
    #  「用户与操作日志」已并入「个人设置」，不再是独立菜单，
    #  因此系统管理组只留一个入口。
    ADMIN_LINKS = [
        ("个人设置", "/profile", "user"),
    ]
    @app.context_processor
    def inject_nav():
        path = request.path or "/"
        def hit(prefix):
            """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
            return path.startswith(prefix.rstrip("/"))
        badges = {}
        if current_user.is_authenticated:
            from app.services import dashboard as dash
            badges = dash.nav_badges()
        def build(links):
            return [{"label": label, "href": href, "icon":
    icon,
                     "active": hit(href), "badge":
    badges.get(href, 0)}
                    for label, href, icon in links]
        groups = [
            {"title": "评价流程", "links": build(FLOW_LINKS)},
            {"title": "系统管理", "links": build(ADMIN_LINKS)},
        ]
        #  面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
        if path == "/":
            page_title = "工作台"
        elif path.startswith("/profile"):
            page_title = "个人设置"
        elif path.startswith(("/users", "/logs")):
            page_title = "账号与日志"
        elif path.startswith("/search"):
            page_title = "搜索结果"
        else:
            page_title = "工作台"
            for label, href, _icon in FLOW_LINKS:
                if hit(href):
                    page_title = label
                    break
        #  层级引导用：项目是归档单位，图斑才是评价对象。
        #  下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
        from app.models import Project, MinePatch
        projects = Project.query.order_by(Project.id).all()
        return {
            "nav_groups": groups,
            "home_active": path == "/",
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                "page_title": page_title,
                "today": date.today().strftime("%Y-%m-%d"),
                "search_q": request.args.get("q", ""),
                "role_labels": ROLE_LABELS,
                "project_count": len(projects),
                "patch_count": MinePatch.query.count(),
                "projects_without_patch": [p for p in projects
                                           if p.patches.count()
    == 0],
            }
    # 操作审计：统一在响应阶段记录成功的写操作（操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    @app.errorhandler(CSRFError)
    def csrf_error(e):
        """令牌过期或来源不符时给一句人话，并把人送回上一页。
        原来校验一失败就是默认英文 400 页，或者（对手写表单）干脆什么都不发生，
        用户只会觉得「点了没反应」。回跳目标只取本站地址，避免开放重定向。
        """
        from flask import flash, redirect, url_for
        flash("页面停留时间过久或提交来源校验未通过，请重新提交一次", "error")
        referrer = request.referrer or ""
        if referrer.startswith(request.host_url):
            return redirect(referrer)
        return redirect(url_for("auth.index"))
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
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    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 模板改动立即生效，不必重启服务。
    # Flask 在非 debug 下默认把模板缓存到进程里，改完模板刷新浏览器还是旧页面，
    # 容易被误当成「改了没生效」。本系统只在本机运行，这点开销可以忽略。
    TEMPLATES_AUTO_RELOAD = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024            # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
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    default=True)                                                    301
    created_at = db.Column(db.DateTime,                              302
    default=datetime.utcnow)                                         303
    # ---- 个人资料（用户可在「个人设置」里自行维护）----                304
    display_name = db.Column(db.String(64), nullable=True)           305
    # 昵称                                                           306
    email = db.Column(db.String(128), nullable=True)                 307
    phone = db.Column(db.String(32), nullable=True)                  308
    gender = db.Column(db.String(8), nullable=True)                  309
    # 男 / 女 / 保密                                                 310
    avatar = db.Column(db.String(256), nullable=True)                311
    # 头像，相对 static 的路径                                       312
    @property                                                        313
    def is_admin(self):                                              314
        return self.role == "admin"                                  315
    @property                                                        316
    def shown_name(self):                                            317
        """界面上显示的名字：优先用昵称，没填就用账号名。"""          318
        return self.display_name or self.username                    319
    @property                                                        320
    def initial(self):                                               321
        """没有头像时用首字占位。"""                                  322
        return (self.shown_name or "?")[0].upper()                   323
    def to_dict(self):                                               324
        return {"id": self.id, "username": self.username,            325
                "display_name": self.display_name,                   326
    "gender": self.gender,                                           327
                "role": self.role, "active": self.active}            328
class OperationLog(db.Model):                                        329
    """操作日志。"""                                                 330
    __tablename__ = "operation_logs"                                 331
    id = db.Column(db.Integer, primary_key=True)                     332
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"         333
    ), nullable=True)                                                334
    action = db.Column(db.String(128), nullable=False)               335
    # 操作类型                                                       336
    target_type = db.Column(db.String(64), nullable=True)            337
    # 对象类型                                                       338
    target_id = db.Column(db.Integer, nullable=True)                 339
    # 对象主键                                                       340
    detail = db.Column(db.Text, nullable=True)                       341
    # 操作详情                                                       342
    created_at = db.Column(db.DateTime,                              343
    default=datetime.utcnow, index=True)                             344
class Project(db.Model):                                             345
    """评价项目（项目与矿山档案）。"""                               346
    __tablename__ = "projects"                                       347
    id = db.Column(db.Integer, primary_key=True)                     348
    name = db.Column(db.String(128), nullable=False)                 349
    code = db.Column(db.String(64), nullable=True)                   350
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    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    patches = db.relationship("MinePatch",
    backref="project", lazy="dynamic",
                              cascade="all, delete-orphan")
class MinePatch(db.Model):
    """矿山图斑。"""
    __tablename__ = "mine_patches"
    id = db.Column(db.Integer, primary_key=True)
    project_id = db.Column(db.Integer,
    db.ForeignKey("projects.id"), nullable=False)
    name = db.Column(db.String(128), nullable=False)
    admin_region = db.Column(db.String(128), nullable=True)
    # 所在行政区
    center_x = db.Column(db.Float, nullable=True)
    # 中心经度 / 横坐标
    center_y = db.Column(db.Float, nullable=True)
    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
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    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)  # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)    # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
    __tablename__ = "standard_rules"
    id = db.Column(db.Integer, primary_key=True)
    standard_id = db.Column(db.Integer,
    db.ForeignKey("standards.id"), nullable=False)
    item = db.Column(db.String(128), nullable=False)
    # 判定项
    operator = db.Column(db.String(8), nullable=False)
    # 比较符：>  >=  <  <=  ==
    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")   # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录。"""
    __tablename__ = "constraint_checks"
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    id = db.Column(db.Integer, primary_key=True)                    451
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        452
    ches.id"), nullable=False)                                      453
    policy_pass = db.Column(db.Boolean, nullable=True)              454
    # 政策约束                                                      455
    location_pass = db.Column(db.Boolean, nullable=True)            456
    # 区位约束                                                      457
    pollution_pass = db.Column(db.Boolean, nullable=True)           458
    # 土壤污染                                                      459
    geo_pass = db.Column(db.Boolean, nullable=True)                 460
    # 地质灾害                                                      461
    farmland_pass = db.Column(db.Boolean, nullable=True)            462
    # 耕地属性                                                      463
    passed = db.Column(db.Boolean, nullable=True)                   464
    # 总体是否通过                                                  465
    reason = db.Column(db.Text, nullable=True)                      466
    # 不通过原因                                                    467
    checked_at = db.Column(db.DateTime,                             468
    default=datetime.utcnow)                                        469
class MicroUnit(db.Model):                                          470
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""                  471
    __tablename__ = "micro_units"                                   472
    id = db.Column(db.Integer, primary_key=True)                    473
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        474
    ches.id"), nullable=False)                                      475
    unit_type = db.Column(db.String(32), nullable=False)            476
    # 单元类型                                                      477
    area = db.Column(db.Float, nullable=True)                       478
    # 面积                                                          479
    ratio = db.Column(db.Float, nullable=True)                      480
    # 面积占比（自动计算）                                          481
    description = db.Column(db.Text, nullable=True)                 482
    # 现场描述                                                      483
class Indicator(db.Model):                                          484
    """指标定义：指标树，含角色与启用单元类型。"""                  485
    __tablename__ = "indicators"                                    486
    id = db.Column(db.Integer, primary_key=True)                    487
    code = db.Column(db.String(32), unique=True,                    488
    nullable=False)  # 指标编码                                    489
    name = db.Column(db.String(64), nullable=False)                 490
    # 指标名称                                                      491
    level = db.Column(db.Integer, nullable=False)                   492
    # 1/2/3 级                                                      493
    parent_code = db.Column(db.String(32), nullable=True)           494
    # 父指标编码                                                    495
    # 角色：土壤污染指标同时作为限制性指标与评价指标               496
    is_constraint = db.Column(db.Boolean, default=False)            497
    is_evaluative = db.Column(db.Boolean, default=True)             498
    unit = db.Column(db.String(32), nullable=True)                  499
    # 单位                                                          500
```

```
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
    nullable=False)        # 指标编码
    weight = db.Column(db.Float, nullable=False)
    # 权重值
class ExpertMatrix(db.Model):
    """专家判断矩阵：多专家打分的原始矩阵。"""
    __tablename__ = "expert_matrices"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=True)
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵
    unit_type = db.Column(db.String(32), nullable=True)
    # 微地质单元类型
    l1_name = db.Column(db.String(64), nullable=True)
    # 所属一级指标（二级矩阵用）
    expert_name = db.Column(db.String(64), nullable=True)
    # 专家姓名
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 矩阵对应指标层级
```

```
    matrix_json = db.Column(db.Text, nullable=False)
    # 判断矩阵（JSON 二维数组）
    cr = db.Column(db.Float, nullable=True)
    # 该专家矩阵的一致性比例
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class IndicatorData(db.Model):
    """指标实测数据：按微地质单元组织。"""
    __tablename__ = "indicator_data"
    id = db.Column(db.Integer, primary_key=True)
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni
    ts.id"), nullable=False, unique=True)
    soil_ratio = db.Column(db.Float, nullable=True)
    # 土石比
    thickness = db.Column(db.Float, nullable=True)
    # 土层厚度 cm
    ph = db.Column(db.Float, nullable=True)
    organic_matter = db.Column(db.Float, nullable=True)
    # 有机质 g/kg
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null
    heavy_metals = db.Column(db.Text, nullable=True)
    coverage = db.Column(db.Float, nullable=True)
    # 植被覆盖度 %
    shannon = db.Column(db.Float, nullable=True)
    # Shannon-Wiener 多样性指数
    slope = db.Column(db.Float, nullable=True)
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡/阴阳坡/阴坡
    def heavy_metals_dict(self):
        """把重金属 JSON 解析成字典。
        字段是手工可改的文本，被脏写或导入半个 JSON 时不能让整页 500：
        解析失败按「未检测」处理（空字典），并把原文留在库里供排查。
        """
        if not self.heavy_metals:
            return {}
        try:
            data = json.loads(self.heavy_metals)
        except (ValueError, TypeError):
            return {}
        return data if isinstance(data, dict) else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
```

```
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
# 邮箱格式：够用的轻量校验。
# WTForms 自带的 Email() 需要额外安装 email_validator,
# 为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
```

```
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。
    校验消息用中文写明：登录框带 novalidate，浏览器不会代填提示，
    字段为空时若不自己给消息，用户点了「登录」会以为没反应。
    """
    username = StringField("账户名称",
                           validators=[DataRequired(message=
    "请输入账户名称"),
                                       Length(max=64,

    message="账户名称不能超过 64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired(messag
    e="请输入密码")])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format(),
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
```

```
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称"),
                                       Length(min=2, max=64,
                                              message="账号名称2–64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired("请设置密码"),
                                         Length(min=6,
                                                max=64,
                                                message="密码至少 6 位")])
    confirm = PasswordField("确认密码",
                            validators=[DataRequired(),
                                        EqualTo("password",
                                                message="两次输入的密码不一致")])
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_reg = SubmitField("注册并登录")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
    old_password = PasswordField("当前密码",
    validators=[DataRequired()])
    new_password = PasswordField("新密码",
                                 validators=[DataRequired(),
    Length(min=6, max=64,
                                             message="新密码至少 6 位")])
    confirm = PasswordField("确认新密码",
                            validators=[DataRequired(),
    EqualTo("new_password",
                                            message="两次输入的新密码不一致")])
    submit_pwd = SubmitField("修改密码")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
```

```
全部数值取自论文，作为系统计算的唯一基准：
- 论文 ４．２．２ 节：指标标准化理想极值与取值依据
- 论文 ４．２．３ 节 / 表 ４：适宜性等级划分
- 论文表 ３：按微地质单元分别确定的一级、二级指标权重
- 论文 ４．２．１ 节：平均随机一致性指标 ＲＩ
"""
# ----------------------------------------------------------------------
    ----------
# 一、微地质单元类型
# ----------------------------------------------------------------------
    ----------
# 论文 ４．１ 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
# 每个单元类型启用的指标类别（论文表 ３ 隐含：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                    # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                        # 不设地形指标
}
# ----------------------------------------------------------------------
    ----------
# 二、指标标准化理想极值（论文 ４．２．２ 节）
# ----------------------------------------------------------------------
    ----------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （ｃｍ）
IDEAL_PH_RANGE = (5.0, 9.0)     # ｐＨ 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            # 有机质理想值
    （ｇ/ｋｇ），全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （％）
IDEAL_SHANNON = 0.7             # Ｓｈａｎｎｏｎ-Ｗｉｅｎｅｒ 多样性指数参照生态系统
# 重金属极值 （ｍｇ/ｋｇ），《绿化种植土壤》ＣＪ/Ｔ ３４０—２０１６ 土壤重金属含量 ＩＶ 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 ４．２．１ 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 ４．２．２ 节引《新一轮退耕还林还草总体方案》发改西部〔２０１４〕１７７２ 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 ４．２．２ 节："阳坡、阴阳坡、阴坡分别赋值 0．2、0．6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# ｐＨ 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
```

```
PH_TOL_HIGH = 10.0
# ---------------------------------------------------------------------
    ----------------
# 三、适宜性等级划分（论文表 4）
# ---------------------------------------------------------------------
    ----------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ---------------------------------------------------------------------
    ----------------
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）
# ---------------------------------------------------------------------
    ----------------
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6:
    1.24,
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}
# 一致性比例阈值（论文：当 CR<0.1 时，判断矩阵满足一致性要求）
CR_THRESHOLD = 0.1
# ---------------------------------------------------------------------
    ----------------
# 五、默认权重方案（论文表 3）
# ---------------------------------------------------------------------
    ----------------
# 论文 4.2.2 节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，
# 分别计算评价指标权重"（论文摘要、图 2 流程与表 5 均作"碎石边坡"，
# 系统统一采用"碎石边坡"）。权重由 10 位专家打分后经 AHP 方法确定。
# 下表为论文表 3 所载的归一化权重，作为系统默认方案。
DEFAULT_WEIGHTS = {
```

```
        "岩质边坡"：  {
            "ｌ１"：  {"植被指标"：  ０．４５，  "地形指标"：  ０．５５},
            "ｌ２"：  {
                "植被指标"：  {"植被覆盖度"：  ０．６７，  "生物多样性"：  ０．３３},
                "地形指标"：  {"坡度"：  ０．６４，  "坡向"：  ０．３６},
            },
        },
        "碎石边坡"：  {
            "ｌ１"：  {"土壤指标"：  ０．４４，  "植被指标"：  ０．３１，  "地形指标"：  ０．２５},
            "ｌ２"：  {
                "土壤指标"：  {"土石比"：  ０．２５，  "土层厚度"：  ０．２５，  "ｐＨ"：  ０．１７，
                              "有机质"：  ０．１７，  "污染指标"：  ０．１７},
                "植被指标"：  {"植被覆盖度"：  ０．６４，  "生物多样性"：  ０．３６},
                "地形指标"：  {"坡度"：  ０．６７，  "坡向"：  ０．３３},
            },
        },
        "平台"：  {
            "ｌ１"：  {"土壤指标"：  ０．５４，  "植被指标"：  ０．４６},
            "ｌ２"：  {
                "土壤指标"：  {"土石比"：  ０．２５，  "土层厚度"：  ０．２５，  "ｐＨ"：  ０．１７，
                              "有机质"：  ０．１７，  "污染指标"：  ０．１７},
                "植被指标"：  {"植被覆盖度"：  ０．６１，  "生物多样性"：  ０．３９},
            },
        },
}
#  - - -
```

```
"""
from sqlalchemy import text
#  （表名， 列名， 列定义）
REQUIRED_COLUMNS = [
    #  遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    #  指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    #  多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
    #  用户个人资料（账户页）
    ("users", "display_name", "VARCHAR(64)"),
    ("users", "email", "VARCHAR(128)"),
    ("users", "phone", "VARCHAR(32)"),
    ("users", "gender", "VARCHAR(8)"),
    ("users", "avatar", "VARCHAR(256)"),
]
#  职务体系由「管理员 / 评价员 / 查看者」三种收敛为「管理员 / 操作员」两种，
#  旧数据里的后两种统一归为操作员。
ROLE_MIGRATION = {"evaluator": "operator", "viewer":
    "operator"}
def _add_columns(db):
    added = []
    with db.engine.connect() as conn:
        for table, column, ddl in REQUIRED_COLUMNS:
            rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
            if not rows:
                continue                            #
    表还不存在，create_all 会建
            existing = {r[1] for r in rows}
            if column in existing:
                continue
            conn.execute(text(
                "ALTER TABLE %s ADD COLUMN %s %s" % (table,
    column, ddl)))
            conn.commit()
            added.append("%s.%s" % (table, column))
    return added
def _migrate_roles(db):
    """把旧角色名统一为新的两种，返回改动的行数。"""
    changed = 0
    with db.engine.connect() as conn:
        rows = conn.execute(text("PRAGMA
    table_info(users)")).fetchall()
```

```
        if not rows:                                                    951
            return 0                                                    952
        for old, new in ROLE_MIGRATION.items():                         953
            r = conn.execute(text("UPDATE users SET role =              954
    :new "                                                              955
                                  "WHERE role = :old"),                 956
                                 {"new": new, "old": old})              957
            changed += r.rowcount or 0                                  958
        conn.commit()                                                   959
    return changed                                                      960
def ensure_schema(db):                                                  961
    """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""                    962
    try:                                                                963
        changed = _add_columns(db)                                      964
        n = _migrate_roles(db)                                          965
        if n:                                                           966
            changed.append("用户角色归一为管理员/操作员 %d 条" % n)              967
        return changed                                                  968
    except Exception as e:                                  #           969
    迁移失败不应阻断启动                                                  970
        print("[schema] 结构检查跳过：%s" % e)                             971
        return []                                                       972
# ==== app/utils/uploads.py ====                                        973
# -*- coding: utf-8 -*-                                                 974
"""上传文件处理（遥感影像、批量导入）。"""                                    975
import os                                                               976
import re                                                               977
import uuid                                                             978
from datetime import datetime                                           979
from werkzeug.utils import secure_filename                              980
def _safe_ext(filename):                                                981
    return os.path.splitext(filename)[1].lower().lstrip(".")            982
def ext_allowed(filename, allowed):                                     983
    return _safe_ext(filename) in allowed                               984
def save_upload(file_storage, subdir, allowed_ext,                      985
    upload_root):                                                       986
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。    987
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。                 988
    """                                                                 989
    if file_storage is None or not file_storage.filename:               990
        return None, "未选择文件"                                        991
    ext = _safe_ext(file_storage.filename)                              992
    if ext not in allowed_ext:                                          993
        return None, "不支持的文件类型：.%s" % ext                        994
    target_dir = os.path.join(upload_root, subdir)                      995
    os.makedirs(target_dir, exist_ok=True)                              996
    # 仅保留安全字符，并加时间戳与随机串防重名                                997
    base = secure_filename(os.path.splitext(file_storage.fil            998
    ename)[0]) or "file"                                                999
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or                1000
```

```
    "file"                                                                1001
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M1002
    %S"),                                                                 1003
                                uuid.uuid4().hex[:6], ext)                1004
    abs_path = os.path.join(target_dir, name)                             1005
    file_storage.save(abs_path)                                           1006
    rel = os.path.join("uploads", subdir,                                 1007
    name).replace("\\", "/")                                              1008
    return rel, None                                                      1009
def remove_upload(rel_path, upload_root):                                 1010
    """删除已保存的上传文件（相对 static 的路径）。"""                    1011
    if not rel_path:                                                      1012
        return                                                            1013
    abs_path = os.path.join(upload_root,                                  1014
    rel_path.replace("uploads/", "", 1))                                  1015
    try:                                                                  1016
        if os.path.isfile(abs_path):                                      1017
            os.remove(abs_path)                                           1018
    except OSError:                                                       1019
        pass                                                              1020
# ==== app/utils/charts.py ====                                           1021
# -*- coding: utf-8 -*-                                                   1022
"""轻量图表。                                                            1023
用纯 SVG 输出，不依赖任何绘图库或前端脚本：                              1024
- 服务端渲染，页面与打印（PDF）中都能稳定显示                            1025
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）              1026
- 单色方案，符合界面「一个视图一个强调色」的约定                        1027
"""                                                                       1028
from html import escape                                                   1029
ACCENT = "#1f3a5f"                                                        1030
ACCENT_SOFT = "#7f93ad"                                                   1031
INK = "#23262b"                                                           1032
MUTED = "#6b7280"                                                         1033
LINE = "#dfe3e8"                                                          1034
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分                            1035
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",                      1036
    "#bcc9da", "#dbe3ec"]                                                 1037
def _fmt(v, digits=4):                                                    1038
    try:                                                                  1039
        return ("%%.%df" % digits) % float(v)                             1040
    except (TypeError, ValueError):                                       1041
        return str(v)                                                     1042
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,                1043
    value_w=76):                                                          1044
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                     1045
    if not items:                                                         1046
        return '<p class="muted">暂无数据</p>'                            1047
    n = len(items)                                                        1048
    height = n * (bar_h + gap) + 10                                       1049
    max_v = max((abs(float(v)) for _, v, _ in items),                     1050
```

```
    default=1) or 1                                                    1051
    body_w = width - label_w - value_w                                 1052
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                   1053
    height="%d" '                                                      1054
              'role="img" aria-label="条形图" '                        1055
              'style="max-width:%dpx;font-family:inherit;">'           1056
    % (width, height, height, width)]                                  1057
    for i, (label, value, disp) in enumerate(items):                   1058
        y = i * (bar_h + gap) + 5                                      1059
        frac = (abs(float(value)) / max_v) if max_v else 0             1060
        w = max(2.0, body_w * frac)                                    1061
        fill = GRAYS[i % len(GRAYS)]                                   1062
        parts.append(                                                  1063
            '<text x="%d" y="%d" text-anchor="end"                     1064
    font-size="12" fill="%s">%s</text>'                                1065
            % (label_w - 10, y + bar_h * 0.72, INK,                    1066
    escape(str(label))))                                               1067
        parts.append(                                                  1068
            '<rect x="%d" y="%d" width="%.1f" height="%d"              1069
    rx="3" fill="%s"/>'                                                1070
            % (label_w, y, w, bar_h, fill))                            1071
        parts.append(                                                  1072
            '<text x="%d" y="%d" font-size="12" fill="%s" '            1073
            'style="font-variant-numeric:tabular-nums;">%s</           1074
    text>'                                                             1075
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,               1076
    escape(str(disp))))                                                1077
    parts.append('</svg>')                                             1078
    return "".join(parts)                                              1079
def donut(items, size=210, thickness=34):                              1080
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]                1081
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。                  1082
    """                                                                1083
    vals = [(l, float(v), d) for l, v, d in items if                   1084
    float(v) > 0]                                                      1085
    if not vals:                                                       1086
        return '<p class="muted">暂无数据</p>'                          1087
    total = sum(v for _, v, _ in vals) or 1                            1088
    r = (size - thickness) / 2                                         1089
    cx = cy = size / 2                                                 1090
    import math                                                        1091
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                      1092
    height="%d" role="img" '                                           1093
             'aria-label="权重分布环形图">' % (size, size, size,       1094
    size)]                                                             1095
    angle = -math.pi / 2                                               1096
    for i, (label, v, disp) in enumerate(vals):                        1097
        sweep = 2 * math.pi * (v / total)                              1098
        a0, a1 = angle, angle + sweep                                  1099
        x0, y0 = cx + r * math.cos(a0), cy + r *                       1100
```

```
    math.sin(a0)                                                    1101
        x1, y1 = cx + r * math.cos(a1), cy + r *                    1102
    math.sin(a1)                                                    1103
        large = 1 if sweep > math.pi else 0                         1104
        parts.append(                                               1105
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f           1106
    %.2f" '                                                         1107
            'fill="none" stroke="%s" stroke-width="%d"/>'           1108
           % (x0, y0, r, r, large, x1, y1, GRAYS[i %                1109
    len(GRAYS)], thickness))                                        1110
        angle = a1                                                  1111
    parts.append('<text x="%d" y="%d" text-anchor="middle"          1112
    font-size="12" '                                                1113
                 'fill="%s">合计</text>' % (cx, cy - 2,             1114
    MUTED))                                                         1115
    parts.append('<text x="%d" y="%d" text-anchor="middle"          1116
    font-size="15" '                                                1117
                 'fill="%s" style="font-variant-numeric:tabu        1118
    lar-nums;">%s</text>'                                           1119
                 % (cx, cy + 18, INK, _fmt(total, 4)))              1120
    parts.append('</svg>')                                          1121
    legend = ['<table style="width:auto;margin-top:10px;"><t        1122
    head><tr>'                                                      1123
              '<th scope="col">项目</th><th                         1124
    scope="col">数值</th>'                                          1125
              '<th scope="col">占比</th></tr></thead><tbody>']      1126
    for i, (label, v, disp) in enumerate(vals):                     1127
        legend.append(                                              1128
            '<tr><td style="text-align:left;">'                     1129
            '<span style="display:inline-block;width:10px;he        1130
    ight:10px;'                                                     1131
            'border-radius:2px;background:%s;margin-right:6p        1132
    x;"></span>%s</td>'                                             1133
            '<td class="mono">%s</td><td                            1134
    class="mono">%.1f%%</td></tr>'                                  1135
           % (GRAYS[i % len(GRAYS)], escape(str(label)),            1136
    escape(str(disp)),                                              1137
               100.0 * v / total))                                  1138
    legend.append('</tbody></table>')                               1139
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p        1140
    x;align-items:center;">'                                        1141
            + "".join(parts) + "".join(legend) + '</div>')          1142
# ==== app/utils/browser.py ====                                    1143
# -*- coding: utf-8 -*-                                             1144
"""HTML 转 PDF（报告导出）。                                        1145
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、     1146
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，  1147
避免中文路径在 file:// URL 下出问题。                               1148
"""                                                                 1149
import os                                                           1150
```

```
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
```

```
            ]
            p = subprocess.run(cmd, capture_output=True,
      text=True,
                               timeout=timeout,
      encoding="utf-8", errors="replace")
            if not os.path.isfile(pdf_path) or
      os.path.getsize(pdf_path) < 800:
                return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
      "").strip()[:200]
            with open(pdf_path, "rb") as f:
                return f.read(), None
      except subprocess.TimeoutExpired:
            return None, "生成 PDF 超时，请稍后重试"
      except Exception as e:
            return None, "生成 PDF 失败：%s" % e
      finally:
            shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
      """由判断矩阵上三角构造完整互反矩阵。
      参数
      ----
      upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
            例：3 阶时传入 [[a12, a13], [a23]]。
            对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
      """
      upper = np.asarray(upper, dtype=float)
      k = upper.shape[0]
      n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
      A = np.ones((n, n), dtype=float)
      idx = 0
      for i in range(n):
            for j in range(i + 1, n):
                A[i, j] = upper[idx]
                A[j, i] = 1.0 / upper[idx]
                idx += 1
      return A
def weights_and_consistency(A):
      """由判断矩阵 A 计算权重与一致性指标。
      返回
      ----
```

```
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
```

```
                                "expected": expected, "deviation":  1301
    deviation}                                                      1302
    return best                                                     1303
def aggregate_matrices(matrices):                                   1304
    """多专家判断矩阵聚合：几何平均（保持互反性）。                  1305
    参数                                                            1306
    ----                                                            1307
    matrices : list[np.ndarray]，多个同阶判断矩阵。                 1308
    """                                                             1309
    stack = np.stack([np.asarray(m, dtype=float) for m in           1310
    matrices])                                                      1311
    return np.prod(stack, axis=0) ** (1.0 / len(stack))             1312
def judge_from_weights(weights):                                    1313
    """由权重向量构造完全一致判断矩阵并检验。                        1314
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量  1315
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。                       1316
    """                                                             1317
    w = np.asarray(weights, dtype=float)                            1318
    n = w.shape[0]                                                  1319
    A = np.outer(w, 1.0 / w)                                        1320
    r = weights_and_consistency(A)                                  1321
    r["matrix"] = A                                                 1322
    return r                                                        1323
# ==== app/services/standardization.py ====                       1324
# -*- coding: utf-8 -*-                                           1325
"""指标标准化（的一部分）。                                        1326
按论文 4.2.2 节式(8) 实现：                                        1327
    x' = (x - x_min) / (x_max - x_min)                              1328
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。        1329
所有理想极值取自 app.utils.constants，可配置。                     1330
"""                                                                 1331
from app.utils.constants import (                                   1332
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,              1333
    IDEAL_ORGANIC,                                                  1334
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,               1335
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,           1336
    PH_TOL_LOW, PH_TOL_HIGH,                                        1337
)                                                                   1338
def std_benefit(value, ideal):                                      1339
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""  1340
    if value is None:                                               1341
        return None                                                 1342
    return max(0.0, min(1.0, value / ideal))                        1343
def std_cost(value, extreme):                                       1344
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""      1345
    if value is None:                                               1346
        return None                                                 1347
    return max(0.0, min(1.0, 1.0 - value / extreme))                1348
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,                    1349
    tol_hi=PH_TOL_HIGH):                                            1350
```

```
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
    HTS 结构），
                未提供时用默认值。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
```

```
    """                                                                 1401
    from app.utils.constants import DEFAULT_WEIGHTS                     1402
    if weights is None:                                                 1403
        weights = DEFAULT_WEIGHTS[unit_type]                            1404
    l1_enabled = weights["l1"].keys()                                   1405
    l2w = weights["l2"]                                                 1406
    result = {}                                                         1407
    # ---- 土壤指标 ----                                                1408
    if "土壤指标" in l1_enabled:                                        1409
        result["土壤指标"] = {                                          1410
            "土石比": (data.get("soil_ratio"),                          1411
                       std_benefit(data.get("soil_ratio"),              1412
IDEAL_SOIL_RATIO),                                                      1413
                       l2w["土壤指标"]["土石比"]),                      1414
            "土层厚度": (data.get("thickness"),                         1415
                         std_benefit(data.get("thickness"),             1416
IDEAL_THICKNESS),                                                       1417
                         l2w["土壤指标"]["土层厚度"]),                  1418
            "pH": (data.get("ph"),                                      1419
                   std_mid_ph(data.get("ph"),                           1420
*IDEAL_PH_RANGE),                                                       1421
                   l2w["土壤指标"]["pH"]),                              1422
            "有机质": (data.get("organic_matter"),                      1423
                       std_benefit(data.get("organic_matter")1424
, IDEAL_ORGANIC),                                                       1425
                       l2w["土壤指标"]["有机质"]),                      1426
            "污染指标": (data.get("heavy_metals") or {},                1427
                         pollution_score(data.get("heavy_meta1428
ls") or {}),                                                            1429
                         l2w["土壤指标"]["污染指标"]),                  1430
        }                                                               1431
    # ---- 植被指标 ----                                                1432
    if "植被指标" in l1_enabled:                                        1433
        result["植被指标"] = {                                          1434
            "植被覆盖度": (data.get("coverage"),                        1435
                           std_benefit(data.get("coverage"),            1436
IDEAL_COVERAGE),                                                        1437
                           l2w["植被指标"]["植被覆盖度"]),              1438
            "生物多样性": (data.get("shannon"),                         1439
                           std_benefit(data.get("shannon"),             1440
IDEAL_SHANNON),                                                         1441
                           l2w["植被指标"]["生物多样性"]),              1442
        }                                                               1443
    # ---- 地形指标 ----                                                1444
    if "地形指标" in l1_enabled:                                        1445
        result["地形指标"] = {                                          1446
            "坡度": (data.get("slope"),                                 1447
                     std_slope(data.get("slope")),                      1448
                     l2w["地形指标"]["坡度"]),                          1449
            "坡向": (data.get("aspect"),                                1450
```

```
                                aspect_score(data.get("aspect")),
                                l2w["地形指标"]["坡向"]),
            }
        return result
# ==== app/services/weight_service.py ====
# -*- coding: utf-8 -*-
"""权重方案的存取与生效权重解析。
系统有两套权重来源：
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），
   是**权威取值来源**，案例复算与报告结论以此为准；
2. 通过层次分析法在本系统中实际算出的权重，保存在
   weight_schemes / weight_items 两张表中，可标记为「生效」。
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：
以论文表 3 为底，用生效方案中已保存的条目覆盖。
没有任何生效方案时，即完全使用论文表 3 的权重。
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，
而不是算完即弃。
"""
import copy
from app.models import db, WeightScheme, WeightItem
from app.utils.constants import DEFAULT_WEIGHTS
from app.services.indicator_service import TREE
#  方案名称：每个微地质单元类型固定一个可生效的方案
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"
def _code_maps():
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""
    l1, l2 = {}, {}
    for row in TREE:
        code, name, level, parent = row[0], row[1], row[2],
    row[3]
        if level == 1:
            l1[code] = name
        elif level == 2:
            l2[code] = (l1.get(parent, parent), name)
    return l1, l2
def effective_weights(unit_type):
    """返回该单元类型当前生效的权重。
    返回 (weights, source_label, scheme)
        weights      嵌套权重结构，可直接交给标准化与评价引擎
        source_label 权重来源说明，用于界面与报告标注
        scheme       生效的权重方案对象，无自定义方案时为 None
    """
    base = DEFAULT_WEIGHTS.get(unit_type)
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,

    name=ACTIVE_NAME).first()
    if scheme is None:
```

```
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.
    id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
    merged["l2"][l1_name]:
                merged["l2"][l1_name][l2_name] = w
                applied += 1
    if applied == 0:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme
def save_weights(unit_type, code_weight_pairs, note=None):
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）
    分次录入而无需一次填完。
    """
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,

    name=ACTIVE_NAME).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
    name=ACTIVE_NAME,
                              level=1, is_default=True)
        db.session.add(scheme)
        db.session.flush()
    if note:
        scheme.level = scheme.level or 1
    if not scheme.is_default:
        scheme.is_default = True
    written = 0
    for code, w in code_weight_pairs:
        if w is None or w <= 0:
            continue
        item = WeightItem.query.filter_by(scheme_id=scheme.i
    d,

    indicator_code=code).first()
        if item is None:
            item = WeightItem(scheme_id=scheme.id,
```

```
        indicator_code=code)                                                    1551
                db.session.add(item)                                            1552
            item.weight = float(w)                                              1553
            written += 1                                                        1554
    db.session.commit()                                                         1555
    return scheme, written                                                      1556
def reset_weights(unit_type):                                                   1557
    """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""                1558
    schemes = WeightScheme.query.filter_by(unit_type=unit_ty                    1559
    pe,                                                                         1560
                                                                                1561
    name=ACTIVE_NAME).all()                                                     1562
    n = 0                                                                       1563
    for s in schemes:                                                           1564
        WeightItem.query.filter_by(scheme_id=s.id).delete(                      1565
            synchronize_session=False)                                          1566
        db.session.delete(s)                                                    1567
        n += 1                                                                  1568
    db.session.commit()                                                         1569
    return n                                                                    1570
def scheme_items(unit_type):                                                    1571
    """返回该单元类型已保存的自定义权重条目 {指标编码： 权重}。"""               1572
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ                    1573
    e,                                                                          1574
                                                                                1575
    name=ACTIVE_NAME).first()                                                   1576
    if scheme is None:                                                          1577
        return {}, None                                                         1578
    items = {i.indicator_code: i.weight                                         1579
             for i in WeightItem.query.filter_by(scheme_id=s                    1580
    cheme.id).all()}                                                            1581
    return items, scheme                                                        1582
def summary():                                                                  1583
    """各单元类型的权重来源概览，供权重首页展示。"""                            1584
    rows = []                                                                   1585
    for unit_type in DEFAULT_WEIGHTS:                                           1586
        weights, label, scheme = effective_weights(unit_type                    1587
    )                                                                           1588
        rows.append({                                                           1589
            "unit_type": unit_type,                                             1590
            "label": label,                                                     1591
            "custom": scheme is not None,                                       1592
        })                                                                      1593
    return rows                                                                 1594
# ==== app/services/evaluation_service.py ====                                  1595
# -*- coding: utf-8 -*-                                                         1596
"""适宜性评价引擎。                                                             1597
按论文式(９) 实现：                                                             1598
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）                 1599
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比                          1600
```

```
- 等级判定按论文表 4。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None):
    """计算单个微地质单元的评价因子 F'。
    返回
    ----
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:
    [...] }
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    std_result = std.standardize_unit(unit_type, data,
    weights)
    l1_weights = weights["l1"]
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
    raw,
                    "std_value": None, "weight": w2,
    "contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
                "contribution": contribution,
            })
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
    # 一级指标权重按实际参与计算的指标归一化。
    # 论文表 3 的权重本身已归一（和为 1），此处为恒等变换；
    # 若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。
    lw_sum = sum(l1_weights[k] for k in l1_scores if k in
    l1_weights)
    if lw_sum > 0:
        unit_score = sum(l1_weights[k] * l1_scores[k]
                         for k in l1_scores if k in
```

```
    l1_weights) / lw_sum
    else:
        unit_score = 0.0
    return {"l1_scores": l1_scores, "unit_score":
    unit_score, "details": details}
def score_mine(units, weights_map=None):
    """计算整个矿山的评价因子 F 与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    weights_map : dict，{单元类型: 权重结构}。未提供的单元类型
        使用论文表 3 的默认权重。权重可来自 AHP 计算并经
        weight_service 保存生效，从而真正参与评价。
    返回
    ----
    dict:{ final_score, grade, grade_info, unit_results }
    """
    weights_map = weights_map or {}
    total = 0.0
    unit_results = []
    for u in units:
        r = score_unit(u["unit_type"], u["data"],
                       weights_map.get(u["unit_type"]))
        weighted = u["ratio"] * r["unit_score"]
        total += weighted
        unit_results.append({
            "unit_type": u["unit_type"],
            "ratio": u["ratio"],
            "unit_score": r["unit_score"],
            "weighted": weighted,
            "l1_scores": r["l1_scores"],
            "details": r["details"],
        })
    grade_info = grade_of(total)
    return {
        "final_score": total,
        "grade": grade_info["grade"],
        "grade_info": grade_info,
        "unit_results": unit_results,
    }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from docx import Document
```

```
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
from app.utils.timefmt import to_local, now_utc
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
        run.font.name = "Times New Roman"
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")
    return h
def _table(doc, headers, rows, widths=None):
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
        _set_run(hdr[i].paragraphs[0].add_run(), htext,
    10.5, bold=True)
    for row in rows:
        cells = t.add_row().cells
        for i, val in enumerate(row):
            _set_run(cells[i].paragraphs[0].add_run(),
    str(val), 10.5)
    return t
def build_report(patch, mine_result):
    """生成评价报告 Word 文档，返回 BytesIO。
    参数
    ----
    patch : MinePatch，图斑对象
    mine_result : evaluation_service.score_mine 的返回 dict
    """
    doc = Document()
    # 页边距
    for section in doc.sections:
```

```
        section.top_margin = Cm(2.5)                                    1751
        section.bottom_margin = Cm(2.5)                                 1752
        section.left_margin = Cm(2.5)                                   1753
        section.right_margin = Cm(2.5)                                  1754
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)                  1755
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0",             1756
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)                 1757
    _para(doc, "生成时间：%s" % to_local(now_utc(), "%Y-%m-%d              1758
    %H:%M:%S"),                                                         1759
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,                 1760
    space_after=12)                                                     1761
    # 图斑信息                                                           1762
    _heading(doc, "一、图斑基本信息", 1)                                   1763
    info = [                                                            1764
        ("图斑名称", patch.name),                                        1765
        ("所在行政区", patch.admin_region or "—"),                       1766
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is      1767
    not None else "—"),                                                 1768
        ("开采矿种", patch.mineral_type or "—"),                         1769
        ("开采方式", patch.mining_method or "—"),                        1770
    ]                                                                   1771
    _table(doc, ["项目", "内容"], info)                                   1772
    # 评价结论                                                           1773
    _heading(doc, "二、评价结论", 1)                                      1774
    gi = mine_result["grade_info"]                                      1775
    concl = [                                                           1776
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),         1777
        ("适宜性等级", mine_result["grade"]),                            1778
        ("判定依据", gi["meaning"]),                                     1779
        ("推荐措施", gi["measure"]),                                     1780
    ]                                                                   1781
    _table(doc, ["项目", "结果"], concl)                                  1782
    # 各单元得分                                                         1783
    _heading(doc, "三、各微地质单元得分", 1)                               1784
    rows = []                                                           1785
    for u in mine_result["unit_results"]:                               1786
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"]             1787
    * 100),                                                             1788
                     "%.4f" % u["unit_score"], "%.4f" %                 1789
    u["weighted"]))                                                     1790
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"],      1791
    rows)                                                               1792
    # 完整得分明细                                                       1793
    _heading(doc, "四、得分明细", 1)                                      1794
    for u in mine_result["unit_results"]:                               1795
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"],                1796
    u["ratio"] * 100),                                                  1797
              size=11, bold=True, space_after=4)                        1798
        detail_rows = []                                                1799
        for d in u["details"]:                                          1800
```

```
                raw = "—" if d["raw"] is None else d["raw"]
                std = "—" if d["std_value"] is None else "%.4f"
    % d["std_value"]
                w = "%.4f" % d["weight"]
                c = "—" if d["contribution"] is None else
    "%.4f" % d["contribution"]
                detail_rows.append((d["l1"], d["l2"], raw, std,
    w, c))
            _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重",
    "贡献"],
                   detail_rows)
            # 一级指标得分
            l1_row = []
            for k, v in u["l1_scores"].items():
                l1_row.append("%s=%.4f" % (k, v))
            _para(doc, "一级指标得分：" + ",".join(l1_row), size=10.5,
    space_after=10)
    buf = io.BytesIO()
    doc.save(buf)
    buf.seek(0)
    return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
    # ---- 二级指标：土壤 ----
    ("soil_ratio", "土石比", 2, "soil", "—",
```

```
        str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
         "论文 4.2.2 节:理想值 0.8,根据矿山修复实际情况确定", False),
        ("thickness", "土层厚度", 2, "soil", "cm",
        str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
         "论文 4.2.2 节:理想值 20 cm", False),
        ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
        "mid", "碎石边坡,平台",
         "论文 4.2.2 节:pH 用适中指标的标准化方法,理想区间 %g~%g" %
        IDEAL_PH_RANGE, False),
        ("organic_matter", "有机质", 2, "soil", "g/kg",
        str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
         "《全国第二次土壤普查养分等级划分标准》二等(较丰富)最低值 30 g/kg", False),
        ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
        "cost", "碎石边坡,平台",
         "同时作为限制性指标与评价指标;8 种重金属等权合成(各 0.125)", True),
        # ---- 二级指标:植被 ----
        ("coverage", "植被覆盖度", 2, "veg", "%",
        str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
         "论文 4.2.2 节:参照生态系统取值 75%", False),
        ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
        "benefit", ALL_UNITS,
         "Shannon-Wiener 多样性指数,参照生态系统取值 0.7", False),
        # ---- 二级指标:地形 ----
        ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
        SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
         "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号:25°
        为陡坡重要分界", False),
        ("aspect", "坡向", 2, "terrain", "—",
        "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
         "论文 4.2.2 节:阴坡及半阴坡水分条件更优,故赋值高于阳坡", False),
]
# 污染指标下的三级指标(8 种重金属)
for _m, _limit in HEAVY_METAL_LIMIT.items():
        TREE.append((
              "hm_" + _m, _m, 3, "pollution", "mg/kg",
        str(_limit), "cost",
              "碎石边坡,平台",
              "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
        _limit,
              False,
        ))
def seed_indicators():
        """写入/更新指标定义(幂等)。返回写入条数。"""
        code_map = {
              "benefit": "越大越优", "cost": "越小越优",
              "mid": "适中型", "discrete": "离散赋值",
        }
        n = 0
        for (code, name, level, parent, unit, ideal, itype,
        units, desc, is_con) in TREE:
```

```
            ind = Indicator.query.filter_by(code=code).first()
            if ind is None:
                ind = Indicator(code=code)
                db.session.add(ind)
            ind.name = name
            ind.level = level
            ind.parent_code = parent
            ind.unit = unit
            ind.ideal_value = ideal
            ind.ideal_type = itype
            ind.enabled_units = units
            ind.description = desc
            ind.is_constraint = is_con
            ind.is_evaluative = (level in (2, 3))
            n += 1
        db.session.commit()
        return n
def ideal_type_label(code):
        return {"benefit": "越大越优", "cost": "越小越优",
                "mid": "适中型", "discrete": "离散赋值"}.get(code,
        code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
        单元类型， 土石比， 土层厚度， pH， 有机质，
        Hg， As， Ni， Cu， Cd， Pb， Cr， Zn，
        植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
        UNIT_TYPES
# 表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
        "单元类型": "unit_type",
        "土石比": "soil_ratio",
        "土层厚度": "thickness",
        "pH": "ph", "PH": "ph", "ph": "ph",
        "有机质": "organic_matter",
        "植被覆盖度": "coverage",
        "Shannon": "shannon", "shannon": "shannon", "生物多样性":
        "shannon",
        "坡度": "slope",
```

```
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    #  示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
    for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
    in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
    None)
                        for i in range(len(header)) if
    header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
```

```
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict, {单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict], 每项 {unit_type, fields{...},
    heavy_metals{}}
      errors : list[str], 逐行问题说明
    """
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    # 表头校验
    unknown = [h for h in (rows[0].keys() if rows else [])
    if h not in FIELD_ALIASES]
    if unknown:
        errors.append("存在无法识别的列，将被忽略：%s" %
    "、".join(map(str, unknown)))
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头
        rec = {}
        for k, v in raw.items():
            key = FIELD_ALIASES.get(str(k).strip())
            if key:
                rec[key] = v
        unit_type = str(rec.get("unit_type") or "").strip()
        if not unit_type:
            errors.append("第 %d 行：缺少「单元类型」" % idx)
            continue
        if unit_type not in unit_type_map:
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
                          % (idx, unit_type,
    "、".join(unit_type_map) or "无"))
            continue
        if unit_type not in UNIT_TYPES:
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"
                          % (idx, unit_type))
            continue
```

```
            fields, heavy = {}, {}
            ok = True
            for key, val in rec.items():
                if key in ("unit_type",):
                    continue
                if key.startswith("hm_"):
                    n, err = _num(val)
                    if err:
                        errors.append("第 %d 行:%s %s" % (idx,
    key[3:], err))
                        ok = False
                    elif n is not None:
                        heavy[key[3:]] = n
                    continue
                if key == "aspect":
                    s = str(val).strip() if val is not None
    else ""
                    if s == "":
                        fields["aspect"] = None
                    elif s in ASPECT_TYPES:
                        fields["aspect"] = s
                    else:
                        errors.append("第 %d 行:坡向「%s」无效（应为 %s）"
                                      % (idx, s,
    "、".join(ASPECT_TYPES)))
                        ok = False
                    continue
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行:%s %s" % (idx, key,
    err))
                    ok = False
                else:
                    fields[key] = n
            if ok:
                parsed.append({"unit_type": unit_type,
    "fields": fields,
                               "heavy_metals": heavy})
        return parsed, errors
# ==== app/services/workflow.py ====
# -*- coding: utf-8 -*-
"""评价流程前置条件（定量评价与报告输出的流程门禁）。
论文 4.1 节确定的评价流程存在严格的先后关系：
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
```

```
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                            EvaluationResult)
#  指标数据非空字段（用于判断某单元是否已录入数据）
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
                "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
        return True
    return bool(ind.heavy_metals_dict())
def patch_progress(patch_id):
    """检查某图斑的评价流程完成情况。
    返回 dict：
        units       {done, count, ratio_sum, detail}
        constraint  {done, passed, detail}
        data        {done, ready, missing, detail}
        blocking    未满足的前置步骤列表（按流程顺序）
        ready       是否可进入定量评价
    """
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    #  ----  步骤一：微地质单元  ----
    units_step = {
        "done": bool(units),
        "count": len(units),
        "ratio_sum": ratio_sum,
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"
                   % (len(units), ratio_sum * 100))
                  if units else "尚未划分微地质单元",
    }
    #  ----  步骤二：限制性条件判定  ----
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)
    \
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
        constraint_step = {"done": True, "passed": False,
```

```
                                "detail": "限制性条件判定未通过（%s）" %
(cc.reason or "")}
# ---- 步骤三：指标数据 ----
missing = []
for u in units:
    ind = IndicatorData.query.filter_by(unit_id=u.id).fi
rst()
    if not _has_data(ind):
        missing.append(u.unit_type)
if not units:
    data_step = {"done": False, "ready": False,
"missing": [],
                 "detail": "尚无可录入数据的微地质单元"}
elif missing:
    data_step = {"done": False, "ready": False,
"missing": missing,
                 "detail": "以下微地质单元尚未录入指标数据：%s"
                           % "、".join(missing)}
else:
    data_step = {"done": True, "ready": True,
"missing": [],
                 "detail": "全部 %d 个微地质单元的指标数据已录入" %
len(units)}
# 占比合计必须落在 １００% 附近才算通过。只挡「偏小」是不够的：
# 库里若出现合计 １．５ 的记录（手工改库、导入异常），F = Σ(Si×F'i)
# 会被放大到 １．５ 倍，所以上下都要挡。
ratio_ok = 0.999 <= ratio_sum <= 1.001 if units else
False
if units and not ratio_ok:
    units_step["detail"] += "（占比合计不等于 １００%，请检查面积录入）"
# ---- 汇总未满足的前置条件（按论文 ４．１ 节的流程顺序：限制性判定优先） ----
# reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
# 缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
blocking = []
if not constraint_step["done"]:
    blocking.append({
        "step": "constraint", "label": "限制性条件判定未执行",
        "reason": "五项限制性条件尚未逐项判定。",
        "fix_url": "/constraint/%d" % patch_id,
"fix_label": "去执行判定",
    })
elif not constraint_step["passed"]:
    blocking.append({
        "step": "constraint", "label": "限制性条件判定未通过",
        "reason": cc.reason or "",
        "fix_url": "/constraint/%d" % patch_id,
"fix_label": "去复核判定结果",
    })
if not units_step["done"]:
    blocking.append({
```

```
                "step": "units", "label": "未划分微地质单元",                2201
                "reason": "该图斑下没有任何微地质单元。",                    2202
                "fix_url": "/unit/%d" % patch_id, "fix_label":    2203
    "去划分",                                                               2204
            })                                                              2205
        elif not ratio_ok:                                                  2206
            blocking.append({                                               2207
                "step": "units", "label": "面积占比合计不为 100%",           2208
                "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),           2209
                "fix_url": "/unit/%d" % patch_id, "fix_label":    2210
    "去核对面积",                                                           2211
            })                                                              2212
        if units_step["done"] and not data_step["ready"]:                   2213
            blocking.append({                                               2214
                "step": "data", "label": "指标数据未录入完整",               2215
                "reason": data_step["detail"],                              2216
                "fix_url": "/indicator/", "fix_label": "去录入",            2217
            })                                                              2218
        return {                                                            2219
            "units": units_step,                                            2220
            "constraint": constraint_step,                                  2221
            "data": data_step,                                              2222
            "blocking": blocking,                                           2223
            "ready": not blocking,                                          2224
        }                                                                   2225
def latest_result(patch_id):                                                2226
    """取该图斑最近一次的评价记录。"""                                      2227
    return EvaluationResult.query.filter_by(patch_id=patch_i    2228
    d) \                                                                    2229
        .order_by(EvaluationResult.id.desc()).first()                       2230
def report_progress(patch_id):                                              2231
    """报告输出的前置条件：评价流程齐备，且已有评价记录。                   2232
    报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，              2233
    导出时读取已保存的评价结果，而不是在导出环节重新计算。                  2234
    否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，        2235
    同一份评价出现两个数字，结论无法追溯。                                  2236
    返回 dict：{ready, blocking, result, flow}                              2237
        result 为最近一次评价记录（EvaluationResult），无记录时为空。       2238
    """                                                                     2239
    flow = patch_progress(patch_id)                                         2240
    blocking = list(flow["blocking"])                                       2241
    record = latest_result(patch_id)                                        2242
    if not blocking and record is None:                                     2243
        blocking.append({                                                   2244
            "step": "evaluate", "label": "尚未执行适宜性评价",              2245
            "reason": "该图斑还没有评价记录。",                             2246
            "fix_url": "/evaluate/%d" % patch_id,                           2247
            "fix_label": "去执行评价",                                      2248
        })                                                                  2249
    return {                                                                2250
```

```
            "ready": not blocking,
            "blocking": blocking,
            "result": record,
            "flow": flow,
        }
# ==== app/services/dashboard.py ====
# -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）
#    key          步骤标识
#    no           步骤编号（01~08）
#    title        名称
#    icon         图标名（见 templates/_macros.html 的 ICONS）
#    desc         一句话说明
#    href         入口地址
STEPS = [
    dict(key="project", no="01", title="项目与矿山档案",
    icon="folder",
          href="/project/",
          desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),
    dict(key="standard", no="02", title="政策与标准知识库",
    icon="book",
          href="/standard/",
          desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),
    dict(key="constraint", no="03", title="限制性指标判定",
    icon="shield",
          href="/constraint/",
          desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),
    dict(key="unit", no="04", title="微地质单元", icon="layers",
          href="/unit/",
          desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),
    dict(key="ahp", no="05", title="指标体系与权重",
    icon="sliders",
          href="/ahp/",
          desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),
    dict(key="indicator", no="06", title="指标数据采集",
    icon="database",
          href="/indicator/",
          desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),
    dict(key="evaluate", no="07", title="适宜性定量评价",
    icon="target",
```

```
            href="/evaluate/",                                              2301
            desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),          2302
        dict(key="report", no="08", title="报告与结果输出",                  2303
        icon="file",                                                        2304
            href="/report/",                                                2305
            desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),          2306
]                                                                           2307
def _has_data(ind):                                                         2308
    return wf._has_data(ind)                                                2309
#  系统的九个功能模块（登录页介绍用，含不在评价流程内的账号与权限管理）          2310
MODULES = [                                                                 2311
    ("01", "账号与权限管理", "自助注册与登录注销、管理员与操作员两种职务、写操作自动留痕"),  2312
    ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),      2313
    ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),          2314
    ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),          2315
    ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),          2316
    ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),     2317
    ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),        2318
    ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),          2319
    ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),       2320
]                                                                           2321
def snapshot():                                                             2322
    """一次算出工作台需要的全部数据（同一请求内只算一次）。                      2323
    返回 dict：                                                             2324
        stats      {projects, patches, unit_types, standards}               2325
        steps      八个步骤，附 available / pending / note                   2326
        badges     {key: 待办数量}，仅含有待办的项                           2327
        progress   {patches, units, data_units, ready,                      2328
    evaluated}                                                              2329
    """                                                                     2330
    cached = getattr(g, "_dash_snapshot", None)                             2331
    if cached is None:                                                      2332
        cached = _compute()                                                 2333
        g._dash_snapshot = cached                                           2334
    return cached                                                           2335
def _compute():                                                             2336
    projects = Project.query.count()                                        2337
    patches = MinePatch.query.all()                                         2338
    patch_ids = [p.id for p in patches]                                     2339
    units = MicroUnit.query.filter(                                         2340
        MicroUnit.patch_id.in_(patch_ids)).all() if                         2341
    patch_ids else []                                                       2342
    unit_by_patch = {}                                                      2343
    for u in units:                                                         2344
        unit_by_patch.setdefault(u.patch_id, []).append(u)                  2345
    unit_ids = [u.id for u in units]                                        2346
    data_unit_ids = set()                                                   2347
    if unit_ids:                                                            2348
        for ind in IndicatorData.query.filter(                              2349
                IndicatorData.unit_id.in_(unit_ids)).all():                 2350
```

```
            if _has_data(ind):
                data_unit_ids.add(ind.unit_id)
    passed_constraint = set()
    if patch_ids:
        for c in ConstraintCheck.query.filter(
                ConstraintCheck.patch_id.in_(patch_ids)).all
    ():
            if c.passed:
                passed_constraint.add(c.patch_id)
    evaluated = set()
    if patch_ids:
        for r in EvaluationResult.query.filter(
                EvaluationResult.patch_id.in_(patch_ids)).al
    l():
            evaluated.add(r.patch_id)
    # ---- 各步骤的待办数量 ----
    pending = {
        "constraint": sum(1 for p in patches if p.id not in
    passed_constraint),
        "unit": sum(1 for p in patches if not
    unit_by_patch.get(p.id)),
        "indicator": 0,
        "evaluate": 0,
    }
    ready_patches = 0
    for p in patches:
        us = unit_by_patch.get(p.id, [])
        if us and any(u.id not in data_unit_ids for u in
    us):
            pending["indicator"] += 1
        pr = wf.patch_progress(p.id)
        if pr["ready"]:
            ready_patches += 1
            if p.id not in evaluated:
                pending["evaluate"] += 1
    # ---- 步骤可用性：前置数据不具备时置灰 ----
    avail = {
        "project": True,
        "standard": True,
        "constraint": bool(patches),
        "unit": bool(patches),
        "ahp": True,
        "indicator": bool(units),
        "evaluate": ready_patches > 0,
        "report": bool(evaluated),
    }
    notes = {
        "constraint": "需先建立矿山图斑",
        "unit": "需先建立矿山图斑",
        "indicator": "需先划分微地质单元",
```

```
            "evaluate": "需前置步骤齐备的图斑",
            "report": "需先执行适宜性评价",
        }
        steps = []
        for s in STEPS:
            steps.append(dict(
                s,
                available=avail[s["key"]],
                pending=pending.get(s["key"], 0),
                note=None if avail[s["key"]] else
        notes.get(s["key"], ""),
            ))
        unit_types = len({u.unit_type for u in units})
        stats = {
            "projects": projects,
            "patches": len(patches),
            "unit_types": unit_types,
            "standards": Standard.query.count(),
        }
        return {
            "stats": stats,
            "steps": steps,
            "badges": {k: v for k, v in pending.items() if v},
            "progress": {
                "patches": len(patches),
                "units": len(units),
                "data_units": len(data_unit_ids),
                "ready": ready_patches,
                "evaluated": len(evaluated),
            },
        }
    def nav_badges():
        """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""
        snap = snapshot()
        return {
            "/constraint/": snap["badges"].get("constraint", 0),
            "/unit/": snap["badges"].get("unit", 0),
            "/indicator/": snap["badges"].get("indicator", 0),
            "/evaluate/": snap["badges"].get("evaluate", 0),
        }
    # ==== app/services/audit.py ====
    # -*- coding: utf-8 -*-
    """操作审计。
    在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
    且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
    动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
    """
    from flask import request
    from flask_login import current_user
    from app.models import db, OperationLog
```

```
#  端点  →  可读动作名
ACTION_NAMES  =  {
      "auth.login":  "登录系统",
      "auth.register":  "注册账号",
      "auth.logout":  "退出登录",
      "auth.add_user":  "新增账号",
      "auth.edit_user":  "编辑账号",
      "auth.delete_user":  "删除账号",
      "auth.reset_password":  "重置用户密码",
      "auth.profile":  "修改个人资料",
      "auth.upload_avatar":  "更新头像",
      "auth.delete_avatar":  "移除头像",
      "project.add":  "新建项目",
      "project.edit":  "编辑项目",
      "project.delete":  "删除项目",
      "project.patch_add":  "新建矿山图斑",
      "project.patch_edit":  "编辑矿山图斑",
      "project.patch_delete":  "删除矿山图斑",
      "project.upload_image":  "上传遥感影像",
      "unit.add":  "添加微地质单元",
      "unit.edit":  "编辑微地质单元",
      "unit.delete":  "删除微地质单元",
      "standard.add":  "添加标准 / 政策文件",
      "standard.edit":  "编辑标准 / 政策文件",
      "standard.delete":  "删除标准 / 政策文件",
      "standard.rules":  "维护标准规则",
      "standard.rule_delete":  "删除标准规则",
      "constraint.check":  "执行限制性条件判定",
      "ahp.save_expert":  "录入专家判断矩阵",
      "ahp.delete_expert":  "删除专家判断矩阵",
      "ahp.aggregate":  "汇总多专家矩阵",
      "indicator.edit":  "保存指标数据",
      "indicator.import_data":  "批量导入指标数据",
      "indicator.update_meta":  "维护指标定义",
      "evaluate.run":  "执行适宜性评价",
}
#  无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS  =  {"static",  "auth.index"}
def  action_name(endpoint):
      #  个人设置页的两个表单共用一个端点，按提交的动作区分
      if  endpoint  ==  "auth.profile":
            return  "修改密码"  if  request.form.get("action")  ==
      "password"  \
                  else  "修改个人资料"
      if  endpoint  in  ACTION_NAMES:
            return  ACTION_NAMES[endpoint]
      return  "写操作  %s  %s"  %  (request.method,  request.path)
def  extract_target_id():
      """从视图参数中取出被操作对象的主键，取不到则返回  None。"""
      args  =  request.view_args  or  {}
```

```
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx / 5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += ";结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
            target_type=(endpoint or "").split(".")[0] or
    None,
            target_id=extract_target_id(),
            detail=detail,
        ))
        db.session.commit()
    except Exception:
        #  审计失败不应影响业务响应
        db.session.rollback()
#  ==== app/views/auth.py ====
#  -*- coding: utf-8 -*-
"""账号与权限 —— 登录、注册、个人资料、用户维护、操作日志。
页面组织：
  /login          登录页（含「点我注册」入口）
  /register       自助注册（默认「操作员」）
  /profile        个人设置，分类切换：info / security / users / logs
                  后两个分类仅管理员可见，由本模块统一管控。
"""
from flask import (Blueprint, render_template, redirect,
    url_for, request,
                   flash, abort, current_app)
from flask_login import login_user, logout_user,
    login_required, current_user
from werkzeug.security import check_password_hash,
    generate_password_hash
from app.forms import (LoginForm, EmptyForm, ProfileForm,
    PasswordForm,
                       RegisterForm, email_ok)
from app.models import db, User, OperationLog
from app.utils import uploads as up
```

```
from app.utils.constants import ROLE_LABELS, DEFAULT_ROLE, GENDERS
bp = Blueprint("auth", __name__)
#  「个人设置」的四个分类；后两个是管理员专属
PROFILE_TABS = ("info", "security", "users", "logs")
ADMIN_TABS = ("users", "logs")
def require_admin():
    """非管理员直接 403，避免误操作基础数据。"""
    if current_user.role != "admin":
        abort(403)
def _intro_data():
    """登录页下方的系统介绍所用的静态内容。"""
    from app.services import dashboard as dash
    return {"modules": dash.MODULES}
@bp.route("/")
def index():
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        return render_template("auth/home.html", dash=dash.snapshot())
    return redirect(url_for("auth.login"))
@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.data).first()
        if user and check_password_hash(user.password_hash, form.password.data):
            if not user.active:
                flash("该账号已停用，请联系管理员", "toast-error")
            else:
                login_user(user, remember=form.remember.data)
                #  用 toast 类别：渲染为页面中上方的浮层，淡入停留后淡出。
                #  登录成功只是告知性反馈，不需要用户停下来阅读。
                flash("登录成功", "toast")
                return redirect(url_for("auth.index"))
        else:
            flash("账户名称或密码错误", "toast-error")
    #  字段为空之类的校验失败，错误留在表单里逐项显示，不走浮层：
    #  那类消息要对着输入框看，一闪而过反而让人以为「点了没反应」。
    return render_template("auth/login.html", form=form, **_intro_data())
@bp.route("/register", methods=["GET", "POST"])
def register():
    """自助注册。新账号一律为「操作员」，权限只能由管理员事后调整。"""
    if current_user.is_authenticated:
```

```
            return redirect(url_for("auth.index"))                    2601
        form = RegisterForm()                                         2602
        if form.validate_on_submit():                                 2603
            username = form.username.data.strip()                     2604
            if User.query.filter_by(username=username).first():       2605
                form.username.errors.append("该账户名称已被使用")          2606
            else:                                                     2607
                user = User(username=username,                        2608
                            role=DEFAULT_ROLE,                        2609
                            password_hash=generate_password_hash      2610
        (form.password.data),                                         2611
                            display_name=(form.display_name.data      2612
        or "").strip() or None,                                       2613
                            gender=(form.gender.data or               2614
        "").strip() or None,                                          2615
                            email=(form.email.data or                 2616
        "").strip() or None,                                          2617
                            phone=(form.phone.data or                 2618
        "").strip() or None)                                          2619
                rel, err = _save_avatar(request.files.get("avata      2620
        r"))                                                          2621
                if err:                                               2622
                    flash(err, "error")                               2623
                else:                                                 2624
                    user.avatar = rel                                 2625
                    db.session.add(user)                              2626
                    db.session.commit()                               2627
                    login_user(user)                                  2628
                    flash("注册成功，欢迎使用", "toast")                  2629
                    return redirect(url_for("auth.index"))            2630
        return render_template("auth/register.html", form=form)       2631
@bp.route("/logout")                                                  2632
@login_required                                                       2633
def logout():                                                         2634
        logout_user()                                                 2635
        flash("已退出登录", "ok")                                       2636
        return redirect(url_for("auth.login"))                        2637
# ------------------------------ 个人设置 ------------------------------2638
        ----                                                          2639
# 任何登录用户在这里维护自己；管理员额外能管理账号与查看日志。                  2640
# 「用户与操作日志」不再是独立菜单，而是本页的管理员分类。                     2641
@bp.route("/profile", methods=["GET", "POST"])                        2642
@login_required                                                       2643
def profile():                                                        2644
        is_admin = current_user.role == "admin"                       2645
        tab = request.values.get("tab") or "info"                     2646
        if tab not in PROFILE_TABS or (tab in ADMIN_TABS and          2647
        not is_admin):                                                2648
            tab = "info"                                              2649
        # 表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名    2650
```

```
    info_form = ProfileForm(prefix="info", obj=current_user)
    pwd_form = PasswordForm(prefix="pwd")
    if request.method == "POST":
        action = request.form.get("action")
        #  校验失败要重新渲染本页，此时必须回到提交表单所在的那个分类，
        #  否则错误提示会出现在另一个分类里，用户根本看不到。
        if action == "password":
            tab = "security"
        elif action == "info":
            tab = "info"
        if action == "info" and info_form.validate_on_submit
    ():
            current_user.display_name =
    (info_form.display_name.data or "").strip() or None
            current_user.gender = (info_form.gender.data or
    "").strip() or None
            current_user.email = (info_form.email.data or
    "").strip() or None
            current_user.phone = (info_form.phone.data or
    "").strip() or None
            db.session.commit()
            flash("个人资料已保存", "ok")
            return redirect(url_for("auth.profile",
    tab="info"))
        if action == "password" and pwd_form.validate_on_sub
    mit():
            if not check_password_hash(current_user.password
    _hash,

    pwd_form.old_password.data):
                pwd_form.old_password.errors.append("当前密码不正确
    ")
            else:
                current_user.password_hash =
    generate_password_hash(
                    pwd_form.new_password.data)
                db.session.commit()
                flash("密码已修改", "ok")
                return redirect(url_for("auth.profile",
    tab="security"))
    ctx = dict(info_form=info_form, pwd_form=pwd_form,
    avatar_form=EmptyForm(),
              tab=tab, role_labels=ROLE_LABELS,
    genders=GENDERS)
    if tab == "users":
        ctx.update(_users_context())
    elif tab == "logs":
        ctx.update(_logs_context())
    return render_template("auth/profile.html", **ctx)
def _users_context():
```

```
    """管理员分类「账号」所需的数据。"""
    return {"users": User.query.order_by(User.id).all(),
            "form": EmptyForm()}
def _logs_context():
    """管理员分类「操作日志」所需的数据。"""
    per_page = 50
    q = OperationLog.query.order_by(OperationLog.id.desc())
    total = q.count()
    pages = max(1, (total + per_page - 1) // per_page)
    # 页码先夹到 [1, pages] 再算 offset：手工把 page 写成天文数字时，
    # (page-1)*per_page 会超出 SQLite 能绑定的整数范围，直接 500。
    try:
        page = int(request.args.get("page", 1))
    except (TypeError, ValueError):
        page = 1
    page = max(1, min(page, pages))
    records = q.offset((page - 1) * per_page).limit(per_page
    ).all()
    users_map = {u.id: u.username for u in User.query.all()}
    return {"records": records, "page": page, "pages":
    pages,
            "total": total, "users_map": users_map}
# ------------------------------ 头像 ------------------------------
--
def _avatar_length_ok(file_storage):
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 *
    1024 * 1024)
    stream = file_storage.stream
    pos = stream.tell()
    stream.seek(0, 2)
    size = stream.tell()
    stream.seek(pos)
    return size <= limit, size
def _save_avatar(file):
    """保存头像文件，返回 (相对路径, 错误信息)。没有文件时返回 (None, None)。"""
    if file is None or not file.filename:
        return None, None
    ok, size = _avatar_length_ok(file)
    if not ok:
        return None, ("头像不能超过 %.0f MB（当前 %.1f MB）"
                      % (current_app.config["MAX_AVATAR_LENG
    TH"] / 1024 / 1024,
                         size / 1024 / 1024))
    return up.save_upload(file, "avatars",
                          current_app.config["ALLOWED_AVATAR
    _EXT"],
                          current_app.config["UPLOAD_FOLDER"
    ])
@bp.route("/profile/avatar", methods=["POST"])
```

```
@login_required                                                    2751
def upload_avatar():                                               2752
    form = EmptyForm()                                             2753
    back = url_for("auth.profile", tab=request.form.get("tab2754
    ") or "info")                                                  2755
    if not form.validate_on_submit():                              2756
        flash("表单校验失败，请重试", "error")                      2757
        return redirect(back)                                      2758
    rel, err = _save_avatar(request.files.get("avatar"))           2759
    if err:                                                        2760
        flash(err, "error")                                        2761
        return redirect(back)                                      2762
    if rel is None:                                                2763
        flash("请选择头像图片", "error")                            2764
        return redirect(back)                                      2765
    # 换新头像时删掉旧文件，避免残留                                2766
    up.remove_upload(current_user.avatar,                          2767
    current_app.config["UPLOAD_FOLDER"])                           2768
    current_user.avatar = rel                                      2769
    db.session.commit()                                            2770
    flash("头像已更新", "ok")                                      2771
    return redirect(back)                                          2772
@bp.route("/profile/avatar/delete", methods=["POST"])              2773
@login_required                                                    2774
def delete_avatar():                                               2775
    form = EmptyForm()                                             2776
    back = url_for("auth.profile", tab=request.form.get("tab2777
    ") or "info")                                                  2778
    if form.validate_on_submit():                                  2779
        up.remove_upload(current_user.avatar,                      2780
    current_app.config["UPLOAD_FOLDER"])                           2781
        current_user.avatar = None                                 2782
        db.session.commit()                                        2783
        flash("头像已移除", "ok")                                  2784
    return redirect(back)                                          2785
# ---------------------------- 账号维护（仅管理员）                2786
    ----------------------------                                   2787
@bp.route("/users")                                                2788
@login_required                                                    2789
def users():                                                       2790
    """旧地址保留，统一并入「个人设置 · 账号」。"""                 2791
    return redirect(url_for("auth.profile", tab="users"))          2792
@bp.route("/users/add", methods=["GET", "POST"])                   2793
@login_required                                                    2794
def add_user():                                                    2795
    require_admin()                                                2796
    if request.method == "POST":                                   2797
        username = request.form.get("username", "").strip()        2798
        password = request.form.get("password", "")                2799
        role = request.form.get("role", DEFAULT_ROLE)              2800
```

```
        if role not in ROLE_LABELS:                                    2801
            role = DEFAULT_ROLE                                        2802
        if not username or not password:                               2803
            flash("账户名称和密码不能为空", "error")                    2804
            return redirect(url_for("auth.add_user"))                  2805
        if User.query.filter_by(username=username).first():            2806
            flash("账户名称已存在", "error")                            2807
            return redirect(url_for("auth.add_user"))                  2808
        db.session.add(User(                                           2809
            username=username, role=role,                              2810
            password_hash=generate_password_hash(password),            2811
            display_name=request.form.get("display_name",              2812
    "").strip() or None,                                               2813
            gender=request.form.get("gender", "").strip()              2814
    or None,                                                           2815
            email=request.form.get("email", "").strip() or             2816
    None,                                                              2817
            phone=request.form.get("phone", "").strip() or             2818
    None))                                                             2819
        db.session.commit()                                            2820
        flash("账号已添加", "ok")                                       2821
        return redirect(url_for("auth.profile",                        2822
    tab="users"))                                                      2823
    return render_template("auth/add_user.html",                       2824
    role_labels=ROLE_LABELS,                                           2825
                           genders=GENDERS,                            2826
    default_role=DEFAULT_ROLE,                                         2827
                           form=EmptyForm())                           2828
@bp.route("/users/<int:user_id>/edit", methods=["GET",                 2829
    "POST"])                                                           2830
@login_required                                                        2831
def edit_user(user_id):                                                2832
    require_admin()                                                    2833
    user = db.get_or_404(User, user_id)                                2834
    if request.method == "POST":                                       2835
        role = request.form.get("role", user.role)                     2836
        if role not in ROLE_LABELS:                                    2837
            role = user.role                                           2838
        active = request.form.get("active") == "1"                     2839
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统          2840
        if user.role == "admin" and (role != "admin" or not            2841
    active):                                                           2842
            others = User.query.filter(User.role == "admin",           2843
                                       User.id != user.id,             2844
                                                                       2845
    User.active.is_(True)).count()                                     2846
            if others == 0:                                            2847
                flash("系统至少需保留一名启用状态的管理员", "error")      2848
                return redirect(url_for("auth.edit_user",              2849
    user_id=user.id))                                                  2850
```

```
        # 个人资料字段（用户自己也能在「个人设置」里改，管理员在此代改）
        email = request.form.get("email", "").strip()
        if not email_ok(email):
            flash("邮箱格式不正确", "error")
            return redirect(url_for("auth.edit_user",
    user_id=user.id))
        user.role = role
        user.active = active
        user.display_name = request.form.get("display_name",
    "").strip() or None
        user.gender = request.form.get("gender",
    "").strip() or None
        user.email = email or None
        user.phone = request.form.get("phone", "").strip()
    or None
        new_password = request.form.get("password",
    "").strip()
        if new_password:
            user.password_hash = generate_password_hash(new_
    password)
        db.session.commit()
        flash("账号信息已保存", "ok")
        return redirect(url_for("auth.profile",
    tab="users"))
    return render_template("auth/edit_user.html", user=user,
                           role_labels=ROLE_LABELS,
    genders=GENDERS,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/delete", methods=["POST"])
@login_required
def delete_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if user.id == current_user.id:
        flash("不能删除当前登录的账号", "error")
        return redirect(url_for("auth.profile",
    tab="users"))
    if user.role == "admin":
        others = User.query.filter(User.role == "admin",
                                   User.id != user.id,
                                   User.active.is_(True)).co
    unt()
        if others == 0:
            flash("系统至少需保留一名启用状态的管理员", "error")
            return redirect(url_for("auth.profile",
    tab="users"))
    db.session.delete(user)
    db.session.commit()
    flash("账号「%s」已删除" % user.username, "ok")
    return redirect(url_for("auth.profile", tab="users"))
```

```
# ---------------------------- 操作日志（仅管理员）
    ----------------------------
@bp.route("/logs")
@login_required
def logs():
    """旧地址保留，统一并入「个人设置 · 操作日志」。"""
    return redirect(url_for("auth.profile", tab="logs"))
# ==== app/views/project.py ====
# -*- coding: utf-8 -*-
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""
from datetime import datetime
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, current_app)
from flask_login import login_required, current_user
from app.forms import EmptyForm
from app.models import db, Project, MinePatch,
    ReferenceEcosystem
from app.services import cleanup
from app.utils.parsing import FieldError, parse_float
bp = Blueprint("project", __name__, url_prefix="/project")
@bp.route("/")
@login_required
def index():
    projects = Project.query.order_by(Project.id.desc()).all
    ()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or
    None,
            description=request.form.get("description",
    "").strip() or None,
            created_by=current_user.id,
        )
        if not p.name:
            flash("项目名称不能为空", "error")
        else:
            db.session.add(p)
            db.session.commit()
            flash("项目已创建", "ok")
            return redirect(url_for("project.index"))
```

```
    return render_template("project/edit.html", form=form,
    project=None)
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])
@login_required
def edit(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p.name = request.form.get("name", "").strip()
        p.code = request.form.get("code", "").strip() or
    None
        p.description = request.form.get("description",
    "").strip() or None
        p.status = request.form.get("status", "进行中")
        db.session.commit()
        flash("项目已保存", "ok")
        return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form,
    project=p)
@bp.route("/<int:pid>/delete", methods=["POST"])
@login_required
def delete(pid):
    p = db.get_or_404(Project, pid)
    # 项目下的图斑由 ORM 级联删除，但图斑下面的指标数据、限制性判定、
    # 评价结果、得分明细、参照生态系统不会跟着走，必须先逐个图斑清理，
    # 否则每删一个项目都在库里留下一批孤儿数据（见 app/services/cleanup.py）。
    cleanup.delete_project(p)
    flash("项目已删除", "ok")
    return redirect(url_for("project.index"))
# ---------------------------- 矿山图斑 ----------------------------
    ----
@bp.route("/<int:pid>/patches")
@login_required
def patches(pid):
    p = db.get_or_404(Project, pid)
    return render_template("project/patches.html",
    project=p,
                           form=EmptyForm())
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])
@login_required
def patch_delete(patch_id):
    """删除矿山图斑，及其微地质单元、指标数据与评价记录。"""
    patch = db.get_or_404(MinePatch, patch_id)
    pid = patch.project_id
    # 清理逻辑统一放在 app/services/cleanup.py：项目删除走的是同一个函数，
    # 免得两条删除路径各写一份、漏掉某张子表（曾经就漏过）。
    deleted, _counts = cleanup.delete_patch(patch.id)
    flash("图斑「%s」及其数据已删除" % (deleted.name if deleted else
    patch.name),
```

```
            "ok")                                                    3001
    return redirect(url_for("project.patches", pid=pid))             3002
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])           3003
@login_required                                                      3004
def patch_add(pid):                                                  3005
    p = db.get_or_404(Project, pid)                                  3006
    form = EmptyForm()                                               3007
    if request.method == "POST" and form.validate_on_submit(         3008
    ):                                                               3009
        def fv(name, cast=str):                                      3010
            v = request.form.get(name, "").strip()                   3011
            return cast(v) if v != "" else None                      3012
        try:                                                         3013
            area_ha = parse_float(request.form.get("area_ha"         3014
    ), "图斑面积（公顷）",                                           3015
                                  minimum=0)                         3016
            stop_years = parse_float(request.form.get("stop_         3017
    years"), "停产年限",                                             3018
                                     minimum=0)                      3019
        except FieldError as e:                                      3020
            flash(str(e), "error")                                   3021
            return redirect(url_for("project.patch_add",             3022
    pid=p.id))                                                       3023
        patch = MinePatch(                                           3024
            project_id=p.id,                                         3025
            name=request.form.get("name", "").strip(),               3026
            admin_region=fv("admin_region"),                         3027
            area_ha=area_ha,                                         3028
            mineral_type=fv("mineral_type"),                         3029
            mining_method=fv("mining_method"),                       3030
            stop_years=stop_years,                                   3031
            responsible_entity_missing=bool(request.form.get         3032
    ("responsible_entity_missing")),                                 3033
        )                                                            3034
        if not patch.name:                                           3035
            flash("图斑名称不能为空", "error")                       3036
        else:                                                        3037
            db.session.add(patch)                                    3038
            db.session.flush()                                       3039
            # 参照生态系统默认取论文值                               3040
            db.session.add(ReferenceEcosystem(                       3041
                patch_id=patch.id, coverage_benchmark=75.0,          3042
    shannon_benchmark=0.7))                                          3043
            db.session.commit()                                      3044
            flash("图斑已创建", "ok")                                3045
            return redirect(url_for("project.patches",               3046
    pid=p.id))                                                       3047
    return render_template("project/patch_edit.html",                3048
    form=form, project=p, patch=None)                                3049
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",              3050
```

```
    "POST"])
@login_required
def patch_edit(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        # 数值字段统一走 parse_float：填了非数字给中文提示，而不是 500
        try:
            area_ha = parse_float(request.form.get("area_ha"
    ), "图斑面积（公顷）",
                                  minimum=0)
            stop_years = parse_float(request.form.get("stop_
    years"), "停产年限",
                                     minimum=0)
            cov = parse_float(request.form.get("coverage_ben
    chmark"),
                              "植被覆盖度基准（%）", minimum=0,
    maximum=100)
            sha = parse_float(request.form.get("shannon_benc
    hmark"),
                              "Shannon 基准", minimum=0)
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
        patch.name = request.form.get("name", "").strip()
        patch.admin_region = fv("admin_region")
        patch.area_ha = area_ha
        patch.mineral_type = fv("mineral_type")
        patch.mining_method = fv("mining_method")
        patch.stop_years = stop_years
        patch.responsible_entity_missing =
    bool(request.form.get("responsible_entity_missing"))
        # 参照生态系统
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa
    tch.id).first()
        if ref is None:
            ref = ReferenceEcosystem(patch_id=patch.id)
            db.session.add(ref)
        # 基准值：用户填 0 是合法输入（不能写成 `or 75.0`，那样 0 会被悄悄换成默认值）
        ref.coverage_benchmark = 75.0 if cov is None else
    cov
        ref.shannon_benchmark = 0.7 if sha is None else sha
        ref.dominant_species = fv("dominant_species")
        patch.image_note = fv("image_note")
        db.session.commit()
```

```
            flash("图斑已保存", "ok")                                    3101
            return redirect(url_for("project.patches",                   3102
        pid=patch.project_id))                                           3103
        return render_template("project/patch_edit.html",                3104
        form=form,                                                       3105
                                    project=patch.project,               3106
        patch=patch,                                                     3107
                                    reference=ReferenceEcosystem.quer    3108
        y.filter_by(                                                     3109
                                        patch_id=patch.id).first())      3110
# ------------------------------ 遥感影像挂接                           3111
        ------------------------------                                   3112
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])               3113
@login_required                                                          3114
def upload_image(patch_id):                                              3115
        """上传遥感底图 / 无人机正射影像并挂接到图斑。"""                  3116
        from app.utils import uploads as up                              3117
        patch = db.get_or_404(MinePatch, patch_id)                       3118
        form = EmptyForm()                                               3119
        if not form.validate_on_submit():                                3120
            flash("表单校验失败，请重试", "error")                       3121
            return redirect(url_for("project.patch_edit",                3122
        patch_id=patch.id))                                              3123
        rel, err = up.save_upload(request.files.get("image"),            3124
        "patches",                                                       3125
                                  current_app.config["ALLOWED_IM         3126
        AGE_EXT"],                                                       3127
                                  current_app.config["UPLOAD_FOL         3128
        DER"])                                                           3129
        if err:                                                          3130
            flash(err, "error")                                          3131
            return redirect(url_for("project.patch_edit",                3132
        patch_id=patch.id))                                              3133
        # 替换旧影像时清理原文件，避免残留                               3134
        if patch.image_path:                                             3135
            up.remove_upload(patch.image_path,                           3136
        current_app.config["UPLOAD_FOLDER"])                             3137
        patch.image_path = rel                                           3138
        patch.image_uploaded_at = datetime.utcnow()                      3139
        note = request.form.get("image_note", "").strip()                3140
        if note:                                                         3141
            patch.image_note = note                                      3142
        db.session.commit()                                              3143
        flash("遥感影像已挂接", "ok")                                    3144
        return redirect(url_for("project.patch_edit",                    3145
        patch_id=patch.id))                                              3146
@bp.route("/patch/<int:patch_id>/image/delete",                          3147
      methods=["POST"])                                                  3148
@login_required                                                          3149
def delete_image(patch_id):                                              3150
```

```
    from app.utils import uploads as up
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        up.remove_upload(patch.image_path,
    current_app.config["UPLOAD_FOLDER"])
        patch.image_path = None
        patch.image_uploaded_at = None
        db.session.commit()
        flash("遥感影像已移除", "ok")
    return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
# ==== app/views/unit.py ====
# -*- coding: utf-8 -*-
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import UNIT_TYPES
from app.utils.parsing import FieldError, parse_float
bp = Blueprint("unit", __name__, url_prefix="/unit")
def recalc_ratio(patch_id):
    """按面积重算该图斑各单元面积占比。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    if total > 0:
        for u in units:
            u.ratio = round((u.area or 0.0) / total, 6)
    return total
@bp.route("/")
@login_required
def list_patches():
    """微地质单元入口：列出各图斑及其单元划分情况。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    rows = []
    for p in patches:
        units = MicroUnit.query.filter_by(patch_id=p.id).all
    ()
        rows.append({
            "patch": p,
            "count": len(units),
            "ratio_sum": sum(u.ratio or 0.0 for u in units),
        })
    return render_template("unit/list.html", rows=rows)
@bp.route("/<int:patch_id>")
```

```
@login_required                                                    3201
def index(patch_id):                                               3202
    patch = db.get_or_404(MinePatch, patch_id)                     3203
    units = MicroUnit.query.filter_by(patch_id=patch_id).all       3204
    ()                                                             3205
    total = sum(u.area or 0.0 for u in units)                      3206
    ratio_sum = sum(u.ratio or 0.0 for u in units)                 3207
    return render_template("unit/index.html", patch=patch,         3208
    units=units,                                                   3209
                                total_area=total,                  3210
    ratio_sum=ratio_sum,                                           3211
                                unit_types=UNIT_TYPES,             3212
    form=EmptyForm())                                              3213
@bp.route("/<int:patch_id>/add", methods=["POST"])                 3214
@login_required                                                    3215
def add(patch_id):                                                 3216
    patch = db.get_or_404(MinePatch, patch_id)                     3217
    form = EmptyForm()                                             3218
    if form.validate_on_submit():                                  3219
        unit_type = request.form.get("unit_type")                  3220
        if unit_type not in UNIT_TYPES:                            3221
            flash("微地质单元类型不正确", "error")                  3222
            return redirect(url_for("unit.index",                  3223
    patch_id=patch_id))                                            3224
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。   3225
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。          3226
        if MicroUnit.query.filter_by(patch_id=patch.id,            3227
                                     unit_type=unit_type).fi       3228
    rst():                                                         3229
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,  3230
    "error")                                                       3231
            return redirect(url_for("unit.index",                  3232
    patch_id=patch_id))                                            3233
        area = request.form.get("area", "").strip()                3234
        try:                                                       3235
            area_value = parse_float(area, "面积（公顷）",          3236
    minimum=0)                                                     3237
        except FieldError as e:                                    3238
            flash(str(e), "error")                                 3239
            return redirect(url_for("unit.index",                  3240
    patch_id=patch_id))                                            3241
        u = MicroUnit(patch_id=patch.id,                           3242
    unit_type=unit_type,                                           3243
                      area=area_value,                             3244
                      description=request.form.get("descript       3245
    ion", "").strip() or None)                                     3246
        db.session.add(u)                                          3247
        db.session.flush()                                         3248
        recalc_ratio(patch.id)                                     3249
        db.session.commit()                                        3250
```

```
            flash("微地质单元已添加", "ok")
        return redirect(url_for("unit.index",
        patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
        """单个微地质单元的编辑页。"""
        u = db.get_or_404(MicroUnit, unit_id)
        form = EmptyForm()
        if request.method == "POST" and form.validate_on_submit(
        ):
            new_type = request.form.get("unit_type",
        u.unit_type)
            if new_type not in UNIT_TYPES:
                flash("微地质单元类型不正确", "error")
                return redirect(url_for("unit.edit",
        unit_id=u.id))
            # 改类型时同样不允许与本图斑已有的其他单元重复
            dup = MicroUnit.query.filter_by(patch_id=u.patch_id,


        unit_type=new_type).first()
            if dup is not None and dup.id != u.id:
                flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,
        "error")
                return redirect(url_for("unit.edit",
        unit_id=u.id))
            u.unit_type = new_type
            area = request.form.get("area", "").strip()
            try:
                u.area = parse_float(area, "面积（公顷）", minimum=0)
            except FieldError as e:
                flash(str(e), "error")
                return redirect(url_for("unit.edit",
        unit_id=u.id))
            u.description = request.form.get("description",
        "").strip() or None
            recalc_ratio(u.patch_id)
            db.session.commit()
            flash("微地质单元已保存", "ok")
            return redirect(url_for("unit.index",
        patch_id=u.patch_id))
        return render_template("unit/edit.html", unit=u,
        patch=u.patch,
                               unit_types=UNIT_TYPES, form=form)
@bp.route("/delete/<int:unit_id>", methods=["POST"])
@login_required
def delete(unit_id):
        u = db.get_or_404(MicroUnit, unit_id)
        pid = u.patch_id
        name = u.unit_type
```

```
    # 一并删除该单元的指标数据，避免残留
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch
    ronize_session=False)
    db.session.delete(u)
    db.session.flush()
    recalc_ratio(pid)
    db.session.commit()
    flash("微地质单元「%s」已删除" % name, "ok")
    return redirect(url_for("unit.index", patch_id=pid))
# ==== app/views/standard.py ====
# -*- coding: utf-8 -*-
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, Standard, StandardRule
bp = Blueprint("standard", __name__, url_prefix="/standard")
@bp.route("/")
@login_required
def index():
    standards = Standard.query.order_by(Standard.category,
    Standard.province).all()
    local = [s for s in standards if s.category == "地方标准"]
    policy = [s for s in standards if s.category != "地方标准"]
    return render_template("standard/index.html",
    local=local, policy=policy,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s = Standard(
            province=request.form.get("province",
    "").strip(),
            name=request.form.get("name", "").strip(),
            doc_number=request.form.get("doc_number",
    "").strip() or None,
            publish_date=request.form.get("publish_date",
    "").strip() or None,
            implement_date=request.form.get("implement_date"
    , "").strip() or None,
            category=request.form.get("category", "地方标准"),
            summary=request.form.get("summary", "").strip()
    or None,
        )
        if not s.province or not s.name:
            flash("省份与名称不能为空", "error")
```

```
        else:                                                    3351
            db.session.add(s)                                    3352
            db.session.commit()                                  3353
            flash("标准已添加", "ok")                              3354
            return redirect(url_for("standard.index"))           3355
    return render_template("standard/edit.html", form=form,      3356
    standard=None)                                               3357
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])            3358
@login_required                                                  3359
def edit(sid):                                                   3360
    s = db.get_or_404(Standard, sid)                             3361
    form = EmptyForm()                                           3362
    if request.method == "POST" and form.validate_on_submit(     3363
    ):                                                           3364
        s.province = request.form.get("province",                3365
    "").strip()                                                  3366
        s.name = request.form.get("name", "").strip()            3367
        s.doc_number = request.form.get("doc_number",            3368
    "").strip() or None                                          3369
        s.publish_date = request.form.get("publish_date",        3370
    "").strip() or None                                          3371
        s.implement_date = request.form.get("implement_date"     3372
    , "").strip() or None                                        3373
        s.category = request.form.get("category", "地方标准")      3374
        s.summary = request.form.get("summary", "").strip()      3375
    or None                                                      3376
        db.session.commit()                                      3377
        flash("已保存", "ok")                                     3378
        return redirect(url_for("standard.rules", sid=s.id))     3379
    return render_template("standard/edit.html", form=form,      3380
    standard=s)                                                  3381
@bp.route("/<int:sid>/delete", methods=["POST"])                 3382
@login_required                                                  3383
def delete(sid):                                                 3384
    s = db.get_or_404(Standard, sid)                             3385
    StandardRule.query.filter_by(standard_id=sid).delete()       3386
    db.session.delete(s)                                         3387
    db.session.commit()                                          3388
    flash("已删除", "ok")                                         3389
    return redirect(url_for("standard.index"))                   3390
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])           3391
@login_required                                                  3392
def rules(sid):                                                  3393
    s = db.get_or_404(Standard, sid)                             3394
    form = EmptyForm()                                           3395
    if request.method == "POST" and form.validate_on_submit(     3396
    ):                                                           3397
        r = StandardRule(                                        3398
            standard_id=s.id,                                    3399
            item=request.form.get("item", "").strip(),           3400
```

```
            operator=request.form.get("operator", ">="),
            threshold=request.form.get("threshold",
    "").strip() or None,
            condition_type=request.form.get("condition_type"
    , "必要条件"),
            region=request.form.get("region", "").strip()
    or None,
            note=request.form.get("note", "").strip() or
    None,
        )
        if not r.item:
            flash("判定项不能为空", "error")
        else:
            db.session.add(r)
            db.session.commit()
            flash("规则已添加", "ok")
        return redirect(url_for("standard.rules", sid=s.id))
    rs = StandardRule.query.filter_by(standard_id=s.id).all(
    )
    return render_template("standard/rules.html",
    form=form, standard=s, rules=rs)
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
    r = db.get_or_404(StandardRule, rid)
    sid = r.standard_id
    db.session.delete(r)
    db.session.commit()
    flash("规则已删除", "ok")
    return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
@login_required
def compare():
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""
    rules = StandardRule.query.join(Standard).all()
    groups = {}
    for r in rules:
        groups.setdefault(r.item, []).append(r)
    return render_template("standard/compare.html",
    groups=groups)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""限制性指标判定 —— 一票否决判定。
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
```

```
from app.models import db, MinePatch, ConstraintCheck
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
FIELDS = [
    ("policy_pass", "政策约束"),
    ("location_pass", "区位约束"),
    ("pollution_pass", "土壤污染"),
    ("geo_pass", "地质灾害"),
    ("farmland_pass", "耕地属性"),
]
@bp.route("/")
@login_required
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    latest = {}
    for p in patches:
        latest[p.id] = ConstraintCheck.query.filter_by(patch_id=p.id) \
            .order_by(ConstraintCheck.id.desc()).first()
    return render_template("constraint/index.html",
    patches=patches, latest=latest)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
@login_required
def check(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    record = ConstraintCheck.query.filter_by(patch_id=patch_id) \
        .order_by(ConstraintCheck.id.desc()).first()
    if request.method == "POST" and form.validate_on_submit():
        if record is None:
            record = ConstraintCheck(patch_id=patch.id)
            db.session.add(record)
        for field, _ in FIELDS:
            setattr(record, field, request.form.get(field) == "1")
        record.passed = all(getattr(record, f) for f, _ in FIELDS)
        failed = [name for f, name in FIELDS if not getattr(record, f)]
        record.reason = ("全部通过" if record.passed
                         else "未通过：" + "、".join(failed))
        db.session.commit()
        flash("判定结果已保存", "ok")
        return redirect(url_for("constraint.check", patch_id=patch.id))
```

```
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）
    field_states = [(f, name, None if record is None else
    getattr(record, f))
                    for f, name in FIELDS]
    return render_template("constraint/check.html",
    patch=patch, form=form,
                           fields=FIELDS,
    field_states=field_states,
                           record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
import math
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
# 一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
```

```
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 （指标名称列表， 默认权重列表）。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form,
                                   weight_rows=ws.summary())
```

```
        return redirect(url_for("ahp.matrix",                    3601
    unit_type=unit_type,                                         3602
                                        level=2, l1=form.l1.data))  3603
    return render_template("ahp/index.html", form=form,          3604
                                   weight_rows=ws.summary())     3605
@bp.route("/matrix", methods=["GET", "POST"])                    3606
def matrix():                                                    3607
    unit_type = request.values.get("unit_type")                  3608
    # level 是手改地址栏就能改的参数，写成 abc 不该把页面打成 500   3609
    try:                                                         3610
        level = int(request.values.get("level", 1))              3611
    except (TypeError, ValueError):                              3612
        level = 1                                                3613
    if level not in (1, 2):                                      3614
        level = 1                                                3615
    l1 = normalize_l1(unit_type, level,                          3616
    request.values.get("l1"))                                    3617
    if unit_type not in DEFAULT_WEIGHTS:                         3618
        flash("无效的微地质单元类型", "error")                    3619
        return redirect(url_for("ahp.index"))                    3620
    names, vec = indicator_list(unit_type, level, l1)            3621
    n = len(names)                                               3622
    form = EmptyForm()                                           3623
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j  3624
    default = [[round(vec[i] / vec[j], 4) for j in               3625
    range(n)] for i in range(n)]                                 3626
    def compute(upper):                                          3627
        A = ahp.matrix_from_upper(upper)                         3628
        r = ahp.weights_and_consistency(A)                       3629
        bad = ahp.find_most_inconsistent(A)                      3630
        return {                                                 3631
            "weights": [round(v, 4) for v in r["weights"]],      3632
            "lambda_max": round(r["lambda_max"], 4),             3633
            "ci": r["ci"],                                       3634
            "cr": r["cr"],                                       3635
            "consistent": r["consistent"],                       3636
            "matrix": [[round(A[i, j], 4) for j in               3637
    range(n)] for i in range(n)],                                3638
            "most_inconsistent": bad,                            3639
        }                                                        3640
    result = None                                                3641
    if request.method == "POST" and form.validate_on_submit(     3642
    ):                                                           3643
        action = request.form.get("action", "calc")              3644
        try:                                                     3645
            upper = [float(request.form.get("m_%d_%d" % (i,      3646
    j), "1"))                                                    3647
                     for i in range(n) for j in range(i +        3648
    1, n)]                                                       3649
            # 0 / 负数 / inf / nan 能过 float()，但会让 n 次方根算出 inf  3650
```

```
    或 nan,                                                                3651
            # 保存时直接撞上 NOT NULL 约束（weight 为 nan）→ 500           3652
            if any((not math.isfinite(x)) or x <= 0 for x                  3653
in upper):                                                                 3654
                raise ValueError("矩阵元素必须是正数")                      3655
            result = compute(upper)                                        3656
        except (ValueError, ZeroDivisionError):                            3657
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",  3658
"error")                                                                   3659
            return redirect(url_for("ahp.matrix",                          3660
unit_type=unit_type,                                                       3661
                                        level=level, l1=l1))               3662
        if action == "save":                                               3663
            if not result["consistent"]:                                   3664
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"  3665
                      "请调整后再保存" % result["cr"], "error")             3666
            else:                                                          3667
                pairs = to_code_pairs(unit_type, level, l1,                3668
names,                                                                     3669
                                      result["weights"])                   3670
                scheme, cnt = ws.save_weights(unit_type,                   3671
pairs)                                                                     3672
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"          3673
                      "此后该单元类型的评价将以此权重计算。"                3674
                      % (unit_type, cnt, result["cr"]),                    3675
"ok")                                                                      3676
                return redirect(url_for("ahp.matrix",                      3677
unit_type=unit_type,                                                       3678
                                        level=level, l1=l1))               3679
    elif request.values.get("calc") == "1":                                3680
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性          3681
        result = compute([default[i][j] for i in range(n)                  3682
for j in range(i + 1, n)])                                                 3683
    _, source_label, custom_scheme = ws.effective_weights(un               3684
it_type)                                                                   3685
    return render_template(                                                3686
        "ahp/matrix.html",                                                 3687
        form=form, unit_type=unit_type, level=level, l1=l1,                3688
        names=names, vec=vec, default=default,                             3689
result=result,                                                             3690
        source_label=source_label, custom_scheme=custom_sche               3691
me,                                                                        3692
    )                                                                      3693
@bp.route("/weights/reset", methods=["POST"])                              3694
def weights_reset():                                                       3695
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""          3696
    unit_type = request.form.get("unit_type")                              3697
    form = EmptyForm()                                                     3698
    if form.validate_on_submit():                                          3699
        if unit_type not in DEFAULT_WEIGHTS:                               3700
```

```
                flash("无效的微地质单元类型", "error")
            else:
                n = ws.reset_weights(unit_type)
                if n:
                    flash("已恢复「%s」为论文表 3 默认权重" % unit_type,
    "ok")
                else:
                    flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,
    "ok")
    return redirect(url_for("ahp.index"))
# ---------------------------- 指标维护 ----------------------------
    ----
@bp.route("/indicators")
def indicators():
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""
    if Indicator.query.count() == 0:
        ind_svc.seed_indicators()
    all_ind = Indicator.query.order_by(Indicator.level,
    Indicator.id).all()
    by_parent = {}
    for i in all_ind:
        by_parent.setdefault(i.parent_code, []).append(i)
    return render_template("ahp/indicators.html",
    by_parent=by_parent,
                           all_ind=all_ind,
    unit_types=UNIT_TYPES,
                           type_label=ind_svc.ideal_type_lab
    el,
                           form=EmptyForm())
@bp.route("/indicators/<code>/edit", methods=["GET",
    "POST"])
def indicator_edit(code):
    ind = Indicator.query.filter_by(code=code).first_or_404(
    )
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        name = request.form.get("name", "").strip()
        if not name:
            flash("指标名称不能为空", "error")
            return redirect(url_for("ahp.indicator_edit",
    code=code))
        ind.name = name
        ind.unit = request.form.get("unit", "").strip() or
    None
        ind.ideal_value = request.form.get("ideal_value",
    "").strip() or None
        ind.ideal_type = request.form.get("ideal_type") or
    ind.ideal_type
        units = request.form.getlist("enabled_units")
```

```
            ind.enabled_units = ",".join(units) if units else
        None
            ind.is_constraint = request.form.get("is_constraint"
        ) == "1"
            ind.description = request.form.get("description",
        "").strip() or None
            db.session.commit()
            flash("指标「%s」已保存" % ind.name, "ok")
            return redirect(url_for("ahp.indicators"))
        return render_template("ahp/indicator_edit.html",
        ind=ind,
                               unit_types=UNIT_TYPES,
                               type_label=ind_svc.ideal_type_lab
        el, form=form)
@bp.route("/indicators/reset", methods=["POST"])
def indicators_reset():
        """按论文取值重置指标定义。"""
        form = EmptyForm()
        if form.validate_on_submit():
            n = ind_svc.seed_indicators()
            flash("已按论文取值重置 %d 项指标定义" % n, "ok")
        return redirect(url_for("ahp.indicators"))
# ---------------------------- 多专家矩阵 ------------------------
        -----
def expert_scope():
        """从请求中取（并规整）单元类型、层级与所属一级指标。"""
        unit_type = request.values.get("unit_type") or
        UNIT_TYPES[0]
        if unit_type not in DEFAULT_WEIGHTS:
            unit_type = UNIT_TYPES[0]
        try:
            level = int(request.values.get("level", 1))
        except (TypeError, ValueError):
            level = 1
        if level not in (1, 2):
            level = 1
        l1 = normalize_l1(unit_type, level,
        request.values.get("l1"))
        return unit_type, level, l1
def expert_query(unit_type, level, l1):
        q = ExpertMatrix.query.filter_by(unit_type=unit_type,
        level=level)
        if level == 2:
            q = q.filter_by(l1_name=l1)
        else:
            q = q.filter(ExpertMatrix.l1_name.is_(None))
        return q.order_by(ExpertMatrix.id)
def default_scheme(unit_type, level, l1=None):
        """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""
        name = "%s · %s%s" % (unit_type,
```

```
                                "一级指标" if level == 1 else "二级指标",
                                "(%s)" % l1 if l1 else "")
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          level=level,
    name=name).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
    level=level, name=name,
                              is_default=False)
        db.session.add(scheme)
        db.session.flush()
    return scheme
@bp.route("/experts", methods=["GET", "POST"])
def experts():
    """多专家判断矩阵录入与汇总。"""
    unit_type, level, l1 = expert_scope()
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        action = request.form.get("action")
        if action == "add":
            expert_name = request.form.get("expert_name",
    "").strip() or "未具名专家"
            try:
                upper = [float(request.form.get("m_%d_%d" %
    (i, j), "1"))
                         for i in range(n) for j in range(i
    + 1, n)]
            except ValueError:
                flash("矩阵输入有误，请填入正数", "error")
                return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                        level=level, l1=l1))
            # 0 / 负数 / inf / nan 都能过 float()，但会让 n 次方根算出
    inf 或 nan，
            # 于是库里存下 [[1,0],[Infinity,1]] 这种矩阵、CR 还显示
    0.0000。
            if any((not math.isfinite(x)) or x <= 0 for x
    in upper):
                flash("判断矩阵只能填正数（大于 0 的有限数字）", "error")
                return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                        level=level, l1=l1))
            A = ahp.matrix_from_upper(upper)
```

```
            r = ahp.weights_and_consistency(A)
            scheme = default_scheme(unit_type, level, l1)
            db.session.add(ExpertMatrix(
                scheme_id=scheme.id,
                unit_type=unit_type, l1_name=(l1 if level
== 2 else None),
                expert_name=expert_name, level=level,
                matrix_json=json.dumps([list(map(float,
row)) for row in A]),
                cr=round(r["cr"], 6)))
            db.session.commit()
            flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,
r["cr"]), "ok")
        elif action == "save_agg":
            # 把汇总权重落库并生效，使多专家结果真正参与评价计算
            weights = _aggregate(unit_type, level, l1)
            if weights is None:
                flash("暂无可汇总的专家矩阵", "error")
            elif not weights["consistent"]:
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"
                      "请先检查或调整专家矩阵" % weights["cr"],
"error")
            else:
                names_, _ = indicator_list(unit_type,
level, l1)
                pairs = to_code_pairs(unit_type, level, l1,
names_,
                                      weights["weights"])
                _, cnt = ws.save_weights(unit_type, pairs)
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"
                      % (unit_type, cnt, weights["cr"]),
"ok")
        else:
            # action 缺失或值不认识时以前是静默跳回，用户只觉得「点了没反应」
            flash("未识别的操作，请从页面上的按钮重新提交", "error")
        return redirect(url_for("ahp.experts",
unit_type=unit_type,
                                level=level, l1=l1))
    rows = expert_query(unit_type, level, l1).all()
    default = [[round(vec[i] / vec[j], 4) for j in
range(n)] for i in range(n)]
    return render_template("ahp/experts.html",
unit_type=unit_type, level=level,
                           l1=l1, names=names, rows=rows,
default=default,
                           n=n, form=form)
@bp.route("/experts/<int:mid>/delete", methods=["POST"])
def expert_delete(mid):
    row = db.get_or_404(ExpertMatrix, mid)
    unit_type, level, l1 = row.unit_type, row.level,
```

```
    row.l1_name                                                    3901
    form = EmptyForm()                                             3902
    if form.validate_on_submit():                                  3903
        db.session.delete(row)                                     3904
        db.session.commit()                                        3905
        flash("已删除专家矩阵", "ok")                                3906
    return redirect(url_for("ahp.experts",                         3907
    unit_type=unit_type,                                           3908
                                    level=level, l1=l1))           3909
def _aggregate(unit_type, level, l1):                              3910
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""          3911
    rows = expert_query(unit_type, level, l1).all()                3912
    mats = []                                                      3913
    for r in rows:                                                 3914
        try:                                                       3915
            mats.append([list(map(float, line))                    3916
                         for line in json.loads(r.matrix_jso3917
    n)])                                                           3918
        except Exception:                                          3919
            continue                                               3920
    if not mats:                                                   3921
        return None                                                3922
    agg = ahp.aggregate_matrices(mats)                             3923
    res = ahp.weights_and_consistency(agg)                         3924
    return {                                                       3925
        "count": len(mats),                                        3926
        "weights": res["weights"],                                 3927
        "lambda_max": res["lambda_max"],                           3928
        "ci": res["ci"], "cr": res["cr"],                          3929
        "consistent": res["consistent"],                           3930
        "most_inconsistent": ahp.find_most_inconsistent(agg)3931
    ,                                                              3932
        "expert_crs": [r.cr for r in rows if r.cr is not     3933
    None],                                                         3934
    }                                                              3935
@bp.route("/experts/aggregate")                                    3936
def experts_aggregate():                                           3937
    """按几何平均汇总多专家矩阵，并做一致性检验。"""                          3938
    unit_type, level, l1 = expert_scope()                          3939
    if unit_type not in DEFAULT_WEIGHTS:                           3940
        flash("无效的微地质单元类型", "error")                          3941
        return redirect(url_for("ahp.index"))                      3942
    names, vec = indicator_list(unit_type, level, l1)              3943
    rows = expert_query(unit_type, level, l1).all()                3944
    result = _aggregate(unit_type, level, l1)                      3945
    if result is not None:                                         3946
        result["weights"] = [round(v, 4) for v in                  3947
    result["weights"]]                                             3948
        result["lambda_max"] = round(result["lambda_max"],   3949
    4)                                                             3950
```

```
    return render_template("ahp/experts_aggregate.html",
    unit_type=unit_type,
                           level=level, l1=l1, names=names,
    rows=rows,
                           result=result, form=EmptyForm())
# ==== app/views/indicator.py ====
# -*- coding: utf-8 -*-
"""指标数据采集与管理 —— 路由与页面逻辑。
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标
（岩质边坡无土壤指标、平台无地形指标）。
"""
import io
import json
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, send_file)
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_ENABLED_INDICATORS
from app.utils.parsing import FieldError, parse_float
from app.services import import_service as imp
bp = Blueprint("indicator", __name__,
    url_prefix="/indicator")
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)
SOIL_FIELDS = [
    ("soil_ratio", "土石比", ""),
    ("thickness", "土层厚度", "cm"),
    ("ph", "pH", ""),
    ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
    ("coverage", "植被覆盖度", "%"),
    ("shannon", "Shannon-Wiener 多样性指数", ""),
]
TERRAIN_FIELDS = [
    ("slope", "坡度", "°"),
    ("aspect", "坡向", ""),
]
def unit_field_config(unit_type):
    """根据单元类型返回需录入的字段分组。"""
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())
    groups = []
    if "土壤指标" in enabled:
        groups.append(("土壤指标", SOIL_FIELDS, True))  # True
    = 含重金属
    if "植被指标" in enabled:
        groups.append(("植被指标", VEG_FIELDS, False))
    if "地形指标" in enabled:
```

```
        groups.append(("地形指标", TERRAIN_FIELDS, False))
    return groups
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    return render_template("indicator/index.html", patches=patches)
@bp.route("/<int:unit_id>", methods=["GET", "POST"])
def edit(unit_id):
    unit = db.get_or_404(MicroUnit, unit_id)
    ind = IndicatorData.query.filter_by(unit_id=unit.id).first()
    form = EmptyForm()
    groups = unit_field_config(unit.unit_type)
    if request.method == "POST" and form.validate_on_submit():
        # 各字段的合理范围（覆盖度是百分数、pH 0–14、坡度 0–90，其余非负）。
        # 以前这里直接 float(v)，用户填个「约 3.5」看到的就是英文 500 页。
        LIMITS = {
            "soil_ratio": ("土石比", 0, None),
            "thickness": ("土层厚度（cm）", 0, None),
            "ph": ("pH", 0, 14),
            "organic_matter": ("有机质（g/kg）", 0, None),
            "coverage": ("植被覆盖度（%）", 0, 100),
            "shannon": ("Shannon 指数", 0, None),
            "slope": ("坡度（°）", 0, 90),
        }
        try:
            values = {}
            for name, (label, lo, hi) in LIMITS.items():
                values[name] = parse_float(request.form.get(name), label,
                                           minimum=lo, maximum=hi)
            heavy = {}
            for m in HEAVY_METALS:
                v = parse_float(request.form.get("hm_" + m),
                                "重金属 %s（mg/kg）" % m, minimum=0)
                if v is not None:
                    heavy[m] = v
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("indicator.edit", unit_id=unit.id))
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        for name, value in values.items():
```

```
            setattr(ind, name, value)                                    4051
        ind.heavy_metals = json.dumps(heavy,                             4052
    ensure_ascii=False) if heavy else None                               4053
        ind.aspect = request.form.get("aspect", "").strip()              4054
    or None                                                              4055
        db.session.commit()                                              4056
        flash("指标数据已保存", "ok")                                    4057
        return redirect(url_for("indicator.index"))                      4058
    # 组装当前值供模板回显                                               4059
    current = {}                                                         4060
    if ind:                                                              4061
        current = {                                                      4062
            "soil_ratio": ind.soil_ratio, "thickness":                   4063
    ind.thickness,                                                       4064
            "ph": ind.ph, "organic_matter":                              4065
    ind.organic_matter,                                                  4066
            "coverage": ind.coverage, "shannon":                         4067
    ind.shannon,                                                         4068
            "slope": ind.slope, "aspect": ind.aspect,                    4069
        }                                                                4070
        current["heavy"] = ind.heavy_metals_dict()                       4071
    return render_template("indicator/edit.html",                        4072
                           unit=unit, form=form,                         4073
    groups=groups,                                                       4074
                           heavy_metals=HEAVY_METALS,                    4075
                           aspect_types=ASPECT_TYPES,                    4076
    current=current)                                                     4077
# ---------------------------- 批量导入 ----------------------------    4078
    ----                                                                 4079
@bp.route("/import", methods=["GET", "POST"])                            4080
def import_data():                                                       4081
    """批量导入指标数据（CSV / Excel）。"""                              4082
    patches = MinePatch.query.order_by(MinePatch.id.desc()).             4083
    all()                                                                4084
    form = EmptyForm()                                                   4085
    report = None                                                        4086
    if request.method == "POST" and form.validate_on_submit(             4087
    ):                                                                   4088
        patch_id = request.form.get("patch_id", type=int)                4089
        patch = db.session.get(MinePatch, patch_id) if                   4090
    patch_id else None                                                   4091
        if patch is None:                                                4092
            flash("请选择矿山图斑", "error")                             4093
            return redirect(url_for("indicator.import_data")             4094
    )                                                                    4095
        file = request.files.get("datafile")                             4096
        if file is None or not file.filename:                            4097
            flash("请选择要导入的文件", "error")                         4098
            return redirect(url_for("indicator.import_data")             4099
    )                                                                    4100
```

```
    ext = file.filename.rsplit(".", 1)[-1].lower()
    if ext not in ("csv", "xlsx", "xls"):
        flash("仅支持 CSV 或 Excel（.xlsx/.xls）文件", "error")
        return redirect(url_for("indicator.import_data")
)
    units = MicroUnit.query.filter_by(patch_id=patch.id)
.all()
    unit_map = {}
    for u in units:
        unit_map.setdefault(u.unit_type, u)
    if not unit_map:
        flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")
        return redirect(url_for("indicator.import_data")
)
    try:
        parsed, errors = imp.parse_upload(file.filename,
file.stream, unit_map)
    except Exception as e:
        flash("文件解析失败：%s" % e, "error")
        return redirect(url_for("indicator.import_data")
)
    saved = 0
    for rec in parsed:
        unit = unit_map[rec["unit_type"]]
        ind = IndicatorData.query.filter_by(unit_id=unit
.id).first()
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        f = rec["fields"]
        for key in ("soil_ratio", "thickness", "ph",
"organic_matter",
                    "coverage", "shannon", "slope",
"aspect"):
            if key in f:
                setattr(ind, key, f[key])
        if rec["heavy_metals"]:
            ind.heavy_metals = json.dumps(rec["heavy_met
als"],

ensure_ascii=False)
        saved += 1
    db.session.commit()
    report = {"patch": patch.name, "saved": saved,
"errors": errors,
              "total": len(parsed) + len(
                  [e for e in errors if e.startswith("第
")])}
    if saved:
        flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")
```

```
        else:                                                          4151
            flash("没有可导入的数据行，请检查文件内容", "error")                4152
    return render_template("indicator/import.html",                    4153
    patches=patches,                                                   4154
                                form=form, report=report,              4155
    headers=imp.HEADERS)                                               4156
@bp.route("/import/template")                                          4157
def import_template():                                                 4158
    """下载导入模板（CSV，含一行示例）。"""                                  4159
    csv_text = imp.build_template_csv()                                4160
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于             4161
    Excel 正确识别中文                                                     4162
    return send_file(io.BytesIO(data), as_attachment=True,             4163
                     download_name="指标数据导入模板.csv",                 4164
                     mimetype="text/csv")                              4165
# ==== app/views/evaluate.py ====                                      4166
# -*- coding: utf-8 -*-                                                4167
"""适宜性定量评价引擎 —— 路由与页面逻辑。                                    4168
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，                          4169
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。                          4170
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），   4171
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。                       4172
"""                                                                    4173
import json                                                            4174
from flask import Blueprint, render_template, request,                 4175
    redirect, url_for, flash                                           4176
from app.forms import EmptyForm                                        4177
from app.models import (db, MinePatch, MicroUnit,                      4178
    IndicatorData,                                                     4179
                        EvaluationResult, ScoreDetail)                 4180
from app.services import evaluation_service as ev                      4181
from app.services import workflow as wf                                4182
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")           4183
def load_patch_input(patch_id):                                        4184
    """从数据库组装评价引擎输入。"""                                          4185
    units = MicroUnit.query.filter_by(patch_id=patch_id).all           4186
    ()                                                                 4187
    result = []                                                        4188
    for u in units:                                                    4189
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi           4190
    rst()                                                              4191
        data = {                                                       4192
            "soil_ratio": ind.soil_ratio if ind else None,             4193
            "thickness": ind.thickness if ind else None,               4194
            "ph": ind.ph if ind else None,                             4195
            "organic_matter": ind.organic_matter if ind                4196
    else None,                                                         4197
            "heavy_metals": ind.heavy_metals_dict() if ind             4198
    else {},                                                           4199
            "coverage": ind.coverage if ind else None,                 4200
```

```
                "shannon": ind.shannon if ind else None,
                "slope": ind.slope if ind else None,
                "aspect": ind.aspect if ind else None,
            }
            result.append({
                "unit_id": u.id,
                "unit_type": u.unit_type,
                "ratio": u.ratio if u.ratio is not None else
    0.0,
                "data": data,
            })
        return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自 AHP 计算并保存的生效方案；未配置时用论文表 3 的默认值。
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 JSON。
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
            "unit_score": _round(u["unit_score"]),
            "weighted": _round(u["weighted"]),
            "weight_source": sources.get(u["unit_type"]),
            "l1_scores": {k: _round(v) for k, v in
    u["l1_scores"].items()},
            "details": [{
                "l1": d["l1"], "l2": d["l2"],
                "raw": None if d["raw"] is None else
    str(d["raw"]),
                "std_value": _round(d["std_value"]),
```

```
                    "weight": _round(d["weight"]),                    4251
                    "contribution": _round(d["contribution"]),        4252
                } for d in u["details"]],                             4253
            })                                                        4254
    return out                                                        4255
def save_evaluation(patch_id, units):                                 4256
    """执行评价并把结果与逐层明细落库，返回评价记录。                  4257
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用        4258
    论文表 3 的默认权重；来源一并记录。结果 JSON 保留各单元得分、      4259
    一级指标得分与逐条二级指标明细，结果页与报告据此复现              4260
    记录下来的那次评价，不再重新推算。                                4261
    """                                                               4262
    weights, sources = load_weights(units)                            4263
    res = ev.score_mine(units, weights)                               4264
    saved = EvaluationResult(                                         4265
        patch_id=patch_id,                                            4266
        final_score=round(res["final_score"], 4),                     4267
        grade=res["grade"],                                           4268
        measures=res["grade_info"]["measure"],                        4269
        unit_scores=json.dumps(pack_units(res["unit_results"]         4270
    ], sources),                                                      4271
                                        ensure_ascii=False),          4272
    )                                                                 4273
    db.session.add(saved)                                             4274
    db.session.flush()        # 先取得 result.id，再写明细            4275
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}          4276
    for u in res["unit_results"]:                                     4277
        for d in u["details"]:                                        4278
            db.session.add(ScoreDetail(                               4279
                result_id=saved.id,                                   4280
                unit_id=unit_ids.get(u["unit_type"]),                 4281
                l1_name=d["l1"], l2_name=d["l2"],                     4282
                raw_value=None if d["raw"] is None else               4283
    str(d["raw"]),                                                    4284
                std_value=d["std_value"],                             4285
    weight=d["weight"],                                               4286
                contribution=d["contribution"]))                      4287
    db.session.commit()                                               4288
    return saved                                                      4289
def block_reasons(progress):                                          4290
    """把未满足的前置条件转成可读的拦截原因。"""                      4291
    return ["%s：%s" % (b["label"], b["reason"]) for b in             4292
    progress["blocking"]]                                             4293
@bp.route("/")                                                        4294
def index():                                                          4295
    patches = MinePatch.query.order_by(MinePatch.id.desc()).          4296
    all()                                                             4297
    latest = {}                                                       4298
    progress = {}                                                     4299
    for p in patches:                                                 4300
```

```
            latest[p.id] = EvaluationResult.query.filter_by(patc
        h_id=p.id) \
                .order_by(EvaluationResult.id.desc()).first()
            progress[p.id] = wf.patch_progress(p.id)
        return render_template("evaluate/index.html",
        patches=patches,
                               latest=latest, progress=progress)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
        patch = db.get_or_404(MinePatch, patch_id)
        form = EmptyForm()
        progress = wf.patch_progress(patch_id)
        if request.method == "POST" and form.validate_on_submit(
        ):
            if not progress["ready"]:
                for reason in block_reasons(progress):
                    flash("无法执行评价 —— " + reason, "error")
                return redirect(url_for("evaluate.run",
        patch_id=patch_id))
            units = load_patch_input(patch_id)
            saved = save_evaluation(patch.id, units)
            return redirect(url_for("evaluate.result",
        result_id=saved.id))
        # 展示将采用的权重来源，便于在执行前确认
        units_now = load_patch_input(patch_id)
        _, sources = load_weights(units_now) if units_now else
        ({}, {})
        return render_template("evaluate/run.html",
        patch=patch, form=form,
                               progress=progress,
        weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
        r = db.get_or_404(EvaluationResult, result_id)
        patch = db.session.get(MinePatch, r.patch_id)
        unit_scores = json.loads(r.unit_scores) if
        r.unit_scores else []
        return render_template("evaluate/result.html", r=r,
        patch=patch,
                               unit_scores=unit_scores)
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
import json
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
from app.services import report_service as rs
```

```
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.utils.constants import grade_of
from app.utils.timefmt import now_utc
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查
    结果，由调用方拦截并提示先去执行评价。
    """
    patch = db.get_or_404(MinePatch, patch_id)
    prog = wf.report_progress(patch_id)
    if not prog["ready"]:
        return patch, None, None, None, prog
    rec = prog["result"]
    result = {
        "final_score": rec.final_score,
        "grade": rec.grade,
        "grade_info": grade_of(rec.final_score),
        "unit_results": json.loads(rec.unit_scores) if
    rec.unit_scores else [],
    }
    reference = ReferenceEcosystem.query.filter_by(patch_id=
    patch_id).first()
    return patch, reference, result, rec, prog
def _deny(progress, patch_id, endpoint="report.detail"):
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""
    for b in progress["blocking"]:
        flash("暂不能生成报告 —— %s：%s" % (b["label"],
    b["reason"]), "error")
    if progress["blocking"]:
        flash("请先完成上述步骤后再生成报告。", "error")
    return redirect(url_for("report.index"))
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
    chart_units = charts.hbars(units)
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
            acc[k] = acc.get(k, 0.0) + v
            cnt[k] = cnt.get(k, 0) + 1
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
```

```
    items.sort(key=lambda x: -x[1])                                    4401
    chart_weight = charts.donut(items)                                 4402
    return chart_units, chart_weight                                   4403
@bp.route("/")                                                         4404
def index():                                                           4405
    patches = MinePatch.query.order_by(MinePatch.id.desc()).           4406
    all()                                                              4407
    progress = {p.id: wf.report_progress(p.id) for p in                4408
    patches}                                                           4409
    return render_template("report/index.html",                        4410
    patches=patches,                                                   4411
                                progress=progress)                     4412
@bp.route("/<int:patch_id>")                                           4413
def detail(patch_id):                                                  4414
    patch, reference, result, record, prog =                           4415
    _prepare(patch_id)                                                 4416
    if result is None:                                                 4417
        return _deny(prog, patch_id)                                   4418
    chart_units, chart_weight = _charts(result)                        4419
    return render_template("report/detail.html",                       4420
    patch=patch, result=result,                                        4421
                                record=record,                         4422
    reference=reference,                                               4423
                                chart_units=chart_units,               4424
    chart_weight=chart_weight)                                         4425
@bp.route("/<int:patch_id>/word")                                      4426
def word(patch_id):                                                    4427
    patch, reference, result, record, prog =                           4428
    _prepare(patch_id)                                                 4429
    if result is None:                                                 4430
        return _deny(prog, patch_id)                                   4431
    buf = rs.build_report(patch, result)                               4432
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name             4433
    return send_file(buf, as_attachment=True,                          4434
    download_name=filename,                                            4435
                        mimetype="application/vnd.openxmlformat        4436
    s-officedocument."                                                 4437
                            "wordprocessingml.document")               4438
@bp.route("/<int:patch_id>/pdf")                                       4439
def pdf(patch_id):                                                     4440
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""                4441
    patch, reference, result, record, prog =                           4442
    _prepare(patch_id)                                                 4443
    if result is None:                                                 4444
        return _deny(prog, patch_id)                                   4445
    chart_units, chart_weight = _charts(result)                        4446
    html = render_template(                                            4447
        "report/print.html", patch=patch,                              4448
    reference=reference, result=result,                                4449
        chart_units=chart_units, chart_weight=chart_weight,            4450
```

```
            generated_at=now_utc())                                    4451
        data, err = browser.html_to_pdf(html)                          4452
        if err:                                                        4453
            flash("PDF 导出失败：%s" % err, "error")                    4454
            return redirect(url_for("report.detail",                   4455
        patch_id=patch_id))                                            4456
        return send_file(io.BytesIO(data), as_attachment=True,         4457
                         download_name="自然恢复适宜性评价报告_%s.pdf" %    4458
        patch.name,                                                    4459
                         mimetype="application/pdf")                   4460
# ==== app/views/search.py ====                                        4461
# -*- coding: utf-8 -*-                                                4462
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。                    4463
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，                             4464
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。                        4465
"""                                                                    4466
from sqlalchemy import or_                                             4467
from flask import Blueprint, render_template, request                  4468
from app.models import db, Project, MinePatch, Standard                4469
bp = Blueprint("search", __name__)                                     4470
LIMIT = 50              # 每类结果上限，避免一次渲染过多                    4471
@bp.route("/search")                                                   4472
def index():                                                           4473
    q = (request.args.get("q") or "").strip()                          4474
    projects, patches, standards = [], [], []                          4475
    if q:                                                              4476
        like = "%" + q + "%"                                           4477
        projects = Project.query.filter(or_(                           4478
            Project.name.like(like),                                   4479
            Project.code.like(like),                                   4480
            Project.description.like(like),                            4481
        )).order_by(Project.id.desc()).limit(LIMIT).all()              4482
        patches = MinePatch.query.filter(or_(                          4483
            MinePatch.name.like(like),                                 4484
            MinePatch.admin_region.like(like),                         4485
            MinePatch.mineral_type.like(like),                         4486
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()            4487
        standards = Standard.query.filter(or_(                         4488
            Standard.name.like(like),                                  4489
            Standard.doc_number.like(like),                            4490
            Standard.province.like(like),                              4491
            Standard.summary.like(like),                               4492
        )).order_by(Standard.id).limit(LIMIT).all()                    4493
    total = len(projects) + len(patches) + len(standards)              4494
    return render_template("search.html", q=q,                         4495
    projects=projects,                                                 4496
                                patches=patches,                       4497
    standards=standards, total=total)                                  4498
# ==== app/templates/base.html ====                                    4499
{% import "_macros.html" as ui %}<!DOCTYPE html>                       4500
```

```
<html lang="zh-CN">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
    initial-scale=1.0">
  <link rel="icon" href="{{ url_for('static',
    filename='favicon.ico') }}">
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock
    %}</title>
  <style>
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */
    :root {
      --accent: #2c6b5a;                    /* 主色：青绿，千里江山石绿偏青 */
      --accent-hover: #1e5548;
      --accent-soft: #e6f0ec;
      --side-1: #1a4a4a;                    /* 侧边栏：深青墨 */
      --side-2: #0f3338;
      --danger: #a0392a;                    /* 朱砂红 */
      --danger-soft: #fbeae6;
      --ok: #2c6b5a;
      --ok-soft: #e6f0ec;
      --todo: #c08a3e;                      /* 赭石色待办 */
      --ink: #2a2520;                       /* 深褐墨色 */
      --ink-soft: #6b5d4f;                  /* 褐灰 */
      --line: #e0d8c8;                      /* 绢本边线 */
      --surface: #fdfaf2;                   /* 暖白绢本 */
      --canvas: #f5efe0;                    /* 绢本米黄背景 */
      --radius: 10px;
      --shadow: 0 2px 8px rgba(42, 37, 32, .06);
      --header-h: 60px;
      --nav-w: 226px;
    }
    * { box-sizing: border-box; }
    html, body { margin: 0; padding: 0; }
    body {
      font-family: "Microsoft YaHei", "PingFang SC", "Noto
    Serif SC", "SimSun", serif;
      font-size: 14px;
      line-height: 1.6;
      color: var(--ink);
      background: var(--canvas);
    }
    /* ---------- 跳到主要内容（键盘可达性） ---------- */
    .skip-link {
      position: absolute;
      left: 12px;
      top: -60px;
      z-index: 30;
      padding: 8px 14px;
      background: var(--accent);
```

```
    color: #fff;                                                    4551
    border-radius: var(--radius);                                   4552
    text-decoration: none;                                          4553
    transition: top .15s ease-out;                                  4554
}                                                                   4555
.skip-link:focus { top: 12px; }                                     4556
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ----------          4557
   固定为一屏高，页面滚动交给右侧内容区，                           4558
   这样左侧导航栏（含底部用户区）不会随长页面一起被拉到下面去。 */  4559
.app-shell {                                                        4560
    display: flex;                                                  4561
    align-items: stretch;                                           4562
    height: 100vh;                                                  4563
    height: 100dvh;                                                 4564
    overflow: hidden;                                               4565
}                                                                   4566
/* ---------- 侧边栏 ---------- */                                  4567
.app-side {                                                         4568
    flex: 0 0 var(--nav-w);                                         4569
    width: var(--nav-w);                                            4570
    display: flex;                                                  4571
    flex-direction: column;                                         4572
    background: linear-gradient(180deg, var(--side-1) 0%,          4573
var(--side-2) 100%);                                                4574
    color: #fff;                                                    4575
}                                                                   4576
.app-brand {                                                        4577
    display: flex;                                                  4578
    align-items: center;                                            4579
    gap: 11px;                                                      4580
    padding: 17px 16px 15px;                                        4581
    border-bottom: 1px solid rgba(255, 255, 255, .10);              4582
}                                                                   4583
.app-brand__mark {                                                  4584
    flex: none;                                                     4585
    display: grid;                                                  4586
    place-items: center;                                            4587
    width: 36px;                                                    4588
    height: 36px;                                                   4589
    border-radius: 10px;                                            4590
    background: rgba(255, 255, 255, .15);                           4591
}                                                                   4592
.app-brand__name {                                                  4593
    font-size: 14.5px;                                              4594
    font-weight: 600;                                               4595
    line-height: 1.35;                                              4596
    letter-spacing: .2px;                                           4597
}                                                                   4598
.app-brand__sub {                                                   4599
    font-size: 11.5px;                                              4600
```

```
  line-height: 1.35;
  color: rgba(255, 255, 255, .58);
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y: auto; }
.app-nav__group {
  margin: 12px 0 5px;
  padding: 0 10px;
  font-size: 11.5px;
  letter-spacing: .05em;
  color: rgba(255, 255, 255, .48);
}
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 9px 11px;
  margin-bottom: 2px;
  border-radius: 8px;
  color: rgba(255, 255, 255, .80);
  text-decoration: none;
  transition: background-color .16s ease, color .16s ease;
}
.app-nav a:hover { background: rgba(255, 255, 255, .10); color: #fff; }
.app-nav a:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
  background: rgba(255, 255, 255, .17);
  color: #fff;
  font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
  margin-left: auto;
  min-width: 20px;
  padding: 0 6px;
  font-size: 11px;
  font-weight: 600;
  line-height: 18px;
  text-align: center;
  border-radius: 9px;
  background: var(--todo);
  color: #fff;
  font-variant-numeric: tabular-nums;
}
```

```
.app-user {
  display: flex;
  align-items: center;
  gap: 8px;
  padding: 10px 14px;
  border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__me {
  display: flex;
  flex: 1;
  min-width: 0;
  align-items: center;
  gap: 10px;
  padding: 4px 6px;
  border-radius: 8px;
  color: inherit;
  text-decoration: none;
  transition: background-color .16s ease;
}
.app-user__me:hover { background: rgba(255, 255, 255,
.10); }
.app-user__me:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-user__avatar {
  flex: none;
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  border-radius: 50%;
  background: rgba(255, 255, 255, .16);
  font-size: 13px;
  font-weight: 600;
  object-fit: cover;
}
.app-user__name {
  display: block;
  font-size: 13px;
  line-height: 1.3;
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}
.app-user__state {
  display: flex;
  align-items: center;
  gap: 5px;
  font-size: 11.5px;
```

```
    line-height: 1.3;                                                    4701
    color: rgba(255, 255, 255, .60);                                     4702
}                                                                        4703
.app-user__dot {                                                         4704
   width: 6px;                                                           4705
    height: 6px;                                                         4706
    border-radius: 50%;                                                  4707
    background: #6ed49a;                                                 4708
}                                                                        4709
.app-user__out {                                                         4710
    flex: none;                                                          4711
    padding: 6px;                                                        4712
    border-radius: 6px;                                                  4713
    color: rgba(255, 255, 255, .68);                                     4714
    text-decoration: none;                                               4715
}                                                                        4716
.app-user__out:hover { color: #fff; background:                          4717
rgba(255, 255, 255, .12); }                                              4718
/* ---------- 右侧内容区 ----------                                      4719
    整个骨架固定一屏高，滚动只发生在内容区内部：                        4720
    左侧导航栏（含底部用户区）始终留在原位，右侧页面自己滚。 */          4721
.app-content {                                                           4722
    flex: 1;                                                             4723
   min-width: 0;                                                         4724
    display: flex;                                                       4725
    flex-direction: column;                                              4726
    overflow-y: auto;                                                    4727
    overflow-x: hidden;                                                  4728
    /* 细滚动条：默认半透明，悬停时加深，不抢视线 */                    4729
    scrollbar-width: thin;                                               4730
    scrollbar-color: rgba(31, 107, 69, .28) transparent;                 4731
}                                                                        4732
.app-content::-webkit-scrollbar { width: 12px; }                         4733
.app-content::-webkit-scrollbar-track { background:                      4734
transparent; }                                                           4735
.app-content::-webkit-scrollbar-thumb {                                  4736
    background: rgba(31, 107, 69, .26);                                  4737
    border: 3px solid transparent;                                       4738
    background-clip: content-box;                                        4739
    border-radius: 999px;                                                4740
}                                                                        4741
.app-content::-webkit-scrollbar-thumb:hover {                            4742
    background: rgba(31, 107, 69, .46);                                  4743
    background-clip: content-box;                                        4744
}                                                                        4745
/* ---------- 顶栏：面包屑 + 搜索 ----------                             4746
    粘在内容区顶部，滚动时面包屑与搜索框始终可用 */                      4747
.app-header {                                                            4748
    position: sticky;                                                    4749
    top: 0;                                                              4750
```

```
    z-index: 20;
    display: flex;
    align-items: center;
    gap: 14px;
    height: var(--header-h);
    padding: 0 22px;
    background: var(--surface);
    border-bottom: 1px solid var(--line);
}
.crumb {
    display: flex;
    align-items: center;
    gap: 7px;
    font-size: 13px;
    color: var(--ink-soft);
    white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
    display: flex;
    align-items: center;
    gap: 8px;
    width: 268px;
    padding: 7px 12px;
    background: var(--canvas);
    border: 1px solid var(--line);
    border-radius: 9px;
    color: var(--ink-soft);
    transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
    flex: 1;
    min-width: 0;
    border: none;
    outline: none;
    background: transparent;
    font-family: inherit;
    font-size: 13px;
    color: var(--ink);
}
.search-box button {
    display: grid;
```

```
    place-items: center;
    padding: 0;
    border: none;
    background: transparent;
    color: inherit;
    cursor: pointer;
}
.icon-btn {
    display: grid;
    place-items: center;
    width: 34px;
    height: 34px;
    border-radius: 9px;
    border: 1px solid transparent;
    color: var(--ink-soft);
    text-decoration: none;
    transition: background-color .16s ease, color .16s ease,
                border-color .16s ease;
}
.icon-btn:hover {
    color: var(--accent);
    background: var(--accent-soft);
    border-color: rgba(31, 107, 69, .18);
}
.app-main {
    flex: 1;
    min-width: 0;
    padding: 22px 26px 26px;
}
.container { max-width: none; width: 100%; }
/* ---------- 页脚 ---------- */
.app-foot {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 14px;
    padding: 0 26px 18px;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 未登录时的裸页面（登录页） ---------- */
.login-shell {
    display: flex;
    flex-direction: column;
    min-height: 100dvh;
}
/* ---------- 卡片 ---------- */
.card {
    background: var(--surface);
```

```
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  padding: 22px 24px;
  margin-bottom: 18px;
}
.card > h2 {
  margin: 0 0 14px;
  font-size: 16px;
  font-weight: 600;
  color: var(--accent);
  text-wrap: balance;
}
.card > h3 {
  margin: 20px 0 10px;
  font-size: 14px;
  font-weight: 600;
  color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
  width: 100%;
  border-collapse: collapse;
  margin: 10px 0 16px;
  font-variant-numeric: tabular-nums;
}
caption {
  text-align: left;
  color: var(--ink-soft);
  font-size: 12px;
  padding-bottom: 6px;
}
th, td {
  border: 1px solid var(--line);
  padding: 8px 12px;
  text-align: center;
  vertical-align: middle;
}
th {
  background: var(--accent-soft);
  font-weight: 600;
  white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
  display: block;
  margin-bottom: 4px;
```

```
    color: var(--ink-soft);
    font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
    padding: 6px 9px;
    font-family: inherit;
    font-size: 14px;
    color: var(--ink);
    background: #fff;
    border: 1px solid #c6ccd4;
    border-radius: 4px;
    max-width: 100%;
}
input[type="number"] {
    width: 104px;
    text-align: center;
    font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 1px;
    border-color: var(--accent);
}
.checkbox {
    display: inline-flex;
    align-items: center;
    gap: 7px;
    cursor: pointer;
    user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    gap: 6px;
    padding: 8px 18px;
    font-family: inherit;
    font-size: 14px;
    line-height: 1.4;
    color: #fff;
    background: rgba(31, 107, 69, .80);
```

```
  border: 1px solid rgba(31, 107, 69, .30);                    4951
  border-radius: 8px;                                          4952
  cursor: pointer;                                             4953
  text-decoration: none;                                       4954
  transition: background-color .16s ease, border-color         4955
.16s ease,                                                     4956
              color .16s ease, box-shadow .16s ease,           4957
transform .16s ease;                                           4958
}                                                              4959
.btn:hover {                                                   4960
  background: var(--accent);                                   4961
  border-color: var(--accent);                                 4962
  box-shadow: 0 3px 12px rgba(31, 107, 69, .22);               4963
}                                                              4964
.btn:active { transform: translateY(1px); box-shadow:          4965
none; }                                                        4966
.btn:focus-visible { outline: 2px solid var(--accent);         4967
outline-offset: 2px; }                                         4968
.btn--ghost {                                                  4969
  color: var(--accent);                                        4970
  background: rgba(31, 107, 69, .06);                          4971
  border-color: rgba(31, 107, 69, .22);                        4972
}                                                              4973
.btn--ghost:hover {                                            4974
  color: #fff;                                                 4975
  background: var(--accent);                                   4976
  border-color: var(--accent);                                 4977
}                                                              4978
.btn--danger {                                                 4979
  color: var(--danger);                                        4980
  background: rgba(179, 38, 30, .07);                          4981
  border-color: rgba(179, 38, 30, .26);                        4982
}                                                              4983
.btn--danger:hover {                                           4984
  color: #fff;                                                 4985
  background: var(--danger);                                   4986
  border-color: var(--danger);                                 4987
  box-shadow: 0 3px 12px rgba(179, 38, 30, .20);               4988
}                                                              4989
.btn--sm { padding: 5px 13px; font-size: 13px;                 4990
border-radius: 7px; }                                          4991
.btn:disabled { opacity: .45; cursor: not-allowed; }           4992
.btn:disabled:hover {                                          4993
  background: rgba(31, 107, 69, .80);                          4994
  border-color: rgba(31, 107, 69, .30);                        4995
  box-shadow: none;                                            4996
}                                                              4997
.btn--ghost:disabled:hover { background: rgba(31, 107,         4998
69, .06); color: var(--accent); }                              4999
/* ---------- 表格操作列的行内小按钮 ----------                5000
```

```
    取代原来「文字 | 文字 | 文字」的写法：
    圆角矩形 + 半透明底色，悬停时变实色。 */
.row-actions {
  display: inline-flex;
  flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */
  align-items: center;
  gap: 6px;
  white-space: nowrap;
}
.row-actions form { display: inline-flex; margin: 0; }
.link-btn {
  display: inline-flex;
  align-items: center;
  padding: 4px 11px;
  font-family: inherit;
  font-size: 13px;
  line-height: 1.5;
  color: var(--accent);
  background: rgba(31, 107, 69, .07);
  border: 1px solid rgba(31, 107, 69, .20);
  border-radius: 7px;
  cursor: pointer;
  text-decoration: none;
  white-space: nowrap;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.link-btn:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
}
.link-btn:active { transform: translateY(1px); box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.link-btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
```

```
.actions { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
  padding: 10px 14px;
  margin-bottom: 14px;
  border-radius: var(--radius);
  border-left: 4px solid;
}
.flash.error {
  color: var(--danger);
  background: var(--danger-soft);
  border-left-color: var(--danger);
}
.flash.ok {
  color: var(--ok);
  background: var(--ok-soft);
  border-left-color: var(--ok);
}
/* ---------- 浮层提示（登录成功、各类操作完成、登录失败） ---------- */
   外层 toast-stack 固定在页面中上方，内层每条各自淡入 → 停留 → 淡出，
   多条提示会依次排开而不重叠。
   底色是浅绿半透明：既有一层底色把文字托住，又能透出后面的内容，
   不用高饱和色块，提示不该比页面内容更抢眼。
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
  position: fixed;
  top: 16px;
  left: 50%;
  z-index: 100;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 6px;
  transform: translateX(-50%);
  pointer-events: none;
}
.toast-center {
  max-width: min(620px, 82vw);
  padding: 8px 22px;
  border-radius: 999px;
  /* 浅绿、半透明：不遮住背后的内容，也比纯透明更托得住文字 */
  background: rgba(223, 241, 231, .76);
  border: 1px solid rgba(31, 107, 69, .14);
  color: #2f4a3c;
  font-size: 16px;
  font-weight: 500;
  line-height: 1.65;
  letter-spacing: .03em;
```

```
    text-align: center;
    text-wrap: pretty;
    backdrop-filter: blur(6px) saturate(1.1);
    -webkit-backdrop-filter: blur(6px) saturate(1.1);
    box-shadow: 0 4px 16px rgba(23, 32, 27, .07);
    animation: toast-fade 2.6s ease-in-out forwards;
}
/* 失败类提示：同一套观感，换成浅红底，一眼能区分成败 */
.toast-center--error {
    background: rgba(252, 233, 230, .78);
    border-color: rgba(163, 43, 34, .16);
    color: #9c2b23;
}
@keyframes toast-fade {
    0%          { opacity: 0; transform: translateY(-6px); }
    14%, 74%    { opacity: 1; transform: none; }
    100%        { opacity: 0; transform: none; }
}
/* 系统设置里要求「减少动态效果」时，去掉位移，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
    .toast-center { animation: toast-fade-plain 2.6s
linear forwards; }
    @keyframes toast-fade-plain {
        0%          { opacity: 0; }
        14%, 74%    { opacity: 1; }
        100%        { opacity: 0; }
    }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 12.5px; }
.mono { font-family: Consolas, "Courier New",
monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
    display: inline-block;
    padding: 1px 9px;
    font-size: 12.5px;
    border-radius: 10px;
    border: 1px solid;
    white-space: nowrap;
}
```

```
.tag--ok { color: var(--ok); background:                                    5151
var(--ok-soft); border-color: #b8ddc6; }                                    5152
.tag--warn { color: var(--danger); background:                              5153
var(--danger-soft); border-color: #eec4c0; }                                5154
.tag--muted { color: var(--ink-soft); background:                           5155
#f1f3f6; border-color: #dde1e6; }                                           5156
/* ---------- 登录页 ---------- */                                          5157
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，                5158
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */              5159
.body--login {                                                              5160
  background:                                                               5161
    radial-gradient(1100px 640px at 50% 8%,                                 5162
                    #edf6f1 0%, rgba(237, 246, 241, 0)                      5163
72%),                                                                       5164
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);                      5165
}                                                                           5166
.body--login .app-main { padding: 0; }                                      5167
.body--login .container { max-width: none; width: 100%;                     5168
}                                                                           5169
/* 仅供读屏软件的隐藏文本 */                                                 5170
.visually-hidden {                                                          5171
  position: absolute !important;                                            5172
  width: 1px;                                                               5173
  height: 1px;                                                              5174
  margin: -1px;                                                             5175
  padding: 0;                                                               5176
  overflow: hidden;                                                         5177
  clip: rect(0 0 0 0);                                                      5178
  white-space: nowrap;                                                      5179
  border: 0;                                                                5180
}                                                                           5181
/* ---------- 工作台：页首横幅 ---------- */                                5182
.hero {                                                                     5183
  position: relative;                                                       5184
  overflow: hidden;                                                         5185
  min-height: 176px;                                                        5186
  padding: 26px 28px;                                                       5187
  margin-bottom: 16px;                                                      5188
  border-radius: 12px;                                                      5189
  /* 左侧压深保证文字可读，右侧透出远山插画；                                5190
     背景按底部对齐，避免远山被居中裁掉 */                                   5191
  background:                                                               5192
    linear-gradient(100deg, rgba(15, 40, 45, .90) 0%,                       5193
                            rgba(20, 55, 58, .55) 45%,                      5194
                            rgba(20, 55, 58, .12) 100%),                    5195
    url("{{ url_for('static', filename='hero_landscape.j                    5196
pg') }}")                                                                   5197
    center center / cover no-repeat;                                        5198
  color: #fff;                                                              5199
}                                                                           5200
```

```
.hero__body { max-width: 720px; }
.hero__title {
  margin: 0 0 10px;
  font-size: 21px;
  font-weight: 700;
  line-height: 1.4;
  letter-spacing: .3px;
}
.hero__desc {
  margin: 0;
  max-width: 640px;
  font-size: 12.5px;
  line-height: 1.85;
  color: rgba(255, 255, 255, .86);
}
.hero__actions {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  margin-top: 18px;
}
.hero-btn {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  padding: 7px 15px;
  font-size: 13px;
  border-radius: 8px;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease;
}
.hero-btn--solid {
  color: #14432f;
  background: rgba(255, 255, 255, .92);
  border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
  color: #fff;
  background: rgba(255, 255, 255, .14);
  border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255, 255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff; outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
.stat-cards {
```

```
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(190px,
1fr));
    gap: 12px;
    margin-bottom: 22px;
}
.stat-card {
    display: block;
    padding: 15px 17px;
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    color: var(--ink);
    text-decoration: none;
    transition: border-color .16s ease, box-shadow .16s
ease;
}
.stat-card:hover {
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
    display: block;
    font-size: 12.5px;
    color: var(--ink-soft);
}
.stat-card__value {
    display: block;
    margin: 3px 0 2px;
    font-size: 26px;
    font-weight: 700;
    line-height: 1.25;
    color: var(--accent);
    font-variant-numeric: tabular-nums;
}
.stat-card__unit {
    margin-left: 3px;
    font-size: 13px;
    font-weight: 500;
    color: var(--ink-soft);
}
.stat-card__note {
    display: block;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
```

```
    display: flex;                                                    5301
    align-items: baseline;                                            5302
    flex-wrap: wrap;                                                  5303
    gap: 6px 14px;                                                    5304
    margin-bottom: 12px;                                              5305
}                                                                     5306
.section__title { margin: 0; font-size: 16px; }                       5307
.section__desc { margin: 0; font-size: 12.5px; color:                 5308
var(--ink-soft); }                                                    5309
/* ---------- 工作台：流程步骤卡 ---------- */                          5310
.step-grid {                                                          5311
    display: grid;                                                    5312
    grid-template-columns: repeat(auto-fill,                          5313
minmax(238px, 1fr));                                                  5314
    gap: 12px;                                                        5315
}                                                                     5316
.step-card {                                                          5317
    position: relative;                                               5318
    display: block;                                                   5319
    padding: 16px 16px 15px;                                          5320
    background: var(--surface);                                       5321
    border: 1px solid var(--line);                                    5322
    border-radius: var(--radius);                                     5323
    box-shadow: var(--shadow);                                        5324
    color: var(--ink);                                                5325
    text-decoration: none;                                            5326
    transition: border-color .16s ease, box-shadow .16s               5327
ease,                                                                 5328
                transform .16s ease;                                  5329
}                                                                     5330
.step-card:hover {                                                    5331
    border-color: rgba(31, 107, 69, .55);                             5332
    box-shadow: 0 5px 16px rgba(19, 60, 43, .10);                     5333
    transform: translateY(-1px);                                      5334
}                                                                     5335
.step-card:focus-visible { outline: 2px solid                         5336
var(--accent); outline-offset: 2px; }                                 5337
.step-card__no {                                                      5338
    position: absolute;                                               5339
    top: 13px;                                                        5340
    right: 15px;                                                      5341
    font-size: 17px;                                                  5342
    font-weight: 700;                                                 5343
    color: #d7e3db;                                                   5344
    font-variant-numeric: tabular-nums;                               5345
}                                                                     5346
.step-card__icon {                                                    5347
    display: grid;                                                    5348
    place-items: center;                                              5349
    width: 32px;                                                      5350
```

```
    height: 32px;
    margin-bottom: 10px;
    border-radius: 9px;
    background: var(--accent-soft);
    color: var(--accent);
}
.step-card__title {
    display: block;
    margin-bottom: 5px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
.step-card__desc {
    display: block;
    font-size: 12.5px;
    line-height: 1.65;
    color: var(--ink-soft);
    text-wrap: pretty;
}
.step-card__badge {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #fdf1e2;
    color: #9a5a12;
    border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
    background: #fafbfa;
    border-color: #eceff0;
    box-shadow: none;
    cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
    background: #f1f3f2;
    color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
```

```
    background: #f1f3f2;
    color: #8b948f;
}
/* ---------- 结果摘要 ---------- */
.stat-row {
    display: flex;
    flex-wrap: wrap;
    gap: 28px;
    margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
margin: 4px 0 18px; }
.step {
    display: flex; align-items: center; gap: 7px;
    padding: 8px 12px; font-size: 13px;
    background: #fff; border: 1px solid var(--line);
    border-radius: var(--radius);
}
.step__no {
    display: inline-flex; align-items: center;
justify-content: center;
    width: 20px; height: 20px; border-radius: 50%;
    background: #eef1f5; color: var(--ink-soft);
    font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color:
var(--ink-soft); }
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块（中性底色，不用整块警示色压页面） ---------- */
.notice {
    padding: 14px 16px; margin: 0 0 16px;
```

```
    background: var(--surface);
    border: 1px solid var(--line);
    border-left: 3px solid var(--todo);
    border-radius: var(--radius);
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
/* ---------- 前置条件表（与页面主表同款，只把待处理的行标成浅红） ---------- */
.prereq-table { margin: 0 0 18px; }
.prereq-table > caption { padding-bottom: 7px; }
.prereq-row > td { background: var(--danger-soft); }
.prereq-row:hover > td { background: #f6dcd7; }
.prereq-flag { color: var(--danger); font-weight: 600; }
.prereq-row .muted { color: #9a6b62; }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
    border: 1px solid var(--line);
    border-radius: var(--radius);
    background: var(--surface);
    padding: 14px 16px;
    margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
    gap: 6px 14px;
    margin: 0 0 12px;
    padding-bottom: 9px;
    border-bottom: 1px solid var(--line);
}
.patch-panel__name {
    margin: 0;
    font-size: 15px;
    font-weight: 600;
}
.patch-panel__meta {
    font-size: 12.5px;
    color: var(--ink-soft);
}
.patch-panel__meta a { color: var(--accent); }
.patch-panel__tail { margin-left: auto; }
.patch-panel .notice { margin: 0; }
.patch-panel table { margin: 0; }
/* ---------- 窄屏适配 ---------- */
@media (max-width: 960px) {
    /* 上下布局时取消「一屏高 + 内容区内部滚动」，
       否则导航会占掉半屏、内容被挤在剩下的小格子里滚动 */
    .app-shell {
        flex-direction: column;
```

```
          height: auto;                                                    5501
          min-height: 100vh;                                               5502
          min-height: 100dvh;                                              5503
          overflow: visible;                                               5504
        }                                                                  5505
        .app-content { overflow: visible; }                                5506
        .app-header { position: static; }                                  5507
        .app-side { width: auto; flex: none; }                             5508
        .app-brand { padding: 12px 14px; }                                 5509
        .app-nav { padding: 4px 10px 10px; }                               5510
        .app-nav ul { display: flex; flex-wrap: wrap; gap:                 5511
    4px; }                                                                 5512
        .app-nav a { margin-bottom: 0; }                                   5513
        .app-nav__group { margin: 8px 0 4px; }                             5514
        .app-user { padding: 10px 14px; }                                  5515
        .app-header { padding: 0 14px; gap: 10px; }                        5516
        .search-box { width: 180px; }                                      5517
        .app-main { padding: 16px 14px 20px; }                             5518
        .app-foot { padding: 0 14px 14px; }                                5519
    }                                                                      5520
    @media (max-width: 640px) {                                            5521
        .crumb { display: none; }                                          5522
        .search-box { width: auto; flex: 1; }                              5523
    }                                                                      5524
  </style>                                                                 5525
</head>                                                                    5526
<body class="{% if not current_user.is_authenticated                       5527
    %}body--login{% endif %}">                                             5528
  <a class="skip-link" href="#main">跳到主要内容</a>                          5529
  {# 取一次即可（同一请求内重复调用返回相同内容）。                            5530
     类别为 toast / ok / toast-error 的渲染成中上方浮层，出现后自动淡出；      5531
     其余仍是页面顶部的提示条 ——                                            5532
     需要对照着改表单的错误，留在页面上比一闪而过更有用。 #}                   5533
  {% set flash_msgs = get_flashed_messages(with_categories=t                5534
    rue) %}                                                                5535
  <div class="toast-stack">                                                5536
    {% for category, message in flash_msgs if category in                  5537
    ("toast", "ok", "toast-error") %}                                      5538
    <div class="toast-center{% if category == 'toast-error'                5539
    %} toast-center--error{% endif %}"                                     5540
         role="status" aria-live="polite">{{ message                       5541
    }}</div>                                                               5542
    {% endfor %}                                                           5543
  </div>                                                                   5544
  {% if current_user.is_authenticated %}                                   5545
  <div class="app-shell">                                                  5546
    <aside class="app-side">                                               5547
      <div class="app-brand">                                              5548
        <span class="app-brand__mark">{{                                   5549
    ui.icon("mountain", 21) }}</span>                                      5550
```

```
      <span>                                                          5551
        <span class="app-brand__name">矿山自然恢复评价</span><br>      5552
        <span class="app-brand__sub">适宜性评价系统 V1.0</span>       5553
      </span>                                                         5554
    </div>                                                            5555
    <nav class="app-nav" aria-label="功能导航">                       5556
      <ul>                                                            5557
        <li>                                                          5558
          <a href="/"{% if home_active %}                             5559
aria-current="page"{% endif %}>                                       5560
            {{ ui.icon("home") }}<span>工作台</span>                  5561
          </a>                                                        5562
        </li>                                                         5563
      </ul>                                                           5564
      {% for g in nav_groups %}                                       5565
      <p class="app-nav__group">{{ g.title }}</p>                     5566
      <ul>                                                            5567
        {% for item in g.links %}                                     5568
        <li>                                                          5569
          <a href="{{ item.href }}"{% if item.active %}               5570
aria-current="page"{% endif %}>                                       5571
            {{ ui.icon(item.icon) }}<span>{{ item.label               5572
}}</span>                                                             5573
            {% if item.badge %}<span                                  5574
class="app-nav__badge">{{ item.badge }}</span>{% endif                5575
%}                                                                    5576
          </a>                                                        5577
        </li>                                                         5578
        {% endfor %}                                                  5579
      </ul>                                                           5580
      {% endfor %}                                                    5581
    </nav>                                                            5582
    <div class="app-user">                                            5583
      <a class="app-user__me" href="/profile"                         5584
title="个人设置">                                                     5585
        {% if current_user.avatar %}                                  5586
        <img class="app-user__avatar"                                 5587
             src="{{ url_for('static',                                5588
filename=current_user.avatar) }}" alt="">                             5589
        {% else %}                                                    5590
        <span class="app-user__avatar"                                5591
aria-hidden="true">{{ current_user.initial }}</span>                  5592
        {% endif %}                                                   5593
        <span style="min-width:0;">                                   5594
          <span class="app-user__name">{{                            5595
current_user.shown_name }}</span>                                     5596
          <span class="app-user__state">                              5597
            <span class="app-user__dot"                               5598
aria-hidden="true"></span>                                            5599
            {{ role_labels.get(current_user.role,                     5600
```

```
    current_user.role) }}                                                5601
            </span>                                                      5602
          </span>                                                        5603
        </a>                                                             5604
        <a class="app-user__out" href="/logout" title="退出登录"          5605
           aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>         5606
      </div>                                                             5607
    </aside>                                                             5608
    <div class="app-content">                                            5609
      <header class="app-header">                                        5610
        <nav class="crumb" aria-label="当前位置">                        5611
          <a href="/">首页</a>                                           5612
          {{ ui.icon("chevron", 13) }}                                   5613
          <strong>{{ page_title }}</strong>                              5614
        </nav>                                                           5615
        <span class="app-header__spacer"></span>                         5616
        <form class="search-box" method="get"                            5617
    action="/search" role="search">                                      5618
          {{ ui.icon("search", 16) }}                                    5619
          <label for="q" class="visually-hidden">搜索项目、图斑、标准<    5620
    /label>                                                              5621
          <input id="q" type="search" name="q"                           5622
    autocomplete="off"                                                   5623
                 placeholder="搜索项目、图斑、标准" value="{{            5624
    search_q }}">                                                        5625
          <button type="submit" aria-label="搜索">{{                     5626
    ui.icon("chevron", 15) }}</button>                                   5627
        </form>                                                          5628
        {% if current_user.is_admin %}                                   5629
        <a class="icon-btn" href="/profile?tab=logs"                     5630
    title="操作日志"                                                     5631
           aria-label="操作日志">{{ ui.icon("bell", 18) }}</a>           5632
        {% endif %}                                                      5633
      </header>                                                          5634
{% else %}                                                               5635
<div class="login-shell">                                                5636
{% endif %}                                                              5637
    <main id="main" class="app-main">                                    5638
      <div class="container">                                            5639
        {% for category, message in flash_msgs if category               5640
    not in ("toast", "ok", "toast-error") %}                             5641
        <p class="flash {{ category }}" role="status">{{                 5642
    message }}</p>                                                       5643
        {% endfor %}                                                     5644
        {% block content %}{% endblock %}                                5645
      </div>                                                             5646
    </main>                                                              5647
{% if current_user.is_authenticated %}                                   5648
      <footer class="app-foot">                                          5649
        <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０</span>      5650
```

```
            <span>数据截止 {{ today }}</span>
          </footer>
        </div>
      </div>
      {% else %}
      </div>
      {% endif %}
      {% block scripts %}{% endblock %}
</body>
</html>
# ==== app/templates/_macros.html ====
{# 共用宏：图标、流程状态面板 #}
{# ================================================================
   ================
   图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）
   不引图标字体或外部图片，避免额外请求与资源依赖；
   stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。
   ================================================================
   ================ #}
{% set ICONS = {
  "dot": '<circle cx="12" cy="12" r="3"/>',
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5
    17h5"/>',
  "database": '<ellipse cx="12" cy="5.5" rx="8"
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2"/>',
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"
    cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',
  "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0
    2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14
    2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',
  "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5
    20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6
    3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path
    d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
  "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5
    21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4
```

```
      4.4V21"/>',
   "mail": '<rect x="3" y="5" width="18" height="14"
      rx="2.5"/><path d="m3.8 7 8.2 5.8L20.2 7"/>',
   "phone": '<path d="M6.4 3h2.8l1.5 4.4-2 1.4a12.4 12.4 0 0
      0 6.5 6.5l1.4-2L21 14.8v2.8a2.4 2.4 0 0 1-2.6 2.4A16.6
      16.6 0 0 1 3.5 5.6 2.4 2.4 0 0 1 5.9 3z"/>',
   "lock": '<rect x="4" y="10.4" width="16" height="10.6"
      rx="2.6"/><path d="M8 10.4V7.6a4 4 0 0 1 8 0v2.8"/>',
   "briefcase": '<rect x="3" y="7.4" width="18"
      height="13.1" rx="2.6"/><path d="M8.4 7.4V6A2 2 0 0 1
      10.4 4h3.2a2 2 0 0 1 2 2v1.4"/><path d="M3 12.6h18"/>',
   "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
      20.5-4-4"/>',
   "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
      6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
      1-3.6 0"/>',
   "plus": '<path d="M12 5v14M5 12h14"/>',
   "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
      4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
      21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
   "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
      2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
      7.5"/><path d="M5.5 12H15"/>',
   "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
   "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
      10v4.2"/><path d="M12 17.4h.01"/>',
   "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
      20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
      fill="none" stroke="currentColor" stroke-width="1.7"
      stroke-linecap="round" stroke-linejoin="round"
      aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
```

```
        {% elif progress.constraint.done %}未通过
        {% else %}待完成{% endif %}
      </span>
    </div>
    <div class="step {% if progress.units.done %}step--done{%
      else %}step--todo{% endif %}">
      <span class="step__no">2</span>
      <span class="step__name">划分微地质单元</span>
      <span class="step__state">
        {% if progress.units.done %}已完成{% else %}待完成{% endif
      %}
      </span>
    </div>
    <div class="step {% if progress.data.ready %}step--done{%
      else %}step--todo{% endif %}">
      <span class="step__no">3</span>
      <span class="step__name">录入指标数据</span>
      <span class="step__state">
        {% if progress.data.ready %}已完成{% else %}待完成{% endif
      %}
      </span>
    </div>
    <div class="step step--next">
      <span class="step__no">4</span>
      <span class="step__name">执行适宜性评价</span>
      <span class="step__state">当前步骤</span>
    </div>
  </div>
</div>
{% endmacro %}
{# 前置条件未满足时的提示：做成与页面主表同款的表格，只把待处理的那几行标成浅红，
   说明写在 caption 里，缺什么、去哪补都在行内。 #}
{% macro workflow_blocking(progress) %}
{% if progress.blocking %}
<table class="prereq-table">
  <caption>本步骤的前置条件尚未满足，按论文 4.1 节的流程顺序先补上：</caption>
  <thead>
    <tr>
      <th scope="col">前置步骤</th>
      <th scope="col">说明</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for b in progress.blocking %}
    <tr class="prereq-row">
      <td>{{ b.label }}</td>
      <td class="muted">{{ b.reason }}</td>
      <td><a class="btn btn--sm" href="{{ b.fix_url }}">{{
      b.fix_label }}</a></td>
    </tr>
```

```
    {% endfor %}
  </tbody>
</table>
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label }}</span>
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
   只在流程里**第一个**按图斑操作的页面（限制性指标判定）列一次，后面各页
   不再重复；建好图斑才会出现在下面的表里，但离「能进入本页流程」还差别的步骤，
   所以措辞只说到「出现在这里」为止。 #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
<table class="prereq-table">
  <caption>以下项目下尚未建立矿山图斑，本页按图斑组织数据，建好图斑后才会出现在这里：</caption>
  <thead>
    <tr>
      <th scope="col">所属项目</th>
      <th scope="col">矿山图斑</th>
      <th scope="col">状态</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for p in projects_without_patch %}
    <tr class="prereq-row">
      <td>项目「{{ p.name }}」</td>
      <td class="muted">未建立</td>
      <td><span class="prereq-flag">缺少矿山图斑</span></td>
      <td>
        <a class="btn btn--sm" href="/project/{{ p.id }}/patch/add">
          {{ icon("plus", 14) }}去新建图斑
        </a>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endif %}
{% endmacro %}
# ==== app/templates/_login_badge.svg ====
```

```
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg
  " viewBox="0 0 240 240" role="img"
  aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">
 <title>系统徽标</title>
 <g class="lp-ring" style="--k: 2.7">
  <circle cx="162.00" cy="120.00" r="2.60"
  fill="#c4f5ab"/>
  <circle cx="157.84" cy="138.22" r="2.97"
  fill="#b8f5ab"/>
  <circle cx="146.19" cy="152.84" r="3.06"
  fill="#acf5ab"/>
  <circle cx="129.35" cy="160.95" r="2.80"
  fill="#abf5b7"/>
  <circle cx="110.65" cy="160.95" r="2.40"
  fill="#abf5c2"/>
  <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce"/>
  <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da"/>
  <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6"/>
  <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da"/>
  <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce"/>
  <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2"/>
  <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7"/>
  <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab"/>
  <circle cx="157.84" cy="101.78" r="2.23"
  fill="#b8f5ab"/>
 </g>
 <g class="lp-ring" style="--k: 2.15">
  <circle cx="177.29" cy="129.07" r="3.92"
  fill="#a4e28d"/>
  <circle cx="171.68" cy="146.33" r="4.01"
  fill="#9be28d"/>
  <circle cx="161.01" cy="161.01" r="3.87"
  fill="#91e28d"/>
  <circle cx="146.33" cy="171.68" r="3.55"
  fill="#8de292"/>
  <circle cx="129.07" cy="177.29" r="3.17"
  fill="#8de29c"/>
  <circle cx="110.93" cy="177.29" r="2.88"
  fill="#8de2a6"/>
  <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af"/>
  <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9"/>
  <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2"/>
  <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc"/>
  <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc"/>
  <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2"/>
  <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9"/>
  <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af"/>
  <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6"/>
  <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c"/>
  <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292"/>
```

```
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d"/>
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d"/>
    <circle cx="177.29" cy="110.93" r="3.63"
    fill="#a4e28d"/>
  </g>
  <g class="lp-ring" style="--k: 1.85">
    <circle cx="194.00" cy="120.00" r="4.94"
    fill="#91cf72"/>
    <circle cx="191.85" cy="137.71" r="4.92"
    fill="#88cf72"/>
    <circle cx="185.52" cy="154.39" r="4.73"
    fill="#80cf72"/>
    <circle cx="175.39" cy="169.07" r="4.42"
    fill="#78cf72"/>
    <circle cx="162.04" cy="180.90" r="4.06"
    fill="#72cf73"/>
    <circle cx="146.24" cy="189.19" r="3.73"
    fill="#72cf7b"/>
    <circle cx="128.92" cy="193.46" r="3.51"
    fill="#72cf83"/>
    <circle cx="111.08" cy="193.46" r="3.45"
    fill="#72cf8b"/>
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93"/>
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c"/>
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4"/>
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac"/>
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4"/>
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc"/>
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4"/>
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac"/>
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4"/>
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c"/>
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93"/>
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b"/>
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83"/>
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b"/>
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73"/>
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72"/>
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72"/>
    <circle cx="191.85" cy="102.29" r="4.80"
    fill="#88cf72"/>
  </g>
  <g class="lp-ring" style="--k: 1.62">
    <circle cx="209.57" cy="128.82" r="5.68"
    fill="#76bb59"/>
    <circle cx="206.12" cy="146.13" r="5.40"
    fill="#6fbb59"/>
    <circle cx="199.37" cy="162.43" r="5.06"
    fill="#68bb59"/>
    <circle cx="189.57" cy="177.10" r="4.71"
```

```
    fill="#61bb59" />                                                    5951
    <circle cx="177.10" cy="189.57" r="4.41"                             5952
    fill="#5abb59" />                                                    5953
    <circle cx="162.43" cy="199.37" r="4.19"                             5954
    fill="#59bb5f" />                                                    5955
    <circle cx="146.13" cy="206.12" r="4.10"                             5956
    fill="#59bb66" />                                                    5957
    <circle cx="128.82" cy="209.57" r="4.15"                             5958
    fill="#59bb6d" />                                                    5959
    <circle cx="111.18" cy="209.57" r="4.32"                             5960
    fill="#59bb73" />                                                    5961
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a" />            5962
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81" />            5963
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88" />            5964
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f" />            5965
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96" />            5966
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d" />            5967
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4" />            5968
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4" />            5969
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d" />             5970
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96" />             5971
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f" />             5972
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88" />             5973
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81" />             5974
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a" />             5975
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73" />            5976
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d" />            5977
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66" />            5978
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f" />            5979
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59" />            5980
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59" />            5981
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59" />            5982
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59" />            5983
    <circle cx="209.57" cy="111.18" r="5.85"                             5984
    fill="#76bb59" />                                                    5985
  </g>                                                                   5986
  <g class="lp-ring" style="--k: 1.46">                                  5987
    <circle cx="226.00" cy="120.00" r="5.94"                             5988
    fill="#65a843" />                                                    5989
    <circle cx="224.55" cy="137.45" r="5.62"                             5990
    fill="#5fa843" />                                                    5991
    <circle cx="220.26" cy="154.42" r="5.29"                             5992
    fill="#59a843" />                                                    5993
    <circle cx="213.22" cy="170.45" r="5.00"                             5994
    fill="#53a843" />                                                    5995
    <circle cx="203.65" cy="185.11" r="4.77"                             5996
    fill="#4da843" />                                                    5997
    <circle cx="191.79" cy="197.99" r="4.63"                             5998
    fill="#47a843" />                                                    5999
    <circle cx="177.98" cy="208.74" r="4.59"                             6000
```

```
      fill="#43a846"/>                                                    6001
      <circle cx="162.58" cy="217.07" r="4.67"                            6002
      fill="#43a84c"/>                                                    6003
      <circle cx="146.02" cy="222.76" r="4.85"                            6004
      fill="#43a852"/>                                                    6005
      <circle cx="128.75" cy="225.64" r="5.10"                            6006
      fill="#43a858"/>                                                    6007
      <circle cx="111.25" cy="225.64" r="5.42"                            6008
      fill="#43a85e"/>                                                    6009
      <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864"/>            6010
      <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a"/>            6011
      <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870"/>            6012
      <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876"/>            6013
      <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c"/>            6014
      <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882"/>            6015
      <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888"/>            6016
      <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e"/>            6017
      <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894"/>            6018
      <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e"/>            6019
      <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888"/>             6020
      <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882"/>             6021
      <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c"/>             6022
      <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876"/>             6023
      <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870"/>             6024
      <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a"/>             6025
      <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864"/>             6026
      <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e"/>            6027
      <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858"/>            6028
      <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852"/>            6029
      <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c"/>            6030
      <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846"/>            6031
      <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843"/>            6032
      <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843"/>            6033
      <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843"/>            6034
      <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843"/>            6035
      <circle cx="224.55" cy="102.55" r="6.23"                            6036
      fill="#5fa843"/>                                                    6037
    </g>                                                                  6038
    <g class="lp-badge__mark">                                            6039
      <circle cx="120" cy="120" r="27" fill="#14432f"/>                   6040
      <g transform="translate(120 120) scale(1.5)                         6041
      translate(-12 -12)" fill="none" stroke="#ffffff"                    6042
      stroke-width="1.9" stroke-linecap="round"                           6043
      stroke-linejoin="round">                                            6044
        <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>                       6045
        <circle cx="17" cy="6" r="2.2"/>                                  6046
      </g>                                                                6047
    </g>                                                                  6048
</svg>                                                                    6049
# ==== app/templates/error.html ====                                      6050
```

```
{% extends "base.html" %}
{% block title %}{{ code }} · {{ message }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>{{ code }} · {{ message }}</h2>
  <p class="muted">
    请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
  </p>
  <p class="actions" style="margin-top:16px;">
    <a class="btn" href="/">返回首页</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 登录页（仿 Apple 账户页） ==================
     纯白底、居中、大量留白；色彩只用在徽标与按钮上。
     首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
     随滚动逐条弹入、往回滚时弹回。 */
  .lp {
    color: #1d1d1f;
    background:
      linear-gradient(180deg, rgba(245, 239, 224, .88) 0%,
                              rgba(245, 239, 224, .92)
    100%),
      url("../static/hero_qinglv.jpg")
      center top / cover no-repeat fixed;
  }
  /* ---------- 首屏 ---------- */
  .lp-hero {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
    position: relative;
    background:
      linear-gradient(180deg, rgba(15, 40, 45, .72) 0%,
                              rgba(20, 55, 58, .50) 50%,
                              rgba(15, 40, 45, .76) 100%),
      url("../static/hero_qinglv.jpg")
      center center / cover no-repeat;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
```

```
    --p 为 0~1 的滚动进度，由页面脚本写入；
    每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
.lp-badge {
  width: 212px;
  height: 212px;
  margin-bottom: 36px;
  --p: 0;
  animation: lp-in .7s ease both;
}
.lp-badge__svg {
  display: block;
  width: 100%;
  height: 100%;
  overflow: visible;                    /* 散开时圆点不被画布裁掉 */
  opacity: calc(1 - var(--p) * .94);
  filter: blur(calc(var(--p) * 2.6px));
}
.lp-ring {
  transform-box: fill-box;
  transform-origin: center;
  transform: scale(calc(1 + (var(--k) - 1) * var(--p)));
}
.lp-badge__mark {
  transform-box: fill-box;
  transform-origin: center;
  transform: scale(calc(1 + .18 * var(--p)));
}
.lp-title {
  margin: 0 0 18px;
  font-size: clamp(28px, 4.2vw, 42px);
  font-weight: 700;
  line-height: 1.3;
  letter-spacing: -.4px;
  color: #fff;
  text-shadow: 0 2px 12px rgba(0, 0, 0, .3);
  animation: lp-in .7s ease .06s both;
}
.lp-cta {
  padding: 13px 52px;
  font-family: inherit;
  font-size: 17px;
  font-weight: 500;
  color: #fff;
  background: var(--accent);
  border: none;
  border-radius: 980px;
  cursor: pointer;
  animation: lp-in .7s ease .18s both;
  transition: background-color .2s ease, box-shadow .2s
  ease,
```

```
                transform .2s ease;
}
.lp-cta:hover {
  background: var(--accent-hover);
  box-shadow: 0 10px 26px rgba(44, 107, 90, .4);
}
.lp-cta:active { transform: translateY(1px); box-shadow:
  none; }
.lp-cta:focus-visible { outline: 3px solid rgba(31, 107,
  69, .40); outline-offset: 3px; }
.lp-more {
  margin-top: 38px;
  font-size: 13px;
  color: rgba(255, 255, 255, .7);
  text-decoration: none;
  animation: lp-in .7s ease .24s both;
}
.lp-more:hover { color: #fff; }
@keyframes lp-in {
  from { opacity: 0; transform: translateY(12px); }
  to   { opacity: 1; transform: none; }
}
/* ---------- 滚动进场 ----------
   进入视口时从屏外飞入，离开视口时飞回，可反复。
   交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */
.lp-reveal {
  opacity: 0;
  transition: opacity .7s ease,
              transform 1.05s cubic-bezier(.18, 1.04, .3,
  1),
              border-color .62s ease, background-color
  .62s ease;
  transition-delay: 0s;
  will-change: opacity, transform;
}
/* 起始位置整体在屏外：横向按视口宽度取位移，纵向按视口高度取位移，
   再带一点缩小，落位时按贝塞尔曲线的过冲轻微回弹。 */
.lp-reveal[data-dir="left"]  { transform:
  translate3d(-56vw, 5vh, 0) scale(.9); }
.lp-reveal[data-dir="right"] { transform:
  translate3d(56vw, 5vh, 0) scale(.9); }
.lp-reveal[data-dir="up"]    { transform: translate3d(0,
  24vh, 0) scale(.94); }
.lp-reveal.is-in {
  opacity: 1;
  transform: none;
  transition-delay: calc(var(--i, 0) * 110ms);
}
/* 标题颜色由淡到深，强化「浮现」的层次感 */
.lp-reveal h2,
```

```
.lp-reveal h3,
.lp-reveal .lp-mod__name { color: #c2d2c8; transition:
  color .7s ease; }
.lp-reveal.is-in h3,
.lp-reveal.is-in .lp-mod__name { color: var(--accent); }
.lp-reveal.is-in h2 { color: #1d1d1f; }
/* ---------- 系统介绍 ---------- */
.lp-intro {
  max-width: none;
  margin: 0;
  padding: 56px 24px 72px;
  background: transparent;
  /* 卡片从屏外飞入，横向超出的部分裁掉，免得出现横向滚动条；
     用 clip 而不是 hidden，纵向滚动仍由页面本身负责 */
  overflow-x: clip;
}
.lp-intro > * { max-width: 1040px; margin-left: auto;
  margin-right: auto; }
.lp-intro__title {
  margin: 0 auto 26px;
  font-size: clamp(22px, 2.6vw, 30px);
  text-align: center;
  letter-spacing: -.3px;
}
/* 三张特点卡与九个模块卡都按三等分排版，并整体居中 */
.lp-points {
  display: grid;
  grid-template-columns: repeat(3, minmax(0, 1fr));
  gap: 16px;
  margin-bottom: 46px;
}
.lp-point {
  padding: 30px 26px;
  background: rgba(253, 250, 242, .82);
  backdrop-filter: blur(12px);
  -webkit-backdrop-filter: blur(12px);
  border: 1px solid rgba(224, 216, 200, .6);
  border-radius: 22px;
  box-shadow: 0 24px 60px rgba(42, 37, 32, .10), 0 8px
  20px rgba(42, 37, 32, .05);
  transition: transform .25s ease, box-shadow .25s ease;
}
.lp-point:hover {
  transform: translateY(-4px);
  box-shadow: 0 32px 72px rgba(42, 37, 32, .14), 0 12px
  28px rgba(42, 37, 32, .07);
}
.lp-point__icon {
  display: grid;
  place-items: center;
```

```
    width: 40px;
    height: 40px;
    margin-bottom: 16px;
    border-radius: 12px;
    background: rgba(44, 107, 90, .1);
    color: var(--accent);
  }
  .lp-point h3 { margin: 0 0 8px; font-size: 16px; }
  .lp-point p {
    margin: 0;
    font-size: 13.5px;
    line-height: 1.85;
    color: #6e6e73;
    text-wrap: pretty;
  }
  .lp-sub-title {
    margin: 0 auto 20px;
    font-size: 17px;
    text-align: center;
  }
  .lp-mods {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: 12px;
    margin-bottom: 46px;
    align-items: start;
  }
  /* 模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明 */
  .lp-mod {
    position: relative;
    padding: 17px 18px 17px 48px;
    background: rgba(253, 250, 242, .75);
    backdrop-filter: blur(8px);
    -webkit-backdrop-filter: blur(8px);
    border: 1px solid rgba(224, 216, 200, .6);
    border-radius: 14px;
    outline: none;
    transition: border-color .3s ease, box-shadow .3s ease;
  }
  .lp-mod:hover,
  .lp-mod:focus-within,
  .lp-mod:focus {
    border-color: rgba(44, 107, 90, .5);
    box-shadow: 0 8px 22px rgba(42, 37, 32, .08);
  }
  .lp-mod__no {
    position: absolute;
    left: 17px;
    top: 17px;
    font-size: 13px;
```

```
    font-weight: 700;                                                    6301
    color: #c8d6cd;                                                      6302
    font-variant-numeric: tabular-nums;                                  6303
  }                                                                      6304
  .lp-mod__name {                                                        6305
    display: block;                                                      6306
    font-size: 14px;                                                     6307
    font-weight: 600;                                                    6308
  }                                                                      6309
  /* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */                  6310
  .lp-mod__body {                                                        6311
    display: grid;                                                       6312
    grid-template-rows: 0fr;                                             6313
    transition: grid-template-rows .34s ease, margin-top                 6314
    .34s ease;                                                           6315
  }                                                                      6316
  .lp-mod:hover .lp-mod__body,                                           6317
  .lp-mod:focus-within .lp-mod__body {                                   6318
    grid-template-rows: 1fr;                                             6319
    margin-top: 7px;                                                     6320
  }                                                                      6321
  .lp-mod__desc {                                                        6322
    overflow: hidden;                                                    6323
    margin: 0;                                                           6324
    font-size: 12.5px;                                                   6325
    line-height: 1.7;                                                    6326
    color: #6e6e73;                                                      6327
    opacity: 0;                                                          6328
    transition: opacity .3s ease;                                        6329
    text-wrap: pretty;                                                   6330
  }                                                                      6331
  .lp-mod:hover .lp-mod__desc,                                           6332
  .lp-mod:focus-within .lp-mod__desc { opacity: 1; }                     6333
  /* 触摸设备没有悬停，直接把说明摊开 */                                  6334
  @media (hover: none) {                                                 6335
    .lp-mod__body { grid-template-rows: 1fr; margin-top:                 6336
    7px; }                                                               6337
    .lp-mod__desc { opacity: 1; }                                        6338
  }                                                                      6339
  .lp-foot {                                                             6340
    margin: 0;                                                           6341
    padding-top: 26px;                                                   6342
    border-top: 1px solid #e6e6e8;                                       6343
    text-align: center;                                                  6344
    font-size: 12.5px;                                                   6345
    color: #86868b;                                                      6346
  }                                                                      6347
  /* ---------- 登录面板：点击后渐变浮现 ---------- */                   6348
  .lp-auth {                                                             6349
    position: fixed;                                                     6350
```

```
    inset: 0;                                                          6351
    z-index: 60;                                                       6352
    display: grid;                                                     6353
    place-items: center;                                               6354
    padding: 24px;                                                     6355
    opacity: 0;                                                        6356
    visibility: hidden;                                                6357
    transition: opacity .42s ease, visibility 0s linear                6358
    .42s;                                                              6359
}                                                                      6360
.lp-auth.is-open {                                                     6361
    opacity: 1;                                                        6362
    visibility: visible;                                               6363
    transition: opacity .42s ease, visibility 0s;                      6364
}                                                                      6365
/* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */                                  6366
.lp-auth__scrim {                                                      6367
    position: absolute;                                                6368
    inset: 0;                                                          6369
    background:                                                        6370
      radial-gradient(1000px 640px at 50% 0%,                          6371
                      rgba(31, 107, 69, .22), rgba(31, 107,            6372
    69, 0) 70%),                                                       6373
      linear-gradient(180deg, rgba(255, 255, 255, .84),                6374
                              rgba(246, 250, 247, .95));               6375
    backdrop-filter: blur(16px) saturate(1.25);                        6376
    -webkit-backdrop-filter: blur(16px) saturate(1.25);                6377
}                                                                      6378
.lp-auth__panel {                                                      6379
    position: relative;                                                6380
    width: 100%;                                                       6381
    max-width: 396px;                                                  6382
    padding: 36px 36px 26px;                                           6383
    background: #fff;                                                  6384
    border-radius: 20px;                                               6385
    box-shadow: 0 28px 80px rgba(14, 46, 32, .18);                     6386
    opacity: 0;                                                        6387
    transform: translateY(18px) scale(.975);                           6388
    transition: transform .55s cubic-bezier(.16, .84, .34,             6389
    1) .05s,                                                           6390
                opacity .55s ease .05s;                                6391
}                                                                      6392
.lp-auth.is-open .lp-auth__panel { opacity: 1; transform:              6393
    none; }                                                            6394
.lp-auth__close {                                                      6395
    position: absolute;                                                6396
    top: 14px;                                                         6397
    right: 14px;                                                       6398
    display: grid;                                                     6399
    place-items: center;                                               6400
```

```
    width: 30px;                                                                    6401
    height: 30px;                                                                   6402
    color: #86868b;                                                                 6403
    background: transparent;                                                        6404
    border: none;                                                                   6405
    border-radius: 50%;                                                             6406
    cursor: pointer;                                                                6407
    transition: background-color .18s ease, color .18s ease;                        6408
}                                                                                   6409
.lp-auth__close:hover { background: #f5f5f7; color:                                 6410
  #1d1d1f; }                                                                        6411
.lp-auth__title {                                                                   6412
  margin: 0 0 6px;                                                                  6413
  text-align: center;                                                               6414
  font-size: 24px;                                                                  6415
  font-weight: 700;                                                                 6416
  letter-spacing: -.2px;                                                            6417
}                                                                                   6418
.lp-auth__sub {                                                                     6419
  margin: 0 0 26px;                                                                 6420
  text-align: center;                                                               6421
  font-size: 13px;                                                                  6422
  color: #86868b;                                                                   6423
}                                                                                   6424
/* 字段校验错误：贴着输入框显示，不做成会消失的浮层 ——                                6425
   这类消息要对着输入框看，一闪而过会让人以为「点了没反应」 */                        6426
.lp-field__error {                                                                  6427
  margin: 7px 0 0;                                                                  6428
  font-size: 12.5px;                                                                6429
  color: #9c2b23;                                                                   6430
}                                                                                   6431
.lp-field input[aria-invalid="true"] {                                              6432
  background: #fdeeec;                                                              6433
  border-color: rgba(156, 43, 35, .35);                                             6434
}                                                                                   6435
.lp-field { margin-bottom: 15px; }                                                  6436
.lp-field label {                                                                   6437
  display: block;                                                                   6438
  margin-bottom: 7px;                                                               6439
  font-size: 13px;                                                                  6440
  color: #6e6e73;                                                                   6441
}                                                                                   6442
.lp-field input {                                                                   6443
  width: 100%;                                                                      6444
  padding: 13px 16px;                                                               6445
  font-family: inherit;                                                             6446
  font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */          6447
  color: #1d1d1f;                                                                   6448
  background: #f5f5f7;                                                              6449
  border: 1px solid transparent;                                                    6450
```

```
    border-radius: 12px;                                                    6451
    outline: none;                                                          6452
    transition: background-color .18s ease, border-color                    6453
    .18s ease,                                                              6454
                box-shadow .18s ease;                                       6455
  }                                                                         6456
  .lp-field input::placeholder { color: #a1a1a6; }                          6457
  .lp-field input:focus {                                                   6458
    background: #fff;                                                       6459
    border-color: rgba(31, 107, 69, .55);                                   6460
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                           6461
  }                                                                         6462
  .lp-remember {                                                            6463
    display: flex;                                                          6464
    align-items: center;                                                    6465
    gap: 8px;                                                               6466
    margin: 4px 0 20px;                                                     6467
    font-size: 13px;                                                        6468
    color: #6e6e73;                                                         6469
    cursor: pointer;                                                        6470
  }                                                                         6471
  .lp-remember input {                                                      6472
    width: 16px;                                                            6473
    height: 16px;                                                           6474
    margin: 0;                                                              6475
    accent-color: var(--accent);                                            6476
    cursor: pointer;                                                        6477
  }                                                                         6478
  .lp-submit {                                                              6479
    width: 100%;                                                            6480
    padding: 13px 20px;                                                     6481
    font-family: inherit;                                                   6482
    font-size: 16px;                                                        6483
    font-weight: 500;                                                       6484
    color: #fff;                                                            6485
    background: var(--accent);                                              6486
    border: none;                                                           6487
    border-radius: 980px;                                                   6488
    cursor: pointer;                                                        6489
    transition: background-color .2s ease, box-shadow .2s                   6490
    ease,                                                                   6491
                transform .2s ease;                                         6492
  }                                                                         6493
  .lp-submit:hover {                                                        6494
    background: var(--accent-hover);                                        6495
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                         6496
  }                                                                         6497
  .lp-submit:active { transform: translateY(1px);                           6498
    box-shadow: none; }                                                     6499
  .lp-submit:focus-visible { outline: 3px solid rgba(31,                    6500
```

```
      107, 69, .40); outline-offset: 3px; }
    .lp-auth__hint {
      margin: 18px 0 0;
      text-align: center;
      font-size: 12.5px;
      color: #a1a1a6;
    }
    .lp-auth__reg {
      color: var(--accent);
      font-weight: 500;
      text-decoration: none;
      border-bottom: 1px solid rgba(31, 107, 69, .35);
      transition: border-color .18s ease;
    }
    .lp-auth__reg:hover { border-bottom-color: var(--accent);
      }
    /* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */
    @media (prefers-reduced-motion: reduce) {
      .lp-badge, .lp-title, .lp-cta, .lp-more { animation:
      none; }
      .lp-badge__svg { opacity: 1; filter: none; }
      .lp-ring, .lp-badge__mark { transform: none; }
      .lp-reveal { opacity: 1; transform: none !important; }
      .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name
      { color: inherit; }
      .lp-reveal.is-in h2 { color: #1d1d1f; }
      .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name {
      color: var(--accent); }
      .lp-auth, .lp-auth__panel { transition: opacity .2s
      linear; }
      .lp-auth__panel { transform: none; }
    }
    /* 窄屏退成两列、一列，避免卡片被挤变形 */
    @media (max-width: 900px) {
      .lp-points, .lp-mods { grid-template-columns: repeat(2,
      minmax(0, 1fr)); }
    }
    @media (max-width: 620px) {
      .lp-points, .lp-mods { grid-template-columns: minmax(0,
      1fr); }
    }
    @media (max-width: 560px) {
      .lp-hero { padding: 36px 18px 48px; min-height: auto; }
      .lp-badge { width: 160px; height: 160px; margin-bottom:
      26px; }
      .lp-auth__panel { padding: 30px 24px 22px; }
    }
</style>
<div class="lp">
    {# ---------------- 首屏：徽标 + 标题 + 说明 + 登录入口
```

```
    ----------------- #}
<section class="lp-hero">
    <div class="lp-badge" id="lp-badge" aria-hidden="true">
      {% include "_login_badge.svg" %}
    </div>
    <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>
    <button type="button" class="lp-cta" id="lp-open"
            aria-expanded="false" aria-controls="lp-auth">登录
    </button>
    <a class="lp-more" href="#intro">
      向下滑动，了解系统 <span aria-hidden="true">↓</span>
    </a>
</section>
{# ----------------- 系统介绍：随滚动逐条弹入 ----------------- #}
<section class="lp-intro" id="intro">
    <h2 class="lp-intro__title lp-reveal"
    data-dir="up">一套把评价流程固定下来的工具</h2>
    <div class="lp-points">
      {% set pdirs = ['left', 'up', 'right'] %}
      {% set points = [
        ('sliders', '方法可溯',
         '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回
    看。'),
        ('shield', '流程可控',
         '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重
    建或人工辅助的结论。'),
        ('file', '结果可复算',
         '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')
      ] %}
      {% for ic, title, text in points %}
      <div class="lp-point lp-reveal" data-dir="{{
    pdirs[loop.index0 % 3] }}"
           style="--i: {{ loop.index0 }}">
        <span class="lp-point__icon">{{ icon(ic, 20)
    }}</span>
        <h3>{{ title }}</h3>
        <p>{{ text }}</p>
      </div>
      {% endfor %}
    </div>
    <h3 class="lp-sub-title lp-reveal"
    data-dir="up">九个功能模块</h3>
    <div class="lp-mods">
      {% set mdirs = ['left', 'up', 'right'] %}
      {% for no, name, desc in modules %}
      <div class="lp-mod lp-reveal" data-dir="{{
    mdirs[loop.index0 % 3] }}"
           style="--i: {{ loop.index0 % 3 }}" tabindex="0">
        <span class="lp-mod__no">{{ no }}</span>
        <span class="lp-mod__name">{{ name }}</span>
```

```
          <div class="lp-mod__body"><p class="lp-mod__desc">{{
    desc }}</p></div>
        </div>
        {% endfor %}
    </div>
    <p class="lp-foot lp-reveal" data-dir="up">
        历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0 ·
        运行于浏览器，数据保存在本机 ＳＱＬｉｔｅ 数据库
    </p>
</section>
{# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}
<div class="lp-auth" id="lp-auth" aria-hidden="true">
    <div class="lp-auth__scrim" data-lp-close></div>
    <div class="lp-auth__panel" role="dialog"
    aria-modal="true"
          aria-labelledby="lp-auth-title">
        <button type="button" class="lp-auth__close"
    data-lp-close
                aria-label="关闭登录框">
          <svg width="16" height="16" viewBox="0 0 24 24"
    fill="none"
              stroke="currentColor" stroke-width="2"
    stroke-linecap="round"
              aria-hidden="true"><path d="M6 6l12 12M18 6L6
    18"/></svg>
        </button>
        <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>
        <p class="lp-auth__sub">使用系统账号登录</p>
        <form method="post" novalidate>
          {{ form.hidden_tag() }}
          <div class="lp-field">
            <label for="{{ form.username.id }}">账户名称</label>
            {{ form.username(id=form.username.id,
    autocomplete="username",
                             placeholder="请输入账户名称",
                             **({'aria-invalid': 'true'} if
    form.username.errors else {})) }}
            {% for e in form.username.errors %}
            <p class="lp-field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <div class="lp-field">
            <label for="{{ form.password.id }}">密码</label>
            {{ form.password(id=form.password.id,
    autocomplete="current-password",
                             placeholder="请输入密码",
                             **({'aria-invalid': 'true'} if
    form.password.errors else {})) }}
            {% for e in form.password.errors %}
            <p class="lp-field__error">{{ e }}</p>
```

```
            {% endfor %}                                                    6651
          </div>                                                            6652
          <label class="lp-remember" for="{{ form.remember.id               6653
    }}">                                                                    6654
            {{ form.remember(id=form.remember.id) }}                        6655
            <span>记住我（下次自动登录，一个月内有效）</span>                  6656
          </label>                                                          6657
          {{ form.submit(class_="lp-submit") }}                             6658
        </form>                                                             6659
        <p class="lp-auth__hint">                                           6660
          还没有账号？<a class="lp-auth__reg"                               6661
                         href="{{ url_for('auth.register')                  6662
    }}">点我注册</a>                                                        6663
        </p>                                                                6664
        <p class="lp-auth__hint" style="margin-top:10px;">                  6665
          演示账号：admin 密码：admin123                                    6666
        </p>                                                                6667
      </div>                                                                6668
    </div>                                                                  6669
</div>                                                                      6670
{% endblock %}                                                              6671
{% block scripts %}                                                         6672
<script>                                                                    6673
(function () {                                                              6674
  var reduce = window.matchMedia &&                                         6675
    window.matchMedia('(prefers-reduced-motion:                             6676
    reduce)').matches;                                                      6677
  /* ---------- 登录面板 ---------- */                                       6678
  var btn = document.getElementById('lp-open');                             6679
  var auth = document.getElementById('lp-auth');                            6680
  var lastFocus = null;                                                     6681
  function openPanel() {                                                    6682
    lastFocus = document.activeElement;                                     6683
    auth.classList.add('is-open');                                          6684
    auth.setAttribute('aria-hidden', 'false');                              6685
    btn.setAttribute('aria-expanded', 'true');                              6686
    document.body.style.overflow = 'hidden';                                6687
    // 记到地址栏，直接访问 /login#login 也能展开登录框                    6688
    if (location.hash !== '#login') {                                       6689
      history.replaceState(null, '', location.pathname +                    6690
    '#login');                                                              6691
    }                                                                       6692
    var first = auth.querySelector('input');                                6693
    if (first) window.setTimeout(function () {                              6694
    first.focus(); }, 300);                                                 6695
  }                                                                         6696
  function closePanel() {                                                   6697
    auth.classList.remove('is-open');                                       6698
    auth.setAttribute('aria-hidden', 'true');                               6699
    btn.setAttribute('aria-expanded', 'false');                             6700
```

```
    document.body.style.overflow = '';
    if (location.hash === '#login') {
      history.replaceState(null, '', location.pathname);
    }
    if (lastFocus && lastFocus.focus) lastFocus.focus();
  }
  if (btn && auth) {
    btn.addEventListener('click', openPanel);
    Array.prototype.forEach.call(
      auth.querySelectorAll('[data-lp-close]'),
      function (el) { el.addEventListener('click',
    closePanel); }
    );
    document.addEventListener('keydown', function (e) {
      if (e.key === 'Escape' && auth.classList.contains('is-
    open')) closePanel();
    });
    // 登录失败回到本页、或直接带 #login 访问时，直接展开登录框。
    // 失败可能落在两处：字段校验错误在表单里，凭据错误在浮层里。
    if (location.hash === '#login'
        || document.querySelector('.lp-field__error')
        || document.querySelector('.toast-center')) {
      openPanel();
    }
  }
  /* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- */
     原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，
     内容始终停在 opacity：0。改为直接按元素与视口的相对位置判断，
     与徽标用同一次滚动回调，行为可预期、也便于排查。
     判断「进没进视野」用的是**布局位置**，不是 getBoundingClientRect()：
     进场动画本身就是位移（左右两列横向 56vw、中列自下方 24vh 升起），
     用视觉位置判断时，中列的盒子一直比自己的位置低一截，
     于是它总要再往下滚一段才触发，看起来比左右两列慢半拍。
     transform 不影响 offsetTop，所以这里沿 offsetParent 把布局位置累加出来，
     只在加载时算一次；滚动过程中不再测量，也就没有读写交替带来的重排。 */
  var items = document.querySelectorAll('.lp-reveal');
  var badge = document.getElementById('lp-badge');
  var ticking = false;
  var slots = [];
  function layoutTop(el) {
    var top = 0;
    var node = el;
    while (node) {
      top += node.offsetTop || 0;
      node = node.offsetParent;
    }
    return top;
  }
  if (!reduce) {
    for (var k = 0; k < items.length; k++) {
```

```
        slots.push({
          el: items[k],
          top: layoutTop(items[k]),
          height: items[k].offsetHeight
        });
      }
    }
    function applyScroll() {
      ticking = false;
      var vh = window.innerHeight || 800;
      var y = window.scrollY || window.pageYOffset || 0;
      /* 徽标：滚动前约三分之一屏内完成散开淡出 */
      if (badge) {
        var range = vh * 0.34;
        var p = range > 0 ? y / range : 0;
        badge.style.setProperty('--p', Math.min(1,
      Math.max(0, p)).toFixed(3));
      }
      if (reduce || !slots.length) return;
      for (var j = 0; j < slots.length; j++) {
        var s = slots[j];
        var viewTop = s.top - y;
        // 上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内
        var shown = viewTop < vh * 0.88 && viewTop + s.height
      > vh * 0.06;
        if (shown) s.el.classList.add('is-in');
        else s.el.classList.remove('is-in');
      }
    }
    function onScroll() {
      if (!ticking) {
        ticking = true;
        window.requestAnimationFrame(applyScroll);
      }
    }
    // 减少动态效果时不做滚动特效，内容直接显示
    if (reduce) {
      Array.prototype.forEach.call(items, function (el) {
        el.classList.add('is-in');
      });
      if (badge) badge.style.setProperty('--p', '0');
    } else {
      window.addEventListener('scroll', onScroll, { passive:
      true });
      window.addEventListener('resize', onScroll);
      applyScroll();
    }
  })();
</script>
{% endblock %}
```

```
# ==== app/templates/auth/register.html ====
{% extends "base.html" %}
{% block title %}注册账号 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 注册页 ==================
     与登录页同一套观感：浅绿渐变底 + 白色卡片。
     必填只有账户名称与密码，其余收在「选填」分隔线以下，
     这样首次注册不会被一堆输入框挡住。 */
  .rg {
    min-height: 100dvh;
    display: grid;
    place-items: center;
    padding: 40px 20px 56px;
    background:
      radial-gradient(900px 520px at 50% 0%,
                      rgba(31, 107, 69, .14), rgba(31, 107,
    69, 0) 72%),
      #f6faf7;
  }
  .rg-card {
    width: 100%;
    max-width: 470px;
    padding: 38px 40px 30px;
    background: #fff;
    border-radius: 20px;
    box-shadow: 0 26px 70px rgba(14, 46, 32, .13);
    animation: rg-in .5s cubic-bezier(.16, .84, .34, 1)
    both;
  }
  @keyframes rg-in {
    from { opacity: 0; transform: translateY(16px)
    scale(.98); }
    to   { opacity: 1; transform: none; }
  }
  .rg-title {
    margin: 0 0 6px;
    text-align: center;
    font-size: 25px;
    font-weight: 700;
    letter-spacing: -.2px;
  }
  .rg-sub {
    margin: 0 0 28px;
    text-align: center;
    font-size: 13px;
    color: #86868b;
  }
  .rg-field { margin-bottom: 15px; }
  .rg-field > label {
```

```
    display: block;                                                    6851
    margin-bottom: 7px;                                                6852
    font-size: 13px;                                                   6853
    color: #6e6e73;                                                    6854
  }                                                                    6855
  .rg-field > label .rg-opt {                                          6856
    margin-left: 5px;                                                  6857
    font-size: 12px;                                                   6858
    color: #a1a1a6;                                                    6859
  }                                                                    6860
  .rg-field input[type="text"],                                        6861
  .rg-field input[type="password"] {                                   6862
    width: 100%;                                                       6863
    padding: 12px 15px;                                                6864
    font-family: inherit;                                              6865
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */     6866
    color: #1d1d1f;                                                    6867
    background: #f5f5f7;                                               6868
    border: 1px solid transparent;                                     6869
    border-radius: 12px;                                               6870
    outline: none;                                                     6871
    transition: background-color .18s ease, border-color               6872
    .18s ease,                                                         6873
                box-shadow .18s ease;                                  6874
  }                                                                    6875
  .rg-field input::placeholder { color: #a1a1a6; }                     6876
  .rg-field input:focus {                                              6877
    background: #fff;                                                  6878
    border-color: rgba(31, 107, 69, .55);                              6879
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                      6880
  }                                                                    6881
  .rg-field input[aria-invalid="true"] {                               6882
    border-color: rgba(180, 60, 50, .55);                              6883
    background: #fff;                                                  6884
  }                                                                    6885
  .rg-field select {                                                   6886
    width: 100%;                                                       6887
    padding: 12px 15px;                                                6888
    font-family: inherit;                                              6889
    font-size: 15px;                                                   6890
    color: #1d1d1f;                                                    6891
    background: #f5f5f7;                                               6892
    border: 1px solid transparent;                                     6893
    border-radius: 12px;                                               6894
    outline: none;                                                     6895
  }                                                                    6896
  .rg-field select:focus {                                             6897
    background: #fff;                                                  6898
    border-color: rgba(31, 107, 69, .55);                              6899
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                      6900
```

```
}
.rg-err {
  margin: 6px 0 0;
  font-size: 12.5px;
  color: var(--danger);
}
/* 「选填」分隔线 */
.rg-split {
  display: flex;
  align-items: center;
  gap: 12px;
  margin: 24px 0 18px;
  font-size: 12.5px;
  color: #a1a1a6;
}
.rg-split::before,
.rg-split::after {
  content: "";
  flex: 1;
  height: 1px;
  background: #ececee;
}
.rg-pair {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 0 14px;
}
@media (max-width: 480px) { .rg-pair {
  grid-template-columns: 1fr; } }
/* 头像：圆形预览 + 选择文件 */
.rg-avatar {
  display: flex;
  align-items: center;
  gap: 14px;
  margin-bottom: 15px;
}
.rg-avatar__face {
  display: grid;
  place-items: center;
  flex: none;
  width: 54px;
  height: 54px;
  border-radius: 50%;
  background: #eef2f0;
  color: #9aa8a1;
  font-size: 12px;
  overflow: hidden;
  background-size: cover;
  background-position: center;
}
```

```
  .rg-avatar__pick { min-width: 0; }
  .rg-avatar__pick input[type="file"] {
    max-width: 100%;
    font-family: inherit;
    font-size: 12.5px;
    color: #6e6e73;
  }
  .rg-avatar__tip {
    margin: 4px 0 0;
    font-size: 12px;
    color: #a1a1a6;
  }
  .rg-submit {
    width: 100%;
    margin-top: 24px;
    padding: 13px 20px;
    font-family: inherit;
    font-size: 16px;
    font-weight: 500;
    color: #fff;
    background: var(--accent);
    border: none;
    border-radius: 980px;
    cursor: pointer;
    transition: background-color .2s ease, box-shadow .2s
    ease,
                transform .2s ease;
  }
  .rg-submit:hover {
    background: var(--accent-hover);
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);
  }
  .rg-submit:active { transform: translateY(1px);
    box-shadow: none; }
  .rg-submit:focus-visible { outline: 3px solid rgba(31,
    107, 69, .40); outline-offset: 3px; }
  .rg-foot {
    margin: 20px 0 0;
    text-align: center;
    font-size: 13px;
    color: #86868b;
  }
  .rg-foot a { color: var(--accent); text-decoration: none;
    }
  .rg-foot a:hover { text-decoration: underline; }
 @media (prefers-reduced-motion: reduce) {
    .rg-card { animation: none; }
  }
</style>
<div class="rg">
```

```
<div class="rg-card">
  <h1 class="rg-title">创建账号</h1>
  <p class="rg-sub">新账号为「操作员」，职务由管理员调整</p>
  <form method="post" enctype="multipart/form-data"
  novalidate>
    {{ form.hidden_tag() }}
    <div class="rg-field">
      <label for="{{ form.username.id }}">账户名称</label>
      {{ form.username(id=form.username.id,
  autocomplete="username",
                       placeholder="2–64 个字符",
                       **({'aria-invalid': 'true'} if
  form.username.errors else {})) }}
      {% for e in form.username.errors %}<p
  class="rg-err">{{ e }}</p>{% endfor %}
    </div>
    <div class="rg-field">
      <label for="{{ form.password.id }}">密码</label>
      {{ form.password(id=form.password.id,
  autocomplete="new-password",
                       placeholder="至少 6 位",
                       **({'aria-invalid': 'true'} if
  form.password.errors else {})) }}
      {% for e in form.password.errors %}<p
  class="rg-err">{{ e }}</p>{% endfor %}
    </div>
    <div class="rg-field">
      <label for="{{ form.confirm.id }}">确认密码</label>
      {{ form.confirm(id=form.confirm.id,
  autocomplete="new-password",
                      placeholder="再次输入密码",
                       **({'aria-invalid': 'true'} if
  form.confirm.errors else {})) }}
      {% for e in form.confirm.errors %}<p
  class="rg-err">{{ e }}</p>{% endfor %}
    </div>
    <p class="rg-split">以下均为选填</p>
    <div class="rg-avatar">
      <span class="rg-avatar__face" id="rg-face"
  aria-hidden="true">头像</span>
      <div class="rg-avatar__pick">
        <label class="visually-hidden"
  for="rg-avatar">选择头像图片</label>
        <input id="rg-avatar" type="file" name="avatar"
                 accept=".png,.jpg,.jpeg,.gif,.webp">
        <p class="rg-avatar__tip">PNG / JPG / GIF /
  WebP，不超过 4 MB</p>
      </div>
    </div>
    <div class="rg-pair">
```

```
          <div class="rg-field">                                                    7051
            <label for="{{ form.display_name.id }}">昵称</label>                   7052
            {{ form.display_name(id=form.display_name.id,                          7053
                                 placeholder="留空则显示账户名称") }}              7054
          </div>                                                                   7055
          <div class="rg-field">                                                   7056
            <label for="{{ form.gender.id }}">性别</label>                         7057
            {{ form.gender(id=form.gender.id) }}                                   7058
          </div>                                                                   7059
        </div>                                                                     7060
        <div class="rg-field">                                                     7061
          <label for="{{ form.email.id }}">邮箱</label>                            7062
          {{ form.email(id=form.email.id,                                          7063
    autocomplete="email",                                                          7064
                        placeholder="用于联系，可留空",                            7065
                        **({'aria-invalid': 'true'} if                             7066
    form.email.errors else {})) }}                                                 7067
          {% for e in form.email.errors %}<p                                       7068
    class="rg-err">{{ e }}</p>{% endfor %}                                         7069
        </div>                                                                     7070
        <div class="rg-field" style="margin-bottom:0;">                            7071
          <label for="{{ form.phone.id }}">手机号</label>                          7072
          {{ form.phone(id=form.phone.id, autocomplete="tel",                      7073
    placeholder="可留空") }}                                                       7074
        </div>                                                                     7075
        {{ form.submit_reg(class_="rg-submit") }}                                  7076
    </form>                                                                        7077
    <p class="rg-foot">已有账号？<a href="{{                                       7078
    url_for('auth.login') }}">返回登录</a></p>                                     7079
  </div>                                                                           7080
</div>                                                                             7081
{% endblock %}                                                                     7082
{% block scripts %}                                                                7083
<script>                                                                           7084
// 选好头像后立刻在左侧圆框里预览，省得传上去才知道选错了                          7085
(function () {                                                                     7086
  var input = document.getElementById('rg-avatar');                                7087
  var face = document.getElementById('rg-face');                                   7088
  if (!input || !face || !window.FileReader) return;                               7089
  input.addEventListener('change', function () {                                   7090
    var file = input.files && input.files[0];                                      7091
    if (!file) return;                                                             7092
    var reader = new FileReader();                                                 7093
    reader.onload = function (e) {                                                 7094
      face.textContent = '';                                                       7095
      face.style.backgroundImage = 'url("' +                                       7096
    e.target.result + '")';                                                        7097
    };                                                                             7098
    reader.readAsDataURL(file);                                                    7099
  });                                                                              7100
```

```
})();
</script>
{% endblock %}
# ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
{# ---------- 页首横幅 ---------- #}
<section class="hero">
  <div class="hero__body">
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>
    <p class="hero__desc">
      以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》
      建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，
      系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。
    </p>
    <div class="hero__actions">
      <a class="hero-btn hero-btn--ghost"
    href="/indicator/import">
        {{ icon("upload", 16) }}批量导入
      </a>
      <a class="hero-btn hero-btn--solid"
    href="/project/add">
        {{ icon("plus", 16) }}新建评价项目
      </a>
    </div>
  </div>
</section>
{# ---------- 数据概览 ---------- #}
<section class="stat-cards">
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">评价项目</span>
    <span class="stat-card__value">{{ dash.stats.projects
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {%- if projects_without_patch -%}
        其中 {{ projects_without_patch|length }} 个项目尚未建立图斑
      {%- elif dash.progress.patches -%}
        已建立 {{ dash.progress.patches }} 个矿山图斑
      {%- else -%}
        尚未建立矿山图斑
      {%- endif -%}
    </span>
  </a>
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">矿山图斑</span>
    <span class="stat-card__value">{{ dash.stats.patches
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
```

```
        {% if dash.progress.ready %}其中 {{ dash.progress.ready
    }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}
      </span>
  </a>
  <a class="stat-card" href="/unit/">
    <span class="stat-card__label">微地质单元</span>
    <span class="stat-card__value">{{ dash.stats.unit_types
    }}<span class="stat-card__unit">类</span></span>
    <span class="stat-card__note">
      {% if dash.progress.units %}共划分 {{
    dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}
    </span>
  </a>
  <a class="stat-card" href="/standard/">
    <span class="stat-card__label">标准与政策</span>
    <span class="stat-card__value">{{ dash.stats.standards
    }}<span class="stat-card__unit">项</span></span>
    <span class="stat-card__note">地方标准与政策文件</span>
  </a>
</section>
{# ---------- 评价流程 ---------- #}
<section class="section">
  <div class="section__head">
    <h2 class="section__title">评价流程</h2>
    <p class="section__desc">
      按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算
    </p>
  </div>
  <div class="step-grid">
    {% for s in dash.steps %}
    {% if s.available %}
    <a class="step-card" href="{{ s.href }}">
      <span class="step-card__no">{{ s.no }}</span>
      <span class="step-card__icon">{{ icon(s.icon, 18)
    }}</span>
      <span class="step-card__title">{{ s.title }}</span>
      <span class="step-card__desc">{{ s.desc }}</span>
      {% if s.pending %}
      <span class="step-card__badge">待处理 {{ s.pending
    }}</span>
      {% endif %}
    </a>
    {% else %}
    <div class="step-card step-card--locked"
    aria-disabled="true">
      <span class="step-card__no">{{ s.no }}</span>
      <span class="step-card__icon">{{ icon(s.icon, 18)
    }}</span>
      <span class="step-card__title">{{ s.title }}</span>
      <span class="step-card__desc">{{ s.desc }}</span>
```

```
        <span class="step-card__lock">{{ s.note }}</span>
      </div>
      {% endif %}
      {% endfor %}
    </div>
</section>
{# ---------- 系统管理 ---------- #}
{# 个人设置、账号与操作日志的入口都在左侧导航栏，工作台不再重复列出 #}
{% endblock %}
# ==== app/templates/auth/add_user.html ====
{% extends "base.html" %}
{% block title %}新增账号{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>新增账号</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="username">账户名称</label>
    </div>
    <input id="username" type="text" name="username"
    required
           maxlength="64" autocomplete="off"
    style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="password">密码</label>
      <input id="password" type="password" name="password"
    required
           autocomplete="new-password"
    style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="role">职务</label>
      <select id="role" name="role">
        {% for code, label in role_labels.items() %}
        <option value="{{ code }}" {% if code ==
    default_role %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
      <p class="muted" style="margin:6px 0 0;">
        管理员可管理系统、维护基础数据并查看操作日志；操作员可使用全部业务功能。
      </p>
    </div>
    <hr style="margin:22px 0;border:none;border-top:1px
    solid var(--line);">
    <h3 style="margin:0 0 12px;font-size:15px;">个人资料（选填）</h3
    >
    <div class="field">
      <label class="field__label" for="display_name">昵称</lab
```

```
    el>                                                                    7251
      <input id="display_name" type="text"                                 7252
    name="display_name" maxlength="64"                                     7253
           placeholder="留空则显示账户名称">                                 7254
    </div>                                                                 7255
    <div class="field">                                                    7256
      <label class="field__label" for="gender">性别</label>                 7257
      <select id="gender" name="gender">                                   7258
        <option value="">不填</option>                                     7259
        {% for g in genders %}<option value="{{ g }}">{{ g                 7260
    }}</option>{% endfor %}                                                7261
      </select>                                                            7262
    </div>                                                                 7263
    <div class="field">                                                    7264
      <label class="field__label" for="email">邮箱</label>                  7265
      <input id="email" type="text" name="email"                           7266
    maxlength="128">                                                       7267
    </div>                                                                 7268
    <div class="field">                                                    7269
      <label class="field__label" for="phone">手机号</label>                7270
      <input id="phone" type="text" name="phone"                           7271
    maxlength="32">                                                        7272
    </div>                                                                 7273
    <p class="actions" style="margin-top:18px;">                           7274
      <button type="submit" class="btn">保存</button>                      7275
      <a class="btn btn--ghost" href="{{                                   7276
    url_for('auth.profile', tab='users') }}">返回</a>                      7277
    </p>                                                                   7278
  </form>                                                                  7279
</div>                                                                     7280
{% endblock %}                                                             7281
# ==== app/templates/auth/edit_user.html ====                              7282
{% extends "base.html" %}                                                  7283
{% block title %}编辑账号 · {{ user.username }}{% endblock %}               7284
{% block content %}                                                        7285
<div class="card" style="max-width:560px;">                                7286
  <h2>编辑账号 · {{ user.username }}</h2>                                   7287
  <h3 style="margin:0 0 12px;font-size:15px;">个人资料</h3>                 7288
  <p class="muted" style="margin:0 0 14px;font-size:12.5px;"7289
    >                                                                      7290
    以下内容用户也可在「个人设置」中自行修改，此处为管理员代改。                 7291
  </p>                                                                     7292
  <form method="post">                                                     7293
    {{ form.hidden_tag() }}                                                7294
    <div class="field">                                                    7295
      <label class="field__label" for="display_name">昵称</lab7296
    el>                                                                    7297
      <input id="display_name" type="text"                                 7298
    name="display_name" maxlength="64"                                     7299
           value="{{ user.display_name or '' }}"                           7300
```

```
            placeholder="留空则显示账户名称">
</div>
<div class="field">
  <label class="field__label" for="gender">性别</label>
  <select id="gender" name="gender">
    <option value="">不填</option>
    {% for g in genders %}
    <option value="{{ g }}" {% if user.gender == g
%}selected{% endif %}>{{ g }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="email">邮箱</label>
  <input id="email" type="text" name="email"
maxlength="128"
            value="{{ user.email or '' }}">
</div>
<div class="field">
  <label class="field__label" for="phone">手机号</label>
  <input id="phone" type="text" name="phone"
maxlength="32"
            value="{{ user.phone or '' }}">
</div>
<hr style="margin:22px 0;border:none;border-top:1px
solid var(--line);">
<h3 style="margin:0 0 12px;font-size:15px;">职务与权限</h3>
<p class="muted" style="margin:0 0
14px;font-size:12.5px;">
  职务只能由管理员设置：管理员可管理系统、维护基础数据并查看操作日志；
  操作员可使用全部业务功能。
</p>
<div class="field">
  <label class="field__label" for="role">职务</label>
  <select id="role" name="role">
    {% for code, label in role_labels.items() %}
    <option value="{{ code }}" {% if user.role == code
%}selected{% endif %}>{{ label }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="checkbox" for="active">
    <input id="active" type="checkbox" name="active"
value="1"
           {% if user.active %}checked{% endif %}>
    <span>启用该账号</span>
  </label>
  <p class="muted" style="margin:6px 0 0;">
    停用后该账号无法登录，但其历史操作记录会保留。
```

```
            系统至少需保留一名启用状态的管理员。
        </p>
      </div>
      <div class="field">
        <label class="field__label" for="password">重置密码（留空则不修改
      ）</label>
        <input id="password" type="password" name="password"
               autocomplete="new-password"
      style="width:280px;"
               placeholder="留空表示保持原密码">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="{{
      url_for('auth.profile', tab='users') }}">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/auth/profile.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 个人设置（仿 Apple 账户页） ==================
     左栏是身份与分类，右栏是该分类下的功能卡。
     每张卡默认只显示当前值，点开才展开对应的修改表单 ——
     不点就不会误改，页面也不会一上来堆满输入框。 */
  .acct {
    display: grid;
    grid-template-columns: 244px 1fr;
    gap: 40px;
    align-items: start;
  }
  @media (max-width: 900px) {
    .acct { grid-template-columns: 1fr; gap: 24px; }
  }
  /* ---------- 左栏 ---------- */
  .acct-rail__avatar {
    display: grid;
    place-items: center;
    width: 88px;
    height: 88px;
    border-radius: 50%;
    background: #e9edeb;
    color: #97a09b;
    font-size: 32px;
    font-weight: 500;
    object-fit: cover;
```

```
}
.acct-rail__name {
  margin: 16px 0 2px;
  font-size: 16px;
  font-weight: 600;
}
.acct-rail__sub {
  margin: 0 0 24px;
  font-size: 13px;
  color: #86868b;
  word-break: break-all;
}
.acct-tabs {
  display: flex;
  flex-direction: column;
  gap: 2px;
  margin-bottom: 24px;
}
.acct-tabs a {
  padding: 7px 12px;
  border: 1px solid transparent;
  border-radius: 7px;
  font-size: 15px;
  color: #1d1d1f;
  text-decoration: none;
  transition: background-color .16s ease, border-color
  .16s ease;
}
.acct-tabs a:hover { background: #f5f5f7; }
.acct-tabs a.is-active {
  font-weight: 600;
  border-color: #bcc9c1;
  background: #fff;
}
/* 更换头像：折叠在左栏，点开才出现文件选择 */
.acct-avatar-edit > summary {
  display: inline-block;
  padding: 5px 12px;
  font-size: 13px;
  color: var(--accent);
  background: var(--accent-soft);
  border-radius: 7px;
  cursor: pointer;
  list-style: none;
}
.acct-avatar-edit > summary::-webkit-details-marker {
  display: none; }
.acct-avatar-edit > summary:hover { background: #dcebe2; }
.acct-avatar-edit__body { margin-top: 12px; }
.acct-avatar-edit__body .btn { width: 100%;
```

```
    margin-bottom: 8px; }
  .acct-avatar-edit__file {
    margin: 0 0 10px;
    font-size: 12px;
    color: var(--ink-soft);
    word-break: break-all;
  }
  /* ---------- 右栏 ---------- */
  .acct-title {
    margin: 0 0 8px;
    font-size: 26px;
    font-weight: 700;
    letter-spacing: -.3px;
  }
  .acct-lead {
    margin: 0 0 26px;
    font-size: 14px;
    color: #6e6e73;
  }
  /* 固定两列，与 Apple 账户页的 2×2 一致；
     卡片也宽一些，邮箱之类的长值不至于被拆成两行 */
  .acct-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 16px;
    align-items: start;
  }
  @media (max-width: 760px) {
    .acct-grid { grid-template-columns: 1fr; }
  }
  /* 功能卡：details/summary 实现，点开才展开表单 */
  .acct-card {
    background: #fff;
    border: 1px solid #e6e6e8;
    border-radius: 14px;
    transition: border-color .2s ease, box-shadow .2s ease;
  }
  .acct-card[open] {
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
  }
  .acct-card > summary {
    position: relative;
    display: block;
    padding: 18px 56px 18px 20px;
    border-radius: 14px;
    cursor: pointer;
    list-style: none;
  }
  .acct-card > summary::-webkit-details-marker { display:
```

```
    none; }
.acct-card > summary:hover { background: #fafbfa; }
.acct-card > summary:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 2px;
}
.acct-card__title {
  display: block;
  font-size: 15px;
  font-weight: 600;
}
.acct-card__value {
  display: block;
  margin-top: 3px;
  font-size: 14px;
  color: #6e6e73;
  overflow-wrap: anywhere;
}
.acct-card__icon {
  position: absolute;
  top: 17px;
  right: 17px;
  display: grid;
  place-items: center;
  width: 27px;
  height: 27px;
  border-radius: 8px;
  background: var(--accent-soft);
  color: var(--accent);
}
.acct-card__body {
  padding: 4px 20px 20px;
  border-top: 1px solid #f0f1f2;
  margin-top: 2px;
}
.acct-card__body .field:first-child { margin-top: 16px; }
.acct-card__body .btn { width: 100%; }
/* 只读信息列表 */
.acct-kv {
  display: grid;
  grid-template-columns: 88px 1fr;
  gap: 7px 12px;
  margin: 16px 0 0;
  font-size: 13.5px;
}
.acct-kv dt { color: var(--ink-soft); }
.acct-kv dd { margin: 0; }
.field__error {
  margin: 5px 0 0;
  font-size: 12.5px;
```

```
      color: var(--danger);
    }
    /* 管理员分类（账号 / 操作日志）里的表格面板 */
    .acct-panel {
      background: #fff;
      border: 1px solid #e6e6e8;
      border-radius: 14px;
      overflow: hidden;
    }
    .acct-panel__head {
      display: flex;
      align-items: center;
      justify-content: space-between;
      gap: 12px;
      padding: 13px 20px;
      border-bottom: 1px solid #f0f1f2;
    }
    .acct-panel__count {
      margin: 0;
      font-size: 13.5px;
      color: #6e6e73;
    }
    .acct-panel table { margin: 0; }
    .acct-panel th,
    .acct-panel td { border: 0; border-bottom: 1px solid
      #f0f1f2; }
    .acct-panel thead th { background: #f7f9f8; }
    .acct-panel tbody tr:last-child td { border-bottom: 0; }
    .acct-panel tbody tr:hover { background: #fafbfa; }
</style>
<div class="acct">
    {# ---------------- 左栏：身份 + 分类 ---------------- #}
    <aside class="acct-rail">
      {% if current_user.avatar %}
      <img class="acct-rail__avatar"
           src="{{ url_for('static', filename=current_user.ava
      tar) }}"
           alt="{{ current_user.shown_name }} 的头像">
      {% else %}
      <span class="acct-rail__avatar" aria-hidden="true">{{
      current_user.initial }}</span>
      {% endif %}
      <p class="acct-rail__name">{{ current_user.shown_name
      }}</p>
      <p class="acct-rail__sub">
        {{ role_labels.get(current_user.role,
      current_user.role) }}
        {% if current_user.email %} · {{ current_user.email
      }}{% endif %}
      </p>
```

```
    <nav class="acct-tabs" aria-label="个人设置分类">                    7601
      <a href="{{ url_for('auth.profile', tab='info') }}"                7602
         {% if tab == 'info' %}class="is-active"                         7603
    aria-current="page"{% endif %}>个人信息</a>                          7604
      <a href="{{ url_for('auth.profile', tab='security')                7605
    }}"                                                                  7606
         {% if tab == 'security' %}class="is-active"                     7607
    aria-current="page"{% endif %}>登录与安全性</a>                      7608
      {% if current_user.is_admin %}                                     7609
      <a href="{{ url_for('auth.profile', tab='users') }}"               7610
         {% if tab == 'users' %}class="is-active"                        7611
    aria-current="page"{% endif %}>账号</a>                              7612
      <a href="{{ url_for('auth.profile', tab='logs') }}"                7613
         {% if tab == 'logs' %}class="is-active"                         7614
    aria-current="page"{% endif %}>操作日志</a>                          7615
      {% endif %}                                                        7616
    </nav>                                                               7617
    <details class="acct-avatar-edit">                                   7618
      <summary>更换头像</summary>                                        7619
      <div class="acct-avatar-edit__body">                               7620
        <form method="post" action="/profile/avatar"                     7621
              enctype="multipart/form-data">                             7622
          {{ avatar_form.hidden_tag() }}                                 7623
          <input type="hidden" name="tab" value="{{ tab }}">             7624
          <label class="visually-hidden"                                 7625
    for="avatar-file">选择头像图片</label>                               7626
          <input id="avatar-file" type="file" name="avatar"              7627
                 accept=".png,.jpg,.jpeg,.gif,.webp">                    7628
          <p class="acct-avatar-edit__file">PNG / JPG / GIF              7629
    / WebP，不超过 4 MB</p>                                              7630
          <button type="submit" class="btn                               7631
    btn--sm">上传头像</button>                                           7632
        </form>                                                          7633
        {% if current_user.avatar %}                                     7634
        <form method="post" action="/profile/avatar/delete"              7635
              onsubmit="return confirm('确定移除当前头像吗？');">        7636
          {{ avatar_form.hidden_tag() }}                                 7637
          <input type="hidden" name="tab" value="{{ tab }}">             7638
          <button type="submit" class="btn btn--sm                       7639
    btn--ghost">移除头像</button>                                        7640
        </form>                                                          7641
        {% endif %}                                                      7642
      </div>                                                             7643
    </details>                                                           7644
  </aside>                                                               7645
  {# ---------------- 右栏：功能卡 ---------------- #}                   7646
  <div class="acct-main">                                                7647
    {% if tab == 'security' %}                                           7648
    <h1 class="acct-title">登录与安全性</h1>                             7649
    <p class="acct-lead">管理与登录账号、账号安全有关的设置。</p>       7650
```

```
<div class="acct-grid">
  <details class="acct-card"
           {% if pwd_form.old_password.errors or
pwd_form.new_password.errors or pwd_form.confirm.errors
%}open{% endif %}>
    <summary>
      <span class="acct-card__title">密码</span>
      <span class="acct-card__value">修改登录密码</span>
      <span class="acct-card__icon">{{ icon("lock", 15)
}}</span>
    </summary>
    <div class="acct-card__body">
      <form method="post">
        {{ pwd_form.hidden_tag() }}
        <input type="hidden" name="action"
value="password">
        <input type="hidden" name="tab"
value="security">
        <div class="field">
          <label class="field__label" for="{{
pwd_form.old_password.id }}">当前密码</label>
          {{ pwd_form.old_password(id=pwd_form.old_passw
ord.id,

autocomplete="current-password") }}
          {% for e in pwd_form.old_password.errors %}
          <p class="field__error">{{ e }}</p>
          {% endfor %}
        </div>
        <div class="field">
          <label class="field__label" for="{{
pwd_form.new_password.id }}">新密码</label>
          {{ pwd_form.new_password(id=pwd_form.new_passw
ord.id,

autocomplete="new-password") }}
          {% for e in pwd_form.new_password.errors %}
          <p class="field__error">{{ e }}</p>
          {% endfor %}
        </div>
        <div class="field">
          <label class="field__label" for="{{
pwd_form.confirm.id }}">确认新密码</label>
          {{ pwd_form.confirm(id=pwd_form.confirm.id,
                              autocomplete="new-password
") }}
          {% for e in pwd_form.confirm.errors %}
          <p class="field__error">{{ e }}</p>
          {% endfor %}
        </div>
```

```
          <p class="actions">{{ pwd_form.submit_pwd(class_
="btn") }}</p>
        </form>
      </div>
    </details>
    <details class="acct-card">
      <summary>
        <span class="acct-card__title">登录账号</span>
        <span class="acct-card__value">{{
current_user.username }}</span>
        <span class="acct-card__icon">{{ icon("user", 15)
}}</span>
      </summary>
      <div class="acct-card__body">
        <dl class="acct-kv">
          <dt>登录账号</dt><dd class="mono">{{
current_user.username }}</dd>
          <dt>职务</dt>
          <dd>{{ role_labels.get(current_user.role,
current_user.role) }}</dd>
          <dt>状态</dt>
          <dd>
            {% if current_user.active %}
            <span class="tag tag--ok">启用</span>
            {% else %}
            <span class="tag tag--warn">停用</span>
            {% endif %}
          </dd>
          <dt>创建时间</dt>
          <dd class="mono">{{ current_user.created_at |
localdt }}</dd>
        </dl>
      </div>
    </details>
</div>
{% elif tab == 'users' %}
<h1 class="acct-title">账号</h1>
<p class="acct-lead">管理本系统的账号、职务与启用状态。新注册的账号默认为「操作员」。</p
>
{% include "auth/_users.html" %}
{% elif tab == 'logs' %}
<h1 class="acct-title">操作日志</h1>
<p class="acct-lead">记录各账号在本系统中的关键操作。</p>
{% include "auth/_logs.html" %}
{% else %}
<h1 class="acct-title">个人信息</h1>
<p class="acct-lead">管理你的昵称、性别以及可以联系到你的邮箱与手机号。</p>
<form method="post">
  {{ info_form.hidden_tag() }}
  <input type="hidden" name="action" value="info">
```

```
    <input type="hidden" name="tab" value="info">
    <div class="acct-grid">
      <details class="acct-card" {% if
info_form.display_name.errors %}open{% endif %}>
        <summary>
          <span class="acct-card__title">昵称</span>
          <span class="acct-card__value">
            {{ current_user.display_name or "未设置" }}
          </span>
          <span class="acct-card__icon">{{ icon("user",
15) }}</span>
        </summary>
        <div class="acct-card__body">
          <div class="field">
            <label class="field__label" for="{{
info_form.display_name.id }}">
              昵称
            </label>
            {{ info_form.display_name(id=info_form.display
_name.id,

placeholder="留空则显示账户名称") }}
            {% for e in info_form.display_name.errors %}
            <p class="field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <p class="actions">{{ info_form.submit_info(clas
s_="btn") }}</p>
        </div>
      </details>
      <details class="acct-card" {% if
info_form.gender.errors %}open{% endif %}>
        <summary>
          <span class="acct-card__title">性别</span>
          <span class="acct-card__value">{{
current_user.gender or "未设置" }}</span>
          <span class="acct-card__icon">{{ icon("user",
15) }}</span>
        </summary>
        <div class="acct-card__body">
          <div class="field">
            <label class="field__label" for="{{
info_form.gender.id }}">性别</label>
            {{ info_form.gender(id=info_form.gender.id) }}
            {% for e in info_form.gender.errors %}
            <p class="field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <p class="actions">{{ info_form.submit_info(clas
s_="btn") }}</p>
```

```
        </div>                                                    7801
      </details>                                                  7802
      <details class="acct-card" {% if                            7803
info_form.email.errors %}open{% endif %}>                         7804
        <summary>                                                 7805
          <span class="acct-card__title">邮箱</span>              7806
          <span class="acct-card__value">{{                       7807
current_user.email or "未设置" }}</span>                          7808
          <span class="acct-card__icon">{{ icon("mail",           7809
15) }}</span>                                                     7810
        </summary>                                                7811
        <div class="acct-card__body">                             7812
          <div class="field">                                     7813
            <label class="field__label" for="{{                   7814
info_form.email.id }}">邮箱</label>                               7815
              {{ info_form.email(id=info_form.email.id) }}        7816
              {% for e in info_form.email.errors %}               7817
              <p class="field__error">{{ e }}</p>                 7818
              {% endfor %}                                        7819
          </div>                                                  7820
          <p class="actions">{{ info_form.submit_info(clas        7821
s_="btn") }}</p>                                                  7822
        </div>                                                    7823
      </details>                                                  7824
      <details class="acct-card" {% if                            7825
info_form.phone.errors %}open{% endif %}>                         7826
        <summary>                                                 7827
          <span class="acct-card__title">手机号</span>            7828
          <span class="acct-card__value">{{                       7829
current_user.phone or "未设置" }}</span>                          7830
          <span class="acct-card__icon">{{ icon("phone",          7831
15) }}</span>                                                     7832
        </summary>                                                7833
        <div class="acct-card__body">                             7834
          <div class="field">                                     7835
            <label class="field__label" for="{{                   7836
info_form.phone.id }}">手机号</label>                             7837
              {{ info_form.phone(id=info_form.phone.id) }}        7838
              {% for e in info_form.phone.errors %}               7839
              <p class="field__error">{{ e }}</p>                 7840
              {% endfor %}                                        7841
          </div>                                                  7842
          <p class="actions">{{ info_form.submit_info(clas        7843
s_="btn") }}</p>                                                  7844
        </div>                                                    7845
      </details>                                                  7846
    </div>                                                        7847
  </form>                                                         7848
  {% endif %}                                                     7849
</div>                                                            7850
```

```
</div>
{% endblock %}
{% block scripts %}
<script>
// 同时只展开一张卡，避免点开多张后页面被撑得很长
(function () {
  var cards = document.querySelectorAll('.acct-card');
  Array.prototype.forEach.call(cards, function (card) {
    card.addEventListener('toggle', function () {
      if (!card.open) return;
      Array.prototype.forEach.call(cards, function (other) {
        if (other !== card && other.open) other.open =
    false;
      });
    });
  });
})();
</script>
{% endblock %}
# ==== app/templates/auth/_users.html ====
{# 管理员的「账号」分类。由 auth/profile.html 内联引入，不单独成页。 #}
<div class="acct-panel">
  <div class="acct-panel__head">
    <p class="acct-panel__count">共 {{ users|length }}
    个账号</p>
    <a class="btn btn--sm" href="{{ url_for('auth.add_user')
    }}">新增账号</a>
  </div>
  <table>
    <thead>
      <tr>
        <th scope="col">ID</th>
        <th scope="col">账户名称</th>
        <th scope="col">昵称</th>
        <th scope="col">职务</th>
        <th scope="col">状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in users %}
      <tr>
        <td class="num">{{ u.id }}</td>
        <td class="mono">
          {{ u.username }}
          {% if u.id == current_user.id %}<span
    class="muted">（当前登录）</span>{% endif %}
        </td>
        <td style="text-align:left;">
          {% if u.display_name %}{{ u.display_name }}
```

```
            {% else %}<span class="muted">—</span>{% endif %}
          </td>
          <td>
            <span class="tag {{ 'tag--ok' if u.role ==
    'admin' else 'tag--muted' }}">
              {{ role_labels.get(u.role, u.role) }}
            </span>
          </td>
          <td>
            {% if u.active %}<span class="tag
    tag--ok">启用</span>
            {% else %}<span class="tag tag--muted">停用</span>{%
    endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="{{
    url_for('auth.edit_user', user_id=u.id) }}">编辑</a>
              <form method="post"
                    action="{{ url_for('auth.delete_user',
    user_id=u.id) }}"
                    onsubmit="return confirm('确定删除账号「{{
    u.username }}」吗？\n该操作无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
    link-btn--danger"
                        {% if u.id == current_user.id
    %}disabled
                        title="不能删除当前登录的账号"{% endif
    %}>删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
</div>
#  ==== app/templates/auth/_logs.html ====
{#  管理员的「操作日志」分类。由 auth/profile.html 内联引入，不单独成页。 #}
<div class="acct-panel">
  <div class="acct-panel__head">
    <p class="acct-panel__count">
      共 {{ total }} 条记录，当前第 {{ page }} / {{ pages }} 页
    </p>
    {% if pages > 1 %}
    <nav class="row-actions" aria-label="日志分页">
      {% if page > 1 %}
      <a class="link-btn"
         href="{{ url_for('auth.profile', tab='logs',
```

```
      page=page - 1) }}">上一页</a>                                    7951
        {% endif %}                                                    7952
        {% if page < pages %}                                          7953
        <a class="link-btn"                                            7954
            href="{{ url_for('auth.profile', tab='logs',               7955
      page=page + 1) }}">下一页</a>                                    7956
        {% endif %}                                                    7957
      </nav>                                                           7958
      {% endif %}                                                      7959
    </div>                                                             7960
    {% if not records %}                                               7961
    <p class="muted" style="padding:18px 20px;">暂无操作记录。</p>      7962
    {% else %}                                                         7963
    <table>                                                            7964
      <thead>                                                          7965
        <tr>                                                           7966
          <th scope="col">时间</th>                                    7967
          <th scope="col">操作人</th>                                  7968
          <th scope="col">操作</th>                                    7969
          <th scope="col">对象</th>                                    7970
          <th scope="col">说明</th>                                    7971
        </tr>                                                          7972
      </thead>                                                         7973
      <tbody>                                                          7974
        {% for r in records %}                                         7975
        <tr>                                                           7976
          <td class="mono nowrap">{{ r.created_at |                    7977
      localdt("%Y-%m-%d %H:%M:%S") }}</td>                             7978
          <td>{{ users_map.get(r.user_id, "—") }}</td>                 7979
          <td style="text-align:left;">{{ r.action }}</td>             7980
          <td>                                                         7981
            {% if r.target_type %}{{ r.target_type }}{% if             7982
      r.target_id %}#{{ r.target_id }}{% endif %}                      7983
            {% else %}—{% endif %}                                     7984
          </td>                                                        7985
          <td style="text-align:left;">{{ r.detail or "—"              7986
      }}</td>                                                          7987
        </tr>                                                          7988
        {% endfor %}                                                   7989
      </tbody>                                                         7990
    </table>                                                           7991
    {% endif %}                                                        7992
</div>                                                                 7993
#  ==== app/templates/project/index.html ====                         7994
{% extends "base.html" %}                                              7995
{% block title %}项目与矿山档案{% endblock %}                          7996
{% block content %}                                                    7997
<div class="card">                                                     7998
  <h2>项目与矿山档案</h2>                                              7999
  <p class="actions" style="margin-bottom:12px;">                      8000
```

```
    <a class="btn btn--sm" href="/project/add">新建项目</a>
</p>
{% if not projects %}
<p class="muted">
    暂无项目。点击上方「新建项目」开始建立评价项目，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
</p>
{% else %}
<table>
    <caption class="muted">共 {{ projects|length }}
    个项目</caption>
    <thead>
        <tr>
            <th scope="col">项目名称</th>
            <th scope="col">编号</th>
            <th scope="col">状态</th>
            <th scope="col">图斑数</th>
            <th scope="col">操作</th>
        </tr>
    </thead>
    <tbody>
        {% for p in projects %}
        <tr>
            <td style="text-align:left;">{{ p.name }}</td>
            <td class="mono">{{ p.code or "—" }}</td>
            <td><span class="tag tag--muted">{{ p.status
    }}</span></td>
            <td class="num">
                {% if p.patches.count() %}
                    {{ p.patches.count() }}
                {% else %}
                    <span class="tag tag--warn">0</span>
                {% endif %}
            </td>
            <td class="nowrap">
                <span class="row-actions">
                    <a class="link-btn" href="/project/{{ p.id
    }}/patches">{% if p.patches.count() %}矿山图斑{% else %}＋
    新建图斑{% endif %}</a>
                    <a class="link-btn" href="/project/{{ p.id
    }}/edit">编辑</a>
                    <form method="post" action="/project/{{ p.id
    }}/delete"
                          onsubmit="return confirm('确定删除项目「{{
    p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">
                        {{ form.hidden_tag() }}
                        <button type="submit" class="link-btn
    link-btn--danger">删除</button>
                    </form>
```

```
            </span>                                                    8051
          </td>                                                        8052
        </tr>                                                          8053
        {% endfor %}                                                   8054
      </tbody>                                                         8055
    </table>                                                           8056
    {% endif %}                                                        8057
</div>                                                                 8058
{% endblock %}                                                         8059
# ==== app/templates/project/edit.html ====                            8060
{% extends "base.html" %}                                              8061
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%          8062
    endblock %}                                                        8063
{% block content %}                                                    8064
<div class="card" style="max-width:640px;">                            8065
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>                 8066
  <form method="post">                                                 8067
    {{ form.hidden_tag() }}                                            8068
    <div class="field">                                                8069
      <label class="field__label" for="name">项目名称</label>          8070
      <input id="name" type="text" name="name"                         8071
    style="width:100%;" required                                       8072
             value="{{ project.name if project else '' }}">            8073
    </div>                                                             8074
    <div class="field">                                                8075
      <label class="field__label" for="code">项目编号</label>          8076
      <input id="code" type="text" name="code"                         8077
    style="width:240px;"                                               8078
             value="{{ project.code or '' if project else              8079
    '' }}">                                                            8080
    </div>                                                             8081
    {% if project %}                                                   8082
    <div class="field">                                                8083
      <label class="field__label" for="status">状态</label>            8084
      <select id="status" name="status">                               8085
        {% for s in ["进行中", "已完成", "已归档"] %}                  8086
        <option value="{{ s }}" {% if project.status == s              8087
    %}selected{% endif %}>{{ s }}</option>                             8088
        {% endfor %}                                                   8089
      </select>                                                        8090
    </div>                                                             8091
    {% endif %}                                                        8092
    <div class="field">                                                8093
      <label class="field__label" for="description">项目说明</la       8094
    bel>                                                               8095
      <textarea id="description" name="description">{{                 8096
    project.description or '' if project else ''                       8097
    }}</textarea>                                                      8098
    </div>                                                             8099
    <p class="actions" style="margin-top:18px;">                       8100
```

```
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/project/patches.html ====
{% extends "base.html" %}
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>矿山图斑 · {{ project.name }}</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/{{ project.id }}/patch/add">新建图斑</a>
    <a class="btn btn--sm btn--ghost"
    href="/project/">返回项目列表</a>
  </p>
  {% if not project.patches.count() %}
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>
  {% else %}
  <table>
    <caption class="muted">共 {{ project.patches.count() }}
    个图斑</caption>
    <thead>
      <tr>
        <th scope="col">图斑名称</th>
        <th scope="col">行政区</th>
        <th scope="col">面积（公顷）</th>
        <th scope="col">开采矿种</th>
        <th scope="col">微地质单元</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in project.patches %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td>{{ p.admin_region or "—" }}</td>
        <td class="num">{{ p.area_ha if p.area_ha is not
    none else "—" }}</td>
        <td>{{ p.mineral_type or "—" }}</td>
        <td class="num">{{ p.units.count() }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/{{ p.id
    }}">微地质单元</a>
            <a class="link-btn" href="/constraint/{{ p.id
    }}">限制性判定</a>
            <a class="link-btn" href="/project/patch/{{
```

```
      p.id }}/edit">编辑</a>                                                  8151
                <form method="post" action="/project/patch/{{               8152
      p.id }}/delete"                                                       8153
                      onsubmit="return confirm('确定删除图斑「{{            8154
      p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">  8155
                  {{ form.hidden_tag() }}                                   8156
                  <button type="submit" class="link-btn                     8157
      link-btn--danger">删除</button>                                       8158
                </form>                                                     8159
              </span>                                                       8160
            </td>                                                           8161
          </tr>                                                             8162
          {% endfor %}                                                      8163
        </tbody>                                                            8164
      </table>                                                              8165
      {% endif %}                                                           8166
</div>                                                                      8167
{% endblock %}                                                              8168
# ==== app/templates/project/patch_edit.html ====                           8169
{% extends "base.html" %}                                                   8170
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%                 8171
      endblock %}                                                           8172
{% block content %}                                                         8173
<div class="card" style="max-width:720px;">                                 8174
      <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name       8175
      }}</h2>                                                               8176
      <form method="post">                                                  8177
        {{ form.hidden_tag() }}                                             8178
        <div class="field">                                                 8179
          <label class="field__label" for="patch-name">图斑名称</lab8180
        el>                                                                 8181
          <input id="patch-name" type="text" name="name"                    8182
        style="width:100%;" required                                        8183
                 value="{{ patch.name if patch else '' }}">                 8184
        </div>                                                              8185
        <div class="field">                                                 8186
          <label class="field__label" for="admin-region">所在行政区</ 8187
        label>                                                              8188
          <input id="admin-region" type="text"                              8189
        name="admin_region" style="width:100%;"                             8190
                 value="{{ patch.admin_region or '' if patch                8191
        else '' }}">                                                        8192
        </div>                                                              8193
        <div class="field">                                                 8194
          <label class="field__label" for="area-ha">图斑面积（公顷）</la8195
        bel>                                                                8196
          <input id="area-ha" type="number" step="any"                      8197
        name="area_ha"                                                      8198
                 value="{{ patch.area_ha if patch and                       8199
        patch.area_ha is not none else '' }}">                              8200
```

```
</div>
<div class="field">
  <label class="field__label" for="mineral-type">开采矿种</label>
  <input id="mineral-type" type="text" name="mineral_type" style="width:240px;"
         value="{{ patch.mineral_type or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mining-method">开采方式</label>
  <input id="mining-method" type="text" name="mining_method" style="width:240px;"
         value="{{ patch.mining_method or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="stop-years">停产年限</label>
  <input id="stop-years" type="number" step="any" name="stop_years"
         value="{{ patch.stop_years if patch and patch.stop_years is not none else '' }}">
</div>
<div class="field">
  <label class="checkbox" for="responsible-missing">
    <input id="responsible-missing" type="checkbox" name="responsible_entity_missing" value="1"
           {% if not patch or patch.responsible_entity_missing %}checked{% endif %}>
    <span>责任主体缺失</span>
  </label>
</div>
<h3>参照生态系统（指标标准化基准）</h3>
<p class="muted">
  论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。
  该基准与图斑绑定管理，可按不同参照生态系统分别设定。
</p>
<div class="field">
  <label class="field__label" for="coverage-benchmark">植被覆盖度基准（%）</label>
  <input id="coverage-benchmark" type="number" step="any" name="coverage_benchmark"
         value="{{ reference.coverage_benchmark if reference else 75 }}">
</div>
<div class="field">
  <label class="field__label" for="shannon-benchmark">Shannon-Wiener 多样性指数基准</label>
```

```
        <input id="shannon-benchmark" type="number"
      step="any" name="shannon_benchmark"
                value="{{ reference.shannon_benchmark if
      reference else 0.7 }}">
      </div>
      <div class="field">
        <label class="field__label" for="dominant-species">建群种
      </label>
        <input id="dominant-species" type="text"
      name="dominant_species" style="width:100%;"
                value="{{ reference.dominant_species or '' if
      reference else '' }}">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/{{
      project.id }}/patches">返回</a>
      </p>
    </form>
</div>
{% if patch %}
<div class="card" style="max-width:720px;">
    <h2>遥感影像挂接</h2>
    <p class="muted">
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。
    </p>
    {% if patch.image_path %}
    <figure style="margin:0 0 16px;">
      <img src="{{ url_for('static', filename=patch.image_path
      ) }}"
           alt="图斑「{{ patch.name }}」已挂接的遥感影像"
           style="max-width:100%;border:1px solid
      var(--line);border-radius:6px;">
      <figcaption class="muted" style="margin-top:6px;">
        {{ patch.image_note or "遥感影像" }}
        {% if patch.image_uploaded_at %}
        · 上传于 {{ patch.image_uploaded_at | localdt }}
        {% endif %}
      </figcaption>
    </figure>
    <form method="post" action="/project/patch/{{ patch.id
      }}/image/delete"
          onsubmit="return confirm('确定移除该遥感影像吗？');">
      {{ form.hidden_tag() }}
      <button type="submit" class="btn btn--sm
      btn--danger">移除影像</button>
    </form>
    <hr style="margin:16px 0;border:none;border-top:1px solid
      var(--line);">
```

```
  {% endif %}                                                                  8301
  <form method="post" action="/project/patch/{{ patch.id                       8302
    }}/image"                                                                  8303
        enctype="multipart/form-data">                                         8304
    {{ form.hidden_tag() }}                                                    8305
    <div class="field">                                                        8306
      <label class="field__label" for="image-file">                            8307
        {{ "替换影像文件" if patch.image_path else "选择影像文件" }}             8308
      </label>                                                                 8309
      <input id="image-file" type="file" name="image"                          8310
             accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t                   8311
    iff" required>                                                             8312
    </div>                                                                     8313
    <div class="field">                                                        8314
      <label class="field__label" for="image-note">影像说明（如来源、          8315
    时相）</label>                                                             8316
      <input id="image-note" type="text" name="image_note"                     8317
    style="width:100%;"                                                        8318
             placeholder="如：2025 年 6 月无人机正射影像"                      8319
             value="{{ patch.image_note or '' }}">                             8320
    </div>                                                                     8321
    <p class="actions">                                                        8322
      <button type="submit" class="btn">{{ "替换影像" if                       8323
    patch.image_path else "上传并挂接" }}</button>                             8324
    </p>                                                                       8325
  </form>                                                                      8326
</div>                                                                         8327
{% endif %}                                                                    8328
{% endblock %}                                                                 8329
# ==== app/templates/unit/list.html ====                                       8330
{% extends "base.html" %}                                                      8331
{% block title %}微地质单元{% endblock %}                                      8332
{% block content %}                                                            8333
<div class="card">                                                             8334
  <h2>微地质单元</h2>                                                          8335
  {% if not rows %}                                                            8336
  <p class="muted">                                                            8337
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。                          8338
  </p>                                                                         8339
  {% else %}                                                                   8340
  <table>                                                                      8341
    <thead>                                                                    8342
      <tr>                                                                     8343
        <th scope="col">所属项目</th>                                          8344
        <th scope="col">矿山图斑</th>                                          8345
        <th scope="col">已划分单元</th>                                        8346
        <th scope="col">占比合计</th>                                          8347
        <th scope="col">操作</th>                                              8348
      </tr>                                                                    8349
    </thead>                                                                   8350
```

```
      <tbody>
        {% for r in rows %}
        <tr>
          <td>{{ r.patch.project.name if r.patch.project else "—" }}</td>
          <td style="text-align:left;">{{ r.patch.name }}</td>
          <td class="num">{{ r.count }} 个</td>
          <td class="num">
            {% if r.count %}
              {{ "%.2f%%"|format(r.ratio_sum * 100) }}
              {% if r.ratio_sum < 0.999 or r.ratio_sum > 1.001 %}
                <span class="warn">（不等于 100%）</span>
              {% endif %}
            {% else %}—{% endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/unit/{{ r.patch.id }}">划分与维护</a>
              <a class="link-btn" href="/indicator/">录入指标数据</a>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/unit/index.html ====
{% extends "base.html" %}
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元 · {{ patch.name }}</h2>
  {% if units %}
  <table>
    <caption class="muted">
      共 {{ units|length }} 个单元；
      面积合计 <span class="num">{{ "%.4f"|format(total_area) }}</span>；
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum * 100) }}</span>
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum > 1.001) %}
        <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>
      {% endif %}
    </caption>
```

```
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积</th>
        <th scope="col">面积占比</th>
        <th scope="col">说明</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ u.area if u.area is not none else "—" }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{ "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ u.description or "—" }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/edit/{{ u.id }}">编辑</a>
            <form method="post" action="/unit/delete/{{ u.id }}"
                  onsubmit="return confirm('确定删除微地质单元「{{ u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">尚未划分微地质单元。请在下方添加。</p>
  {% endif %}
  <h3>添加微地质单元</h3>
  <form method="post" action="/unit/{{ patch.id }}/add">
    {{ form.hidden_tag() }}
    <div class="actions">
      <label class="field__label" for="add-unit-type"
             style="margin:0;">类型</label>
      <select id="add-unit-type" name="unit_type">
        {% for t in unit_types %}<option value="{{ t }}">{{
```

```
      t }}</option>{% endfor %}                                          8451
        </select>                                                        8452
        <label class="field__label" for="add-unit-area"                  8453
      style="margin:0;">面积</label>                                     8454
        <input id="add-unit-area" type="number" step="any"               8455
      name="area">                                                       8456
        <label class="field__label" for="add-unit-desc"                  8457
      style="margin:0;">说明</label>                                     8458
        <input id="add-unit-desc" type="text"                            8459
      name="description" style="width:220px;">                           8460
        <button type="submit" class="btn btn--sm">添加</button>          8461
      </div>                                                             8462
    </form>                                                              8463
    <p class="actions" style="margin-top:20px;">                         8464
      <a class="btn btn--sm btn--ghost" href="/project/{{                8465
      patch.project_id }}/patches">返回图斑列表</a>                      8466
      <a class="btn btn--sm btn--ghost"                                  8467
      href="/indicator/">去录入指标数据</a>                              8468
    </p>                                                                 8469
</div>                                                                   8470
{% endblock %}                                                           8471
# ==== app/templates/unit/edit.html ====                                 8472
{% extends "base.html" %}                                                8473
{% block title %}编辑微地质单元{% endblock %}                            8474
{% block content %}                                                      8475
<div class="card">                                                       8476
    <h2>编辑微地质单元 · {{ patch.name }}</h2>                           8477
    <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>  8478
    <form method="post">                                                 8479
      {{ form.hidden_tag() }}                                            8480
      <div class="field">                                                8481
        <label class="field__label" for="unit-type">微地质单元类型</l    8482
      abel>                                                              8483
        <select id="unit-type" name="unit_type">                         8484
          {% for t in unit_types %}                                      8485
          <option value="{{ t }}" {% if unit.unit_type == t              8486
      %}selected{% endif %}>{{ t }}</option>                             8487
          {% endfor %}                                                   8488
        </select>                                                        8489
      </div>                                                             8490
      <div class="field">                                                8491
        <label class="field__label" for="unit-area">面积</label>         8492
        <input id="unit-area" type="number" step="any"                   8493
      name="area"                                                        8494
               value="{{ unit.area if unit.area is not none              8495
      else '' }}">                                                       8496
      </div>                                                             8497
      <div class="field">                                                8498
        <label class="field__label" for="unit-desc">说明</label>         8499
        <input id="unit-desc" type="text" name="description"             8500
```

```
      style="width:420px;"                                                    8501
                value="{{ unit.description or '' }}">                        8502
      </div>                                                                  8503
      <p class="actions" style="margin-top:16px;">                            8504
        <button type="submit" class="btn">保存</button>                       8505
        <a class="btn btn--ghost" href="/unit/{{                              8506
      unit.patch_id }}">返回</a>                                              8507
      </p>                                                                    8508
    </form>                                                                   8509
</div>                                                                        8510
{% endblock %}                                                                8511
# ==== app/templates/standard/index.html ====                                 8512
{% extends "base.html" %}                                                     8513
{% block title %}政策与标准知识库{% endblock %}                                 8514
{% block content %}                                                           8515
<div class="card">                                                            8516
  <h2>政策与标准知识库</h2>  <p class="actions"                                 8517
    style="margin-bottom:14px;">                                              8518
    <a class="btn btn--sm" href="/standard/add">添加标准</a>                   8519
    <a class="btn btn--sm btn--ghost"                                         8520
    href="/standard/compare">省际口径比对</a>                                  8521
  </p>                                                                        8522
  <h3>地方标准</h3>                                                           8523
  {% if local %}                                                              8524
  <table>                                                                     8525
    <thead>                                                                   8526
      <tr>                                                                    8527
        <th scope="col">省份</th>                                             8528
        <th scope="col">名称</th>                                             8529
        <th scope="col">编号</th>                                             8530
        <th scope="col">规则数</th>                                           8531
        <th scope="col">操作</th>                                             8532
      </tr>                                                                   8533
    </thead>                                                                  8534
    <tbody>                                                                   8535
      {% for s in local %}                                                    8536
      <tr>                                                                    8537
        <td>{{ s.province }}</td>                                             8538
        <td style="text-align:left;">{{ s.name }}</td>                        8539
        <td class="mono">{{ s.doc_number or "—" }}</td>                       8540
        <td class="num">{{ s.rules|length }}</td>                             8541
        <td class="nowrap">                                                   8542
          <span class="row-actions">                                          8543
            <a class="link-btn" href="/standard/{{ s.id                       8544
    }}/rules">规则</a>                                                        8545
            <a class="link-btn" href="/standard/{{ s.id                       8546
    }}/edit">编辑</a>                                                         8547
            <form method="post" action="/standard/{{ s.id                     8548
    }}/delete"                                                                8549
                  onsubmit="return confirm('确定删除「{{                      8550
```

```
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">暂无地方标准。运行 <span class="mono">python
    seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>
  {% endif %}
  <h3>政策文件</h3>
  {% if policy %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">文号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in policy %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id
    }}/delete"
                  onsubmit="return confirm('确定删除「{{
    s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
```

```
            </span>                                                    8601
          </td>                                                        8602
        </tr>                                                          8603
        {% endfor %}                                                   8604
      </tbody>                                                         8605
    </table>                                                           8606
    {% else %}                                                         8607
    <p class="muted">暂无政策文件。</p>                                  8608
    {% endif %}                                                        8609
</div>                                                                 8610
{% endblock %}                                                         8611
# ==== app/templates/standard/edit.html ====                           8612
{% extends "base.html" %}                                              8613
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%         8614
    endblock %}                                                        8615
{% block content %}                                                    8616
<div class="card" style="max-width:800px;">                            8617
    <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                8618
    <form method="post">                                               8619
      {{ form.hidden_tag() }}                                          8620
      <div class="field">                                              8621
        <label class="field__label" for="province">省份</label>        8622
        <input id="province" type="text" name="province"               8623
      style="width:160px;" required                                    8624
               value="{{ standard.province if standard else            8625
      '' }}">                                                          8626
      </div>                                                           8627
      <div class="field">                                              8628
        <label class="field__label" for="std-name">名称</label>        8629
        <input id="std-name" type="text" name="name"                   8630
      style="width:100%;" required                                     8631
               value="{{ standard.name if standard else ''             8632
      }}">                                                             8633
      </div>                                                           8634
      <div class="field">                                              8635
        <label class="field__label" for="doc-number">标准编号 /        8636
      文号</label>                                                     8637
        <input id="doc-number" type="text" name="doc_number"           8638
      style="width:320px;"                                             8639
               value="{{ standard.doc_number or '' if                  8640
      standard else '' }}">                                            8641
      </div>                                                           8642
      <div class="field">                                              8643
        <label class="field__label" for="category">类别</label>        8644
        <select id="category" name="category">                         8645
          {% for c in ["地方标准", "政策文件"] %}                      8646
          <option value="{{ c }}" {% if standard and                   8647
      standard.category == c %}selected{% endif %}>{{ c                8648
      }}</option>                                                      8649
          {% endfor %}                                                 8650
```

```
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="publish-date">发布时间</l
      abel>
        <input id="publish-date" type="text"
      name="publish_date" style="width:200px;"
               placeholder="如 2023-03-30"
               value="{{ standard.publish_date or '' if
      standard else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="implement-date">实施时间<
      /label>
        <input id="implement-date" type="text"
      name="implement_date" style="width:200px;"
               placeholder="如 2023-04-30（未明确可留空）"
               value="{{ standard.implement_date or '' if
      standard else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="summary">特点 /
      判定标准</label>
        <textarea id="summary" name="summary"
      style="min-height:96px;">{{ standard.summary or '' if
      standard else '' }}</textarea>
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        {% if standard %}
        <form method="post" action="/standard/{{ standard.id
      }}/delete" style="display:inline;"
              onsubmit="return confirm('确定删除「{{
      standard.province }} {{ standard.name
      }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn
      btn--danger">删除该标准</button>
        </form>
        {% endif %}
        <a class="btn btn--ghost" href="/standard/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/standard/rules.html ====
{% extends "base.html" %}
{% block title %}结构化规则 · {{ standard.province }}{% endblock
    %}
{% block content %}
```

```
<div class="card">
  <h2>结构化规则 · {{ standard.province }} {{ standard.name }}</h2>
  {% if rules %}
  <table>
    <caption class="muted">共 {{ rules|length }} 条规则</caption>
    <thead>
      <tr>
        <th scope="col">判定项</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">适用分区</th>
        <th scope="col">备注</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rules %}
      <tr>
        <td style="text-align:left;">{{ r.item }}</td>
        <td class="mono nowrap">{{ r.operator }} {{ r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td>{{ r.region or "—" }}</td>
        <td style="text-align:left;">{{ r.note or "—" }}</td>
        <td>
          <form method="post" action="/standard/rule/{{ r.id }}/delete"
                onsubmit="return confirm('确定删除规则「{{ r.item }} {{ r.operator }} {{ r.threshold or '' }}」吗？');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}

  <h3>添加规则</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="actions">
      <div>
        <label class="field__label" for="r-item">判定项</label>
```

```
          <input id="r-item" type="text" name="item"
      placeholder="如：植被覆盖度" required>
        </div>
        <div>
          <label class="field__label" for="r-op">比较符</label>
          <select id="r-op" name="operator">
            {% for o in [">=", ">", "<=", "<", "=="]
      %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
          </select>
        </div>
        <div>
          <label class="field__label" for="r-thr">阈值</label>
          <input id="r-thr" type="text" name="threshold"
      placeholder="如：50" style="width:110px;">
        </div>
        <div>
          <label class="field__label" for="r-cond">条件类型</label
      >
          <select id="r-cond" name="condition_type">
            <option value="必要条件">必要条件</option>
            <option value="充分条件">充分条件</option>
          </select>
        </div>
        <div>
          <label class="field__label" for="r-region">适用分区</lab
      el>
          <input id="r-region" type="text" name="region"
      placeholder="如宁夏分区" style="width:130px;">
        </div>
        <div>
          <label class="field__label" for="r-note">备注</label>
          <input id="r-note" type="text" name="note"
      style="width:220px;">
        </div>
        <div>
          <button type="submit" class="btn">添加规则</button>
        </div>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost"
      href="/standard/">返回标准列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/standard/compare">省际口径比对</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
```

```
{% block content %}
<div class="card">
  <h2>省际口径比对</h2>
  {% if not groups %}
  <p class="muted">
    暂无规则数据。请先在「标准知识库」中为标准录入规则，
    或运行 <span class="mono">python seed_demo.py</span> 载入论文表 1 数据。
  </p>
  {% else %}
  {% for item, rs in groups.items() %}
  <h3>{{ item }}</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">标准 / 文件</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">备注</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rs %}
      <tr>
        <td>{{ r.standard.province }}</td>
        <td style="text-align:left;">{{ r.standard.name }}</td>
        <td class="mono nowrap">{{ r.operator }} {{ r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td style="text-align:left;">{{ r.note or "—" }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endfor %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
```

```
  <h2>限制性指标判定</h2>                                                    8851
  {{ patch_guide(projects_without_patch) }}                                  8852
  {% if not patches %}                                                       8853
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>          8854
  {% else %}                                                                 8855
  <table>                                                                    8856
    <thead>                                                                  8857
      <tr>                                                                   8858
        <th scope="col">所属项目</th>                                        8859
        <th scope="col">矿山图斑</th>                                        8860
        <th scope="col">判定结果</th>                                        8861
        <th scope="col">操作</th>                                            8862
      </tr>                                                                  8863
    </thead>                                                                 8864
    <tbody>                                                                  8865
      {% for p in patches %}                                                 8866
      {% set r = latest.get(p.id) %}                                         8867
      <tr>                                                                   8868
        <td>{{ p.project.name if p.project else "—" }}</td>                  8869
        <td style="text-align:left;">{{ p.name }}</td>                       8870
        <td style="text-align:left;">                                        8871
          {% if r %}                                                         8872
            {% if r.passed %}                                                8873
            <span class="tag tag--ok">通过</span>                            8874
            {% else %}                                                       8875
            <span class="tag tag--warn">不通过</span>                        8876
            {% endif %}                                                      8877
            <span class="muted">{{ r.reason }}</span>                        8878
          {% else %}                                                         8879
          <span class="tag tag--muted">未判定</span>                         8880
          {% endif %}                                                        8881
        </td>                                                                8882
        <td class="nowrap">                                                  8883
          <span class="row-actions">                                         8884
            <a class="link-btn" href="/constraint/{{ p.id                    8885
    }}">执行判定</a>                                                         8886
          </span>                                                            8887
        </td>                                                                8888
      </tr>                                                                  8889
      {% endfor %}                                                           8890
    </tbody>                                                                 8891
  </table>                                                                   8892
  {% endif %}                                                                8893
</div>                                                                       8894
{% endblock %}                                                               8895
# ==== app/templates/constraint/check.html ====                              8896
{% extends "base.html" %}                                                    8897
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}             8898
{% block content %}                                                          8899
<div class="card" style="max-width:760px;">                                  8900
```

```
  <h2>限制性条件判定 · {{ patch.name }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    {% for field, name, val in field_states %}
    <fieldset style="border:1px solid
    var(--line);border-radius:6px;padding:10px
    14px;margin:0 0 10px;">
      <legend style="padding:0 6px;font-weight:600;">{{
    name }}</legend>
      <div class="actions">
        <label class="checkbox">
          <input type="radio" name="{{ field }}" value="1"
                 {% if val is none or val %}checked{% endif
    %}>
          <span>通过</span>
        </label>
        <label class="checkbox">
          <input type="radio" name="{{ field }}" value="0"
                 {% if val is not none and not val
    %}checked{% endif %}>
          <span>不通过</span>
        </label>
      </div>
    </fieldset>
    {% endfor %}
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">保存判定</button>
      <a class="btn btn--ghost" href="/constraint/">返回</a>
    </p>
  </form>
  {% if record %}
  <div style="margin-top:18px;padding:14px;border-radius:6px
    ;border-left:4px solid;
                {% if record.passed %}background:var(--ok-soft
    );border-left-color:var(--ok);
                {% else %}background:var(--danger-soft);border
    -left-color:var(--danger);{% endif %}">
    <p style="margin:0 0 4px;font-weight:600;">
      判定结论：
      {% if record.passed %}
      <span class="ok">通过，可进入定量评价</span>
      {% else %}
      <span class="warn">不通过 —— 建议生态重建或人工辅助</span>
      {% endif %}
    </p>
    <p class="muted" style="margin:0;">{{ record.reason
    }}</p>
  </div>
  {% endif %}
</div>
```

```
{% endblock %}                                                                  8951
# ==== app/templates/ahp/index.html ====                                        8952
{% extends "base.html" %}                                                       8953
{% block title %}AHP 权重计算{% endblock %}                                     8954
{% block content %}                                                             8955
<div class="card" style="max-width:720px;">                                     8956
  <h2>AHP 权重计算 · 选择指标层级</h2>                                          8957
  <form method="post">                                                          8958
    {{ form.hidden_tag() }}                                                     8959
    <div class="field">                                                         8960
      <label class="field__label" for="{{ form.unit_type.id                     8961
    }}">微地质单元类型</label>                                                  8962
      {{ form.unit_type(id=form.unit_type.id) }}                                8963
    </div>                                                                      8964
    <div class="field">                                                         8965
      <label class="field__label" for="{{ form.level.id                         8966
    }}">指标层级</label>                                                        8967
      {{ form.level(id=form.level.id) }}                                        8968
    </div>                                                                      8969
    <div class="field">                                                         8970
      <label class="field__label" for="{{ form.l1.id }}">                       8971
        一级指标（层级选「二级指标」时需选择）                                  8972
      </label>                                                                  8973
      {{ form.l1(id=form.l1.id) }}                                              8974
    </div>                                                                      8975
    <p class="actions" style="margin-top:16px;">                                8976
      {{ form.submit(class_="btn") }}                                           8977
      <a class="btn btn--ghost" href="/ahp/indicators">指标定义维               8978
    护</a>                                                                      8979
      <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a               8980
    >                                                                           8981
    </p>                                                                        8982
  </form>                                                                       8983
</div>                                                                          8984
<div class="card">                                                              8985
  <h2>各微地质单元当前生效权重</h2>                                             8986
  <table>                                                                       8987
    <thead>                                                                     8988
      <tr>                                                                      8989
        <th scope="col">微地质单元</th>                                         8990
        <th scope="col">当前权重来源</th>                                       8991
        <th scope="col">操作</th>                                               8992
      </tr>                                                                     8993
    </thead>                                                                    8994
    <tbody>                                                                     8995
      {% for r in weight_rows %}                                                8996
      <tr>                                                                      8997
        <td style="text-align:left;">{{ r.unit_type }}</td>                     8998
        <td style="text-align:left;">                                           8999
          {% if r.custom %}                                                     9000
```

```
            <span class="tag tag--warn">{{ r.label }}</span>
            {% else %}
            <span class="tag tag--muted">{{ r.label }}</span>
            {% endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/ahp/matrix?unit_type={{ r.unit_type }}&level=1">一级指标权重</a>
              <a class="link-btn" href="/ahp/matrix?unit_type={{ r.unit_type }}&level=1&calc=1">查看一致性</a>
              {% if r.custom %}
              <form method="post" action="/ahp/weights/reset"
                    onsubmit="return confirm('确定把「{{ r.unit_type }}」恢复为论文表 3 的默认权重吗？');">
                <input type="hidden" name="csrf_token" value="{{ csrf_token() }}">
                <input type="hidden" name="unit_type" value="{{ r.unit_type }}">
                <button type="submit" class="link-btn link-btn--danger">恢复默认</button>
              </form>
              {% endif %}
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
</div>
{% endblock %}
# ==== app/templates/ahp/matrix.html ====
{% extends "base.html" %}
{% block title %}判断矩阵录入{% endblock %}
{% block content %}
<div class="card">
  <h2>
    判断矩阵录入
    · {{ unit_type }}
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}
  </h2>
  <p class="muted">
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
    矩阵已按论文表 3 的权重预填。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <table style="width:auto;">
```

```
        <thead>                                                                  9051
          <tr>                                                                   9052
            <th></th>                                                            9053
            {% for name in names %}<th scope="col">{{ name                       9054
    }}</th>{% endfor %}                                                          9055
          </tr>                                                                  9056
        </thead>                                                                 9057
        <tbody>                                                                  9058
          {% for i in range(names|length) %}                                     9059
          <tr>                                                                   9060
            <th scope="row">{{ names[i] }}</th>                                  9061
            {% for j in range(names|length) %}                                   9062
              {% if i == j %}                                                    9063
              <td class="muted num">1</td>                                       9064
              {% elif i < j %}                                                   9065
              <td>                                                               9066
                <label class="muted" style="position:absolute;                   9067
    left:-9999px;"                                                               9068
                       for="m-{{ i }}-{{ j }}">{{ names[i] }}                    9069
    相对 {{ names[j] }} 的重要性</label>                                          9070
                <input id="m-{{ i }}-{{ j }}" type="number"                      9071
    step="0.01" min="0.01"                                                       9072
                       name="m_{{ i }}_{{ j }}" value="{{                        9073
    default[i][j] }}">                                                           9074
              </td>                                                              9075
              {% else %}                                                         9076
              <td class="muted num">{{ default[i][j] }}</td>                     9077
              {% endif %}                                                        9078
            {% endfor %}                                                         9079
          </tr>                                                                  9080
          {% endfor %}                                                           9081
        </tbody>                                                                 9082
    </table>                                                                     9083
    <p class="actions">                                                          9084
      <button type="submit" class="btn" name="action"                            9085
    value="calc">计算权重</button>                                               9086
      {% if result and result.consistent %}                                      9087
      <button type="submit" class="btn" name="action"                            9088
    value="save">保存并生效</button>                                             9089
      {% endif %}                                                                9090
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>                        9091
    </p>                                                                         9092
  </form>                                                                        9093
  <p class="muted" style="margin-top:10px;">                                     9094
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label                     9095
    }}</strong>。                                                                9096
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；          9097
    如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。              9098
  </p>                                                                           9099
  {% if result %}                                                                9100
```

```
<hr style="margin:22px 0;border:none;border-top:1px solid
  var(--line);">
<h3>计算结果</h3>
<table style="width:auto;">
  <thead>
    <tr>
      <th scope="col">指标</th>
      {% for name in names %}<th scope="col">{{ name
  }}</th>{% endfor %}
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">权重</th>
      {% for w in result.weights %}<td class="mono">{{ w
  }}</td>{% endfor %}
    </tr>
  </tbody>
</table>
<div class="stat-row">
  <div>
    <div class="stat__label">最大特征值 λmax</div>
    <div class="stat__value mono">{{ result.lambda_max
  }}</div>
  </div>
  <div>
    <div class="stat__label">一致性指标 CI</div>
    <div class="stat__value mono">{{
  "%.4f"|format(result.ci) }}</div>
  </div>
  <div>
    <div class="stat__label">一致性比例 CR</div>
    <div class="stat__value mono">{{
  "%.4f"|format(result.cr) }}</div>
  </div>
  <div>
    <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>
    <div class="stat__value">
      {% if result.consistent %}<span class="ok">通过</span>
      {% else %}<span class="warn">不通过</span>{% endif %}
    </div>
  </div>
</div>
{% if not result.consistent and result.most_inconsistent
  %}
<p style="margin-top:14px;padding:12px;border-radius:6px;b
  ackground:var(--danger-soft);
            border-left:4px solid var(--danger);">
  <strong class="warn">建议优先调整：</strong>
  第 {{ result.most_inconsistent.i + 1 }} 行、第 {{
```

```
      result.most_inconsistent.j + 1 }} 列的比较                    9151
      （当前 <span class="mono num">{{ result.most_inconsistent.v 9152
      alue|round(3) }}</span>,                                      9153
      理想约 <span class="mono num">{{ result.most_inconsistent.e 9154
      xpected|round(3) }}</span>）                                  9155
    </p>                                                            9156
    {% endif %}                                                     9157
    {% endif %}                                                     9158
</div>                                                              9159
{% endblock %}                                                      9160
# ==== app/templates/ahp/indicators.html ====                       9161
{% extends "base.html" %}                                           9162
{% block title %}指标定义维护{% endblock %}                          9163
{% block content %}                                                 9164
<div class="card">                                                  9165
    <h2>指标体系维护</h2>                                           9166
    <p class="actions" style="margin-bottom:14px;">                 9167
      <a class="btn btn--sm btn--ghost"                             9168
      href="/ahp/">返回权重计算</a>                                 9169
      <a class="btn btn--sm btn--ghost"                             9170
      href="/ahp/experts">多专家矩阵</a>                            9171
      <form method="post" action="/ahp/indicators/reset"            9172
      style="display:inline;"                                       9173
            onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。 9174
      确定继续？');">                                               9175
        {{ form.hidden_tag() }}                                     9176
        <button type="submit" class="btn btn--sm                    9177
      btn--danger">按论文取值重置</button>                          9178
      </form>                                                       9179
    </p>                                                            9180
    {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,   9181
      "三级指标")] %}                                               9182
    {% set level_items = all_ind|selectattr("level",                9183
      "equalto", level)|list %}                                     9184
    {% if level_items %}                                            9185
    <h3>{{ title }}（{{ level_items|length }} 项）</h3>             9186
    <table>                                                         9187
      <thead>                                                       9188
        <tr>                                                        9189
          <th scope="col">指标</th>                                 9190
          <th scope="col">上级指标</th>                             9191
          <th scope="col">单位</th>                                 9192
          <th scope="col">标准化类型</th>                           9193
          <th scope="col">理想取值</th>                             9194
          <th scope="col">启用单元</th>                             9195
          <th scope="col">限制性</th>                               9196
          <th scope="col">操作</th>                                 9197
        </tr>                                                       9198
      </thead>                                                      9199
      <tbody>                                                       9200
```

```
        {% for i in level_items %}                                    9201
        <tr>                                                          9202
          <td style="text-align:left;">                               9203
            <span class="mono muted">{{ i.code                        9204
      }}</span><br>{{ i.name }}                                       9205
          </td>                                                       9206
          <td>{{ (all_ind|selectattr("code", "equalto",              9207
      i.parent_code)|list|first).name if i.parent_code else           9208
      "—" }}</td>                                                     9209
          <td>{{ i.unit or "—" }}</td>                                9210
          <td>{{ type_label(i.ideal_type) }}</td>                     9211
          <td class="mono">{{ i.ideal_value or "—" }}</td>            9212
          <td style="text-align:left;">{{ i.enabled_units or          9213
      "—" }}</td>                                                     9214
          <td>                                                        9215
            {% if i.is_constraint %}<span class="tag                  9216
      tag--warn">是</span>                                            9217
            {% else %}<span class="tag tag--muted">否</span>{%        9218
      endif %}                                                        9219
          </td>                                                       9220
          <td class="nowrap">                                         9221
            <span class="row-actions">                                9222
              <a class="link-btn" href="/ahp/indicators/{{            9223
      i.code }}/edit">编辑</a>                                        9224
            </span>                                                   9225
          </td>                                                       9226
        </tr>                                                         9227
        {% endfor %}                                                  9228
      </tbody>                                                        9229
    </table>                                                          9230
    {% endif %}                                                       9231
    {% endfor %}                                                      9232
</div>                                                                9233
{% endblock %}                                                        9234
# ==== app/templates/ahp/indicator_edit.html ====                     9235
{% extends "base.html" %}                                             9236
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}              9237
{% block content %}                                                   9238
<div class="card" style="max-width:680px;">                           9239
  <h2>编辑指标 · {{ ind.name }}</h2>                                  9240
  <p class="muted">                                                   9241
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{        9242
    ind.level }} 级，                                                 9243
    由论文指标体系确定，不可修改。                                    9244
  </p>                                                                9245
  <form method="post">                                                9246
    {{ form.hidden_tag() }}                                           9247
    <div class="field">                                               9248
      <label class="field__label" for="name">指标名称</label>         9249
      <input id="name" type="text" name="name"                        9250
```

```
style="width:100%;" required                                    9251
            value="{{ ind.name }}">                             9252
</div>                                                          9253
<div class="field">                                             9254
  <label class="field__label" for="unit">计量单位</label>       9255
  <input id="unit" type="text" name="unit"                      9256
style="width:180px;"                                            9257
            value="{{ ind.unit or '' }}">                       9258
</div>                                                          9259
<div class="field">                                             9260
  <label class="field__label" for="ideal-type">标准化类型</la   9261
bel>                                                            9262
  <select id="ideal-type" name="ideal_type">                    9263
    {% for code, label in [("benefit","越大越优"),("cost","越   9264
小越优"),("mid","适中型"),("discrete","离散赋值")] %}          9265
      <option value="{{ code }}" {% if ind.ideal_type ==        9266
code %}selected{% endif %}>{{ label }}</option>                 9267
    {% endfor %}                                                9268
  </select>                                                     9269
</div>                                                          9270
<div class="field">                                             9271
  <label class="field__label" for="ideal-value">理想取值</la   9272
bel>                                                            9273
  <input id="ideal-value" type="text"                           9274
name="ideal_value" style="width:100%;"                          9275
            value="{{ ind.ideal_value or '' }}">                9276
  <p class="muted" style="margin:4px 0 0;">                     9277
    如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。          9278
  </p>                                                          9279
</div>                                                          9280
<fieldset style="border:1px solid                               9281
var(--line);border-radius:6px;padding:10px                      9282
14px;margin:0 0 12px;">                                         9283
  <legend style="padding:0 6px;font-weight:600;">启用该指标的微   9284
地质单元</legend>                                               9285
  {% for t in unit_types %}                                     9286
  <label class="checkbox" style="margin-right:18px;">           9287
    <input type="checkbox" name="enabled_units"                 9288
value="{{ t }}"                                                 9289
           {% if t in ind.units_list() %}checked{%              9290
endif %}>                                                       9291
    <span>{{ t }}</span>                                        9292
  </label>                                                      9293
  {% endfor %}                                                  9294
  <p class="muted" style="margin:8px 0 0;">                     9295
    论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。      9296
  </p>                                                          9297
</fieldset>                                                     9298
<div class="field">                                             9299
  <label class="checkbox" for="is-constraint">                  9300
```

```
          <input id="is-constraint" type="checkbox"                    9301
      name="is_constraint" value="1"                                   9302
                  {% if ind.is_constraint %}checked{% endif %}>        9303
          <span>同时作为限制性指标（参与一票否决判定）</span>                    9304
        </label>                                                       9305
      </div>                                                           9306
      <div class="field">                                              9307
        <label class="field__label" for="description">取值依据说明</    9308
      label>                                                           9309
        <textarea id="description" name="description"                  9310
      style="min-height:88px;">{{ ind.description or ''                9311
      }}</textarea>                                                    9312
      </div>                                                           9313
      <p class="actions" style="margin-top:16px;">                     9314
        <button type="submit" class="btn">保存</button>                9315
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列    9316
      表</a>                                                           9317
      </p>                                                             9318
    </form>                                                            9319
</div>                                                                 9320
{% endblock %}                                                         9321
# ==== app/templates/ahp/experts.html ====                             9322
{% extends "base.html" %}                                              9323
{% block title %}多专家判断矩阵{% endblock %}                              9324
{% block content %}                                                    9325
<div class="card">                                                     9326
  <h2>多专家判断矩阵 · 录入与汇总</h2>                                      9327
  <form method="get" class="actions" style="margin-bottom:14          9328
      px;">                                                            9329
      <div>                                                            9330
        <label class="field__label" for="unit-type">微地质单元类型</l  9331
      abel>                                                            9332
        <select id="unit-type" name="unit_type"                        9333
      onchange="this.form.submit()">                                   9334
          {% for t in ["岩质边坡", "碎石边坡", "平台"] %}                  9335
          <option value="{{ t }}" {% if unit_type == t                 9336
      %}selected{% endif %}>{{ t }}</option>                           9337
          {% endfor %}                                                 9338
        </select>                                                      9339
      </div>                                                           9340
      <div>                                                            9341
        <label class="field__label" for="level">指标层级</label>       9342
        <select id="level" name="level"                                9343
      onchange="this.form.submit()">                                   9344
          <option value="1" {% if level == 1 %}selected{%              9345
      endif %}>一级指标</option>                                       9346
          <option value="2" {% if level == 2 %}selected{%              9347
      endif %}>二级指标</option>                                       9348
        </select>                                                      9349
      </div>                                                           9350
```

```
</form>
<p class="muted">
  当前范围：<strong>{{ unit_type }}</strong> ·
  {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif
  %}。
  若选二级指标，请先到
  <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。
</p>
{% if rows %}
<table>
  <caption class="muted">已录入 {{ rows|length }}
  位专家的矩阵</caption>
  <thead>
    <tr>
      <th scope="col">专家</th>
      <th scope="col">矩阵规模</th>
      <th scope="col">该矩阵 CR</th>
      <th scope="col">一致性</th>
      <th scope="col">录入时间</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for r in rows %}
    <tr>
      <td>{{ r.expert_name }}</td>
      <td class="num">{{ n }} × {{ n }}</td>
      <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is
  not none else "—" }}</td>
      <td>
        {% if r.cr is not none and r.cr < 0.1 %}<span
  class="tag tag--ok">通过</span>
        {% else %}<span class="tag tag--warn">不通过</span>{%
  endif %}
      </td>
      <td class="mono nowrap">{{ r.created_at | localdt
  }}</td>
      <td>
        <form method="post" action="/ahp/experts/{{ r.id
  }}/delete"
              onsubmit="return confirm('确定删除专家「{{
  r.expert_name }}」的矩阵吗？');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn btn--sm
  btn--danger">删除</button>
        </form>
      </td>
    </tr>
    {% endfor %}
  </tbody>
```

```
</table>
<p class="actions" style="margin-bottom:10px;">
  <a class="btn" href="/ahp/experts/aggregate?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">
    汇总计算（几何平均）
  </a>
</p>
{% else %}
<p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>
{% endif %}
<h3>录入一位专家的判断矩阵</h3>
<form method="post">
  {{ form.hidden_tag() }}
  <input type="hidden" name="action" value="add">
  <input type="hidden" name="unit_type" value="{{ unit_type }}">
  <input type="hidden" name="level" value="{{ level }}">
  {% if l1 %}<input type="hidden" name="l1" value="{{ l1 }}">{% endif %}
  <div class="field">
    <label class="field__label" for="expert-name">专家姓名</label>
    <input id="expert-name" type="text" name="expert_name"
           style="width:220px;" placeholder="如：张工">
  </div>
  <table style="width:auto;">
    <thead>
      <tr>
        <th></th>
        {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      {% for i in range(n) %}
      <tr>
        <th scope="row">{{ names[i] }}</th>
        {% for j in range(n) %}
          {% if i == j %}
          <td class="muted num">1</td>
          {% elif i < j %}
          <td>
            <label class="muted" style="position:absolute;left:-9999px;"
                   for="e-{{ i }}-{{ j }}">{{ names[i] }} 相对 {{ names[j] }} 的重要性</label>
            <input id="e-{{ i }}-{{ j }}" type="number" step="0.01" min="0.01"
                   name="m_{{ i }}_{{ j }}" value="{{
```

```
      default[i][j] }}">                                                    9451
                </td>                                                       9452
                {% else %}                                                  9453
                <td class="muted num">{{ default[i][j] }}</td>              9454
                {% endif %}                                                 9455
              {% endfor %}                                                  9456
            </tr>                                                           9457
            {% endfor %}                                                    9458
          </tbody>                                                          9459
      </table>                                                              9460
      <p class="actions" style="margin-top:12px;">                          9461
        <button type="submit" class="btn">录入该专家矩阵</button>           9462
        <a class="btn btn--ghost" href="/ahp/indicators">指标维护<          9463
      /a>                                                                   9464
        <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>             9465
      </p>                                                                  9466
    </form>                                                                 9467
</div>                                                                      9468
{% endblock %}                                                              9469
#  ==== app/templates/ahp/experts_aggregate.html ====                       9470
{% extends "base.html" %}                                                   9471
{% block title %}多专家矩阵汇总{% endblock %}                               9472
{% block content %}                                                         9473
<div class="card">                                                          9474
    <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{%   9475
      else %}二级指标（{{ l1 }}）{% endif %}</h2>                           9476
    {% if not result %}                                                     9477
    <p class="muted">                                                       9478
      该范围内尚未录入专家矩阵。                                            9479
      <a href="/ahp/experts?unit_type={{ unit_type                          9480
      }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif                   9481
      %}">去录入</a>                                                        9482
    </p>                                                                    9483
    {% else %}                                                              9484
    <div class="stat-row">                                                  9485
      <div>                                                                 9486
        <div class="stat__label">参与汇总的专家</div>                       9487
        <div class="stat__value">{{ result.count }} 位</div>                9488
      </div>                                                                9489
      <div>                                                                 9490
        <div class="stat__label">λmax</div>                                 9491
        <div class="stat__value mono">{{ result.lambda_max                  9492
      }}</div>                                                              9493
      </div>                                                                9494
      <div>                                                                 9495
        <div class="stat__label">CI</div>                                   9496
        <div class="stat__value mono">{{                                    9497
      "%.4f"|format(result.ci) }}</div>                                     9498
      </div>                                                                9499
      <div>                                                                 9500
```

```
      <div class="stat__label">CR（要求 &lt; 0.1）</div>
      <div class="stat__value mono">{{ "%.4f"|format(result.cr) }}</div>
    </div>
    <div>
      <div class="stat__label">一致性检验</div>
      <div class="stat__value">{% if result.consistent %}<span class="ok">通过</span>{% else %}<span class="warn">不通过</span>{% endif %}</div>
    </div>
  </div>
  <h3>汇总后权重</h3>
  <table style="width:auto;">
    <thead>
      <tr>
        <th scope="col">指标</th>
        {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      <tr>
        <th scope="row">权重</th>
        {% for w in result.weights %}<td class="mono">{{ w }}</td>{% endfor %}
      </tr>
    </tbody>
  </table>
  <h3>各专家矩阵的一致性</h3>
  <table style="width:auto;">
    <thead>
      <tr><th scope="col">专家</th><th scope="col">该矩阵CR</th><th scope="col">判定</th></tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.expert_name }}</td>
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is not none else "—" }}</td>
        <td>
          {% if r.cr is not none and r.cr < 0.1 %}<span class="ok">通过</span>{% else %}<span class="warn">不通过</span>{% endif %}
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
```

```
  {% if not result.consistent and result.most_inconsistent %}
  <p style="padding:12px;border-radius:6px;background:var(--danger-soft);
            border-left:4px solid var(--danger);">
    <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{ result.most_inconsistent.j + 1 }} 列的比较
    （当前 <span class="mono">{{ result.most_inconsistent.value|round(3) }}</span>，
    理想约 <span class="mono">{{ result.most_inconsistent.expected|round(3) }}</span>）。
    该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。
  </p>
  {% endif %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <form method="post"
          action="/ahp/experts?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"
          style="display:inline;"
          onsubmit="return confirm('保存后，「{{ unit_type }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">
      {{ form.hidden_tag() }}
      <input type="hidden" name="action" value="save_agg">
      <button type="submit" class="btn" {% if not result.consistent %}disabled{% endif %}>
        保存为生效权重
      </button>
    </form>
    <a class="btn btn--ghost" href="/ahp/experts?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">返回录入</a>
    <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/indicator/index.html ====
{% extends "base.html" %}
{% block title %}指标数据采集{% endblock %}
{% block content %}
<div class="card">
  <h2>指标数据采集与管理</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/indicator/import">批量导入（CSV / Excel）</a>
  </p>
  {% if not patches %}
  <p class="muted">
    暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，
```

```
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
  </p>
  {% else %}
  {% for p in patches %}
  <div class="patch-panel">
    <div class="patch-panel__head">
      <h3 class="patch-panel__name">{{ p.name }}</h3>
      <span class="patch-panel__meta">
        所属项目：{{ p.project.name if p.project else "—" }}
      </span>
      <span class="patch-panel__meta">
        微地质单元：{{ p.units.count() }} 个
      </span>
      {% if p.units.count() %}
      <span class="patch-panel__meta patch-panel__tail">
        <a class="link-btn" href="/indicator/import">批量导入本图斑
    数据</a>
      </span>
      {% endif %}
    </div>
    {% if not p.units.count() %}
    <div class="notice">
      <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
      <p style="margin:0;">
        指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
        请先完成划分，再回到本页录入。
      </p>
      <p style="margin:10px 0 0;">
        <a class="btn btn--sm" href="/unit/{{ p.id
    }}">去划分微地质单元</a>
      </p>
    </div>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">微地质单元</th>
          <th scope="col">面积占比</th>
          <th scope="col">数据状态</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for u in p.units %}
        <tr>
          <td>{{ u.unit_type }}</td>
          <td class="num">
            {% if u.ratio is not none %}{{
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
```

```
            </td>                                                                    9651
            <td>                                                                     9652
              {% if u.indicator_data %}                                              9653
              <span class="tag tag--ok">已录入</span>                                 9654
              {% else %}                                                             9655
              <span class="tag tag--muted">未录入</span>                              9656
              {% endif %}                                                            9657
            </td>                                                                    9658
            <td class="nowrap">                                                      9659
              <span class="row-actions">                                             9660
                <a class="link-btn" href="/indicator/{{ u.id                         9661
    }}">录入 / 编辑</a>                                                              9662
              </span>                                                                9663
            </td>                                                                    9664
          </tr>                                                                      9665
          {% endfor %}                                                               9666
        </tbody>                                                                     9667
    </table>                                                                         9668
    {% endif %}                                                                      9669
  </div>                                                                             9670
  {% endfor %}                                                                       9671
  {% endif %}                                                                        9672
</div>                                                                               9673
{% endblock %}                                                                       9674
# ==== app/templates/indicator/edit.html ====                                        9675
{% extends "base.html" %}                                                            9676
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}                    9677
{% block content %}                                                                  9678
<div class="card" style="max-width:820px;">                                          9679
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>                                       9680
  <p class="muted">                                                                  9681
    输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。                9682
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。                  9683
  </p>                                                                               9684
  <form method="post">                                                               9685
    {{ form.hidden_tag() }}                                                          9686
    {% for group_name, fields, has_hm in groups %}                                   9687
    <h3>{{ group_name }}</h3>                                                        9688
    {% for fname, cname, unit_ in fields %}                                          9689
    <div class="field">                                                              9690
      <label class="field__label" for="f-{{ fname }}">                               9691
        {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}                          9692
      </label>                                                                       9693
      {% if fname == "aspect" %}                                                     9694
      <select id="f-{{ fname }}" name="aspect">                                      9695
        <option value="">— 不适用 —</option>                                         9696
        {% for a in aspect_types %}                                                  9697
        <option value="{{ a }}" {% if current.get("aspect")                          9698
    == a %}selected{% endif %}>{{ a }}</option>                                      9699
        {% endfor %}                                                                 9700
```

```
        </select>
        {% else %}
        <input id="f-{{ fname }}" type="number" step="any"
    name="{{ fname }}"
               value="{{ current.get(fname) if
    current.get(fname) is not none else '' }}">
        {% endif %}
    </div>
    {% endfor %}
    {% endfor %}
    {% for group_name, fields, has_hm in groups %}
        {% if has_hm %}
        <h3>重金属含量（mg/kg）</h3>
        <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>
        <table style="width:auto;">
          <thead>
            <tr>
              {% for m in heavy_metals %}<th scope="col">{{ m
    }}</th>{% endfor %}
            </tr>
          </thead>
          <tbody>
            <tr>
              {% for m in heavy_metals %}
              <td>
                <label class="muted" style="position:absolute;
    left:-9999px;"
                       for="hm-{{ m }}">{{ m }} 含量</label>
                <input id="hm-{{ m }}" type="number"
    step="any" name="hm_{{ m }}"
                       value="{{ current.get('heavy',
    {}).get(m) if current.get('heavy', {}).get(m) is not
    none else '' }}">
              </td>
              {% endfor %}
            </tr>
          </tbody>
        </table>
        {% endif %}
    {% endfor %}
    <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/indicator/">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/indicator/import.html ====
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
```

```
{% block content %}
<div class="card" style="max-width:860px;">
  <h2>批量导入指标数据</h2>
  <p class="muted">
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。
    先下载模板按格式填写，可减少出错。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
    href="/indicator/import/template">下载导入模板（CSV）</a>
    <a class="btn btn--sm btn--ghost"
    href="/indicator/">返回指标数据</a>
  </p>
  <form method="post" enctype="multipart/form-data">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="patch-id">目标矿山图斑</lab
    el>
      <select id="patch-id" name="patch_id" required>
        <option value="">— 请选择 —</option>
        {% for p in patches %}
        <option value="{{ p.id }}">{{ p.name }} ({{
    p.units.count() }} 个单元）</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="datafile">数据文件</label
    >
      <input id="datafile" type="file" name="datafile"
             accept=".csv,.xlsx,.xls" required>
    </div>
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">开始导入</button>
    </p>
  </form>
  <h3>模板列说明</h3>
  <p class="muted">
    共 {{ headers|length }} 列。留空表示该指标「不适用」。
    坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
    （按论文注释视为未受污染）。
  </p>
  <p class="mono muted" style="line-height:1.9;">{{
    headers|join(" , ") }}</p>
  <h3>导入规则</h3>
  <ul class="muted" style="padding-left:20px;line-height:1.9
    ;">
    <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
    <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
```

```
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>                    9801
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非     9802
      <span class="mono">10cm</span>）。</li>                                    9803
    </ul>                                                                       9804
</div>                                                                          9805
{% if report %}                                                                 9806
<div class="card" style="max-width:860px;">                                     9807
    <h2>导入结果</h2>                                                           9808
    <div class="stat-row">                                                      9809
      <div>                                                                     9810
        <div class="stat__label">目标图斑</div>                                 9811
        <div style="font-size:15px;">{{ report.patch }}</div>                   9812
      </div>                                                                    9813
      <div>                                                                     9814
        <div class="stat__label">成功导入</div>                                 9815
        <div class="stat__value ok">{{ report.saved }}                          9816
      个单元</div>                                                              9817
      </div>                                                                    9818
      <div>                                                                     9819
        <div class="stat__label">问题行</div>                                   9820
        <div class="stat__value {% if report.errors %}warn{%                    9821
      endif %}">{{ report.errors|length }}</div>                                9822
      </div>                                                                    9823
    </div>                                                                      9824
    {% if report.errors %}                                                      9825
    <h3>未导入的行</h3>                                                         9826
    <ul class="muted" style="padding-left:20px;line-height:1.9                  9827
      ;">                                                                       9828
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}                  9829
    </ul>                                                                       9830
    {% else %}                                                                  9831
    <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>         9832
    {% endif %}                                                                 9833
    <p class="actions" style="margin-top:16px;">                                9834
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>             9835
    </p>                                                                        9836
</div>                                                                          9837
{% endif %}                                                                     9838
{% endblock %}                                                                  9839
# ==== app/templates/evaluate/index.html ====                                   9840
{% extends "base.html" %}                                                       9841
{% block title %}适宜性定量评价{% endblock %}                                   9842
{% block content %}                                                             9843
<div class="card">                                                              9844
    <h2>适宜性定量评价</h2>                                                     9845
    {% if not patches %}                                                        9846
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>      9847
    {% else %}                                                                  9848
    <table>                                                                     9849
      <thead>                                                                   9850
```

```
        <tr>                                                                    9851
          <th scope="col">所属项目</th>                                          9852
          <th scope="col">矿山图斑</th>                                          9853
          <th scope="col">前置条件</th>                                          9854
          <th scope="col">最近评价结果</th>                                      9855
          <th scope="col">操作</th>                                              9856
        </tr>                                                                   9857
      </thead>                                                                  9858
      <tbody>                                                                   9859
        {% for p in patches %}                                                  9860
        {% set pr = progress.get(p.id) %}                                       9861
        <tr>                                                                    9862
          <td>{{ p.project.name if p.project else "—" }}</td>                   9863
          <td style="text-align:left;">{{ p.name }}</td>                        9864
          <td style="text-align:left;">                                         9865
            {% if pr and pr.ready %}                                            9866
            <span class="tag tag--ok">已具备</span>                             9867
            {% elif pr %}                                                       9868
            <span class="tag tag--warn">缺：{{                                  9869
      pr.blocking[0].label }}</span>                                            9870
            <span class="muted">{{ pr.blocking[0].reason                        9871
      }}</span>                                                                 9872
            {% endif %}                                                         9873
          </td>                                                                 9874
          <td>                                                                  9875
            {% if latest.get(p.id) %}                                           9876
            F = <span class="mono num">{{                                       9877
      latest[p.id].final_score }}</span>,                                       9878
            <span class="tag tag--ok">{{ latest[p.id].grade                     9879
      }}</span>                                                                 9880
            {% else %}                                                          9881
            <span class="tag tag--muted">未评价</span>                          9882
            {% endif %}                                                         9883
          </td>                                                                 9884
          <td class="nowrap">                                                   9885
            <span class="row-actions">                                          9886
              <a class="link-btn" href="/evaluate/{{ p.id                       9887
      }}">执行评价</a>                                                          9888
              {% if latest.get(p.id) %}                                         9889
              <a class="link-btn" href="/evaluate/result/{{                     9890
      latest[p.id].id }}">查看结果</a>                                          9891
              {% endif %}                                                       9892
            </span>                                                             9893
          </td>                                                                 9894
        </tr>                                                                   9895
        {% endfor %}                                                            9896
      </tbody>                                                                  9897
    </table>                                                                    9898
    {% endif %}                                                                 9899
</div>                                                                          9900
```

```
{% endblock %}
# ==== app/templates/evaluate/run.html ====
{% extends "base.html" %}
{% import "_macros.html" as ui %}
{% block title %}执行评价 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:880px;">
  <h2>执行适宜性评价 · {{ patch.name }}</h2>
  {{ ui.workflow_steps(progress) }}
  {{ ui.workflow_blocking(progress) }}
  {% if progress.ready %}
  <form method="post">
    {{ form.hidden_tag() }}
    <p class="actions">
      <button type="submit" class="btn">开始评价</button>
      <a class="btn btn--ghost" href="/evaluate/">返回</a>
    </p>
  </form>
  {% else %}
  <p class="actions">
    <button type="button" class="btn" disabled>开始评价</button>
    <a class="btn btn--ghost" href="/evaluate/">返回</a>
  </p>
  {% endif %}
</div>
<div class="card" style="max-width:880px;">
  <h2>当前流程状态</h2>
  <table>
    <thead>
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th
    scope="col">说明</th></tr>
    </thead>
    <tbody>
      <tr>
        <td>限制性条件判定</td>
        <td>
          {% if progress.constraint.passed %}<span
    class="tag tag--ok">已通过</span>
          {% elif progress.constraint.done %}<span
    class="tag tag--warn">未通过</span>
          {% else %}<span class="tag tag--muted">待完成</span>{
    % endif %}
        </td>
        <td style="text-align:left;">{{
    progress.constraint.detail }}</td>
      </tr>
      <tr>
        <td>划分微地质单元</td>
        <td>
          {% if progress.units.done %}<span class="tag
```

```
      tag--ok">已完成</span>                                                    9951
            {% else %}<span class="tag tag--muted">待完成</span>{ 9952
      % endif %}                                                               9953
          </td>                                                                9954
          <td style="text-align:left;">{{                                     9955
      progress.units.detail }}</td>                                            9956
        </tr>                                                                  9957
        <tr>                                                                   9958
          <td>录入指标数据</td>                                                 9959
          <td>                                                                 9960
            {% if progress.data.ready %}<span class="tag                       9961
      tag--ok">已完成</span>                                                   9962
            {% else %}<span class="tag tag--muted">待完成</span>{ 9963
      % endif %}                                                               9964
          </td>                                                                9965
          <td style="text-align:left;">{{                                     9966
      progress.data.detail }}</td>                                             9967
        </tr>                                                                  9968
      </tbody>                                                                 9969
    </table>                                                                   9970
</div>                                                                         9971
{% if weight_sources %}                                                        9972
<div class="card" style="max-width:880px;">                                    9973
    <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">            9974
      <thead>                                                                  9975
        <tr><th scope="col">微地质单元</th><th                                 9976
      scope="col">权重来源</th></tr>                                           9977
      </thead>                                                                 9978
      <tbody>                                                                  9979
        {% for ut, label in weight_sources.items() %}                          9980
        <tr>                                                                   9981
          <td style="text-align:left;">{{ ut }}</td>                           9982
          <td style="text-align:left;">                                        9983
            {% if "自定义" in label %}                                         9984
            <span class="tag tag--warn">{{ label }}</span>                     9985
            {% else %}                                                         9986
            <span class="tag tag--muted">{{ label }}</span>                    9987
            {% endif %}                                                        9988
          </td>                                                                9989
        </tr>                                                                  9990
        {% endfor %}                                                           9991
      </tbody>                                                                 9992
    </table>                                                                   9993
</div>                                                                         9994
{% endif %}                                                                    9995
{% endblock %}                                                                 9996
# ==== app/templates/evaluate/result.html ====                                 9997
{% extends "base.html" %}                                                      9998
{% block title %}评价结果{% endblock %}                                        9999
{% block content %}                                                           10000
```

```
<div class="card">
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">最终评价因子 F</div>
      <div class="stat__value mono">{{ r.final_score }}</div>
    </div>
    <div>
      <div class="stat__label">适宜性等级</div>
      <div class="stat__value ok">{{ r.grade }}</div>
    </div>
    <div style="flex:1;min-width:240px;">
      <div class="stat__label">推荐措施</div>
      <div style="font-size:15px;">{{ r.measures }}</div>
    </div>
  </div>
  {% if unit_scores %}
  <h3>各微地质单元得分</h3>
  <table style="max-width:860px;">
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">单元评价因子 F′</th>
        <th scope="col">加权贡献</th>
        <th scope="col">权重来源</th>
      </tr>
    </thead>
    <tbody>
      {% for u in unit_scores %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ "%.2f%%"|format(u.ratio * 100) }}</td>
        <td class="mono">{{ u.unit_score }}</td>
        <td class="mono">{{ "%.4f"|format(u.ratio * u.unit_score) }}</td>
        <td class="muted">{{ u.weight_source or "论文表 3 默认权重" }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
  <p class="actions" style="margin-top:18px;">
    <a class="btn btn--ghost" href="/report/{{ r.patch_id }}">查看得分明细</a>
    <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>
  </p>
```

```
</div>                                                
```

```
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/report/detail.html ====
{% extends "base.html" %}
{% block title %}得分明细 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
    <h2>得分明细 · {{ patch.name }}</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">最终评价因子 F</div>
        <div class="stat__value mono">{{
      "%.4f"|format(result.final_score) }}</div>
      </div>
      <div>
        <div class="stat__label">适宜性等级</div>
        <div class="stat__value ok">{{ result.grade }}</div>
      </div>
      <div style="flex:1;min-width:240px;">
        <div class="stat__label">推荐措施</div>
        <div style="font-size:15px;">{{
      result.grade_info.measure }}</div>
      </div>
    </div>
    <p class="muted" style="margin-top:12px;">
      本页与导出报告均取自
      <strong>{% if record %}{{ record.created_at |
      localdt("%Y-%m-%d %H:%M:%S") }}{% endif %}
      记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else
      "—" }}），
      不会因事后修改指标数据而变化。
    </p>
    <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF
      报告</a>
      <a class="btn btn--ghost" href="/report/{{ patch.id
      }}/word">导出 Word 报告</a>
      <a class="btn btn--ghost" href="/report/">返回</a>
    </p>
</div>
<div class="card">
    <h2>结果图表</h2>
    <h3>各微地质单元得分对比</h3>
    <div>{{ chart_units|safe }}</div>
    <h3>各一级指标平均得分</h3>
    <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>
```

```
    <div>{{ chart_weight|safe }}</div>
</div>
{% for u in result.unit_results %}
<div class="card">
    <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *
      100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>
    {% if u.weight_source %}
    <p class="muted" style="margin-top:-6px;">权重来源：{{
      u.weight_source }}</p>
    {% endif %}
    <table>
      <thead>
        <tr>
          <th scope="col">一级指标</th>
          <th scope="col">二级指标</th>
          <th scope="col">实测值</th>
          <th scope="col">标准化值</th>
          <th scope="col">权重</th>
          <th scope="col">贡献</th>
        </tr>
      </thead>
      <tbody>
        {% for d in u.details %}
        <tr>
          <td>{{ d.l1 }}</td>
          <td style="text-align:left;">{{ d.l2 }}</td>
          <td class="mono" style="text-align:left;">{{ "—" if
      d.raw is none else d.raw }}</td>
          <td class="mono">{{ "—" if d.std_value is none else
      "%.4f"|format(d.std_value) }}</td>
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
          <td class="mono">{{ "—" if d.contribution is none
      else "%.4f"|format(d.contribution) }}</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="muted" style="margin:0;">
      一级指标得分：
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span
      class="num">{{ "%.4f"|format(v) }}</span>{% if not
      loop.last %}；{% endif %}{% endfor %}
    </p>
</div>
{% endfor %}
{% endblock %}
# ==== app/templates/report/print.html ====
<!DOCTYPE html>
<html lang="zh-CN">
<head>
```

```
<meta charset="UTF-8">
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>
<style>
  @page { size: A4; margin: 18mm 16mm; }
  * { box-sizing: border-box; }
  body {
    margin: 0; color: #23262b; background: #fff;
    font-family: "Microsoft YaHei", "SimSun", sans-serif;
    font-size: 10.5pt; line-height: 1.65;
  }
  h1 { font-size: 17pt; text-align: center; margin: 0 0
    6px; }
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;
       border-bottom: 1px solid #dfe3e8; padding-bottom:
    4px; }
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0
    6px; }
  .sub { text-align: center; color: #6b7280; font-size:
    9.5pt; margin-bottom: 18px; }
  table { width: 100%; border-collapse: collapse; margin:
    8px 0 14px;
          font-variant-numeric: tabular-nums; }
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;
    text-align: center; }
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace;
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} · {{ patch.admin_region
    }}{% endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１.０ ·
    生成时间 {{ generated_at | localdt }}
  </p>
  <h2>一、图斑基本信息</h2>
  <table class="kv">
    <tr><th>图斑名称</th><td>{{ patch.name }}</td></tr>
```

```
    <tr><th>所在行政区</th><td>{{ patch.admin_region or "—"
    }}</td></tr>
    <tr><th>图斑面积（公顷）</th><td>{{ patch.area_ha if
    patch.area_ha is not none else "—" }}</td></tr>
    <tr><th>开采矿种</th><td>{{ patch.mineral_type or "—"
    }}</td></tr>
    <tr><th>开采方式</th><td>{{ patch.mining_method or "—"
    }}</td></tr>
    <tr><th>责任主体缺失</th><td>{{ "是" if patch.responsible_entit
    y_missing else "否" }}</td></tr>
    {% if reference %}
    <tr><th>参照生态系统</th>
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%,
            多样性指数基准 {{ reference.shannon_benchmark
    }}</td></tr>
    {% endif %}
</table>
<h2>二、评价结论</h2>
<table class="kv">
    <tr><th>最终评价因子 F</th><td class="mono"><strong>{{
    "%.4f"|format(result.final_score) }}</strong></td></tr>
    <tr><th>适宜性等级</th><td><strong>{{ result.grade
    }}</strong></td></tr>
    <tr><th>判定依据</th><td>{{ result.grade_info.meaning
    }}</td></tr>
    <tr><th>推荐措施</th><td>{{ result.grade_info.measure
    }}</td></tr>
</table>
<h2>三、各微地质单元得分</h2>
<table>
    <thead>
      <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子
    F'</th><th>加权贡献</th></tr>
    </thead>
    <tbody>
      {% for u in result.unit_results %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
    }}</td>
        <td class="mono">{{ "%.4f"|format(u.unit_score)
    }}</td>
        <td class="mono">{{ "%.4f"|format(u.weighted)
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
</table>
<div class="chart">{{ chart_units|safe }}</div>
<h2>四、一级指标权重分布</h2>
```

```
    <div class="chart">{{ chart_weight|safe }}</div>
    <h2>五、得分明细</h2>
    {% for u in result.unit_results %}
    <h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *
      100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
    <table>
      <thead>
        <tr>
          <th>一级指标</th><th>二级指标</th><th>实测值</th>
          <th>标准化值</th><th>权重</th><th>贡献</th>
        </tr>
      </thead>
      <tbody>
        {% for d in u.details %}
        <tr>
          <td>{{ d.l1 }}</td>
          <td class="left">{{ d.l2 }}</td>
          <td class="mono left">{{ "—" if d.raw is none else
      d.raw }}</td>
          <td class="mono">{{ "—" if d.std_value is none else
      "%.4f"|format(d.std_value) }}</td>
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
          <td class="mono">{{ "—" if d.contribution is none
      else "%.4f"|format(d.contribution) }}</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="note">
      一级指标得分：
      {% for k, v in u.l1_scores.items() %}{{ k }} = {{
      "%.4f"|format(v) }}{% if not loop.last %}；{% endif %}{%
      endfor %}
    </p>
    {% endfor %}
    <h2>六、评价依据说明</h2>
    <ul class="note">
      <li>指标标准化按论文式（８） 执行，按指标性质分为越大越优、越小越优、
          适中型与离散赋值四类。</li>
      <li>单元评价因子按论文式（９） 由二级指标权重加权求得；
          整矿评价因子由各微地质单元按面积占比加权合成。</li>
      <li>适宜性等级按论文表 4 划分：F&lt;０．３ 不适宜、０．３≤F≤０．７ 基本适宜、F≥０．７
      适宜。</li>
      <li>指标权重取自论文表 3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>
    </ul>
    <p class="sig">
      评价人（签字）：＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿　　日期：＿＿＿＿＿＿＿＿＿＿＿＿
    </p>
</body>
</html>
```

```
# ==== app/templates/search.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="card">
  <h2>搜索结果</h2>
  {% if not q %}
  <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区 / 矿种、
    标准名称与文号。</p>
  {% else %}
  <p class="muted">
    关键词「<strong>{{ q }}</strong>」共找到
    <span class="num">{{ total }}</span> 条结果。
  </p>
  {% if not total %}
  <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>
  {% endif %}
  {% endif %}
</div>
{% if projects %}
<div class="card">
  <h2>评价项目（{{ projects|length }}）</h2>
  <table>
    <thead>
      <tr>
        <th scope="col">项目名称</th>
        <th scope="col">编号</th>
        <th scope="col">状态</th>
        <th scope="col">图斑数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in projects %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td class="mono">{{ p.code or "—" }}</td>
        <td><span class="tag tag--muted">{{ p.status
    }}</span></td>
        <td class="num">{{ p.patches.count() }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/project/{{ p.id
    }}/patches">矿山图斑</a>
            <a class="link-btn" href="/project/{{ p.id
    }}/edit">编辑</a>
          </span>
        </td>
      </tr>
```

```
        {% endfor %}                                                                    10401
      </tbody>                                                                          10402
    </table>                                                                            10403
</div>                                                                                  10404
{% endif %}                                                                             10405
{% if patches %}                                                                        10406
<div class="card">                                                                      10407
  <h2>矿山图斑（{{ patches|length }}）</h2>                                             10408
  <table>                                                                               10409
    <thead>                                                                             10410
      <tr>                                                                              10411
        <th scope="col">所属项目</th>                                                   10412
        <th scope="col">图斑名称</th>                                                   10413
        <th scope="col">行政区</th>                                                     10414
        <th scope="col">矿种</th>                                                       10415
        <th scope="col">操作</th>                                                       10416
      </tr>                                                                             10417
    </thead>                                                                            10418
    <tbody>                                                                             10419
      {% for m in patches %}                                                            10420
      <tr>                                                                              10421
        <td>{{ m.project.name if m.project else "—" }}</td>                             10422
        <td style="text-align:left;">{{ m.name }}</td>                                  10423
        <td>{{ m.admin_region or "—" }}</td>                                            10424
        <td>{{ m.mineral_type or "—" }}</td>                                            10425
        <td class="nowrap">                                                             10426
          <span class="row-actions">                                                    10427
            <a class="link-btn" href="/constraint/{{ m.id                               10428
    }}">限制性判定</a>                                                                  10429
            <a class="link-btn" href="/unit/{{ m.id                                     10430
    }}">微地质单元</a>                                                                  10431
            <a class="link-btn" href="/evaluate/{{ m.id                                 10432
    }}">执行评价</a>                                                                    10433
          </span>                                                                       10434
        </td>                                                                           10435
      </tr>                                                                             10436
      {% endfor %}                                                                      10437
    </tbody>                                                                            10438
  </table>                                                                              10439
</div>                                                                                  10440
{% endif %}                                                                             10441
{% if standards %}                                                                      10442
<div class="card">                                                                      10443
  <h2>标准与政策文件（{{ standards|length }}）</h2>                                     10444
  <table>                                                                               10445
    <thead>                                                                             10446
      <tr>                                                                              10447
        <th scope="col">类别</th>                                                       10448
        <th scope="col">省份</th>                                                       10449
        <th scope="col">名称</th>                                                       10450
```

```
            <th scope="col">文号</th>                                                    10451
            <th scope="col">操作</th>                                                    10452
          </tr>                                                                          10453
        </thead>                                                                         10454
        <tbody>                                                                          10455
          {% for s in standards %}                                                       10456
          <tr>                                                                           10457
            <td>{{ s.category }}</td>                                                    10458
            <td>{{ s.province }}</td>                                                    10459
            <td style="text-align:left;">{{ s.name }}</td>                               10460
            <td class="mono">{{ s.doc_number or "—" }}</td>                              10461
            <td class="nowrap">                                                          10462
              <span class="row-actions">                                                 10463
                <a class="link-btn" href="/standard/{{ s.id                              10464
      }}/rules">规则</a>                                                                 10465
              </span>                                                                    10466
            </td>                                                                        10467
          </tr>                                                                          10468
          {% endfor %}                                                                   10469
        </tbody>                                                                         10470
      </table>                                                                           10471
</div>                                                                                   10472
{% endif %}                                                                              10473
{% endblock %}                                                                           10474
#  ==== seed_demo.py ====                                                               10475
#  -*- coding: utf-8 -*-                                                                10476
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。                                    10477
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。                                10478
运行：python seed_demo.py                  幂等，只补写缺失的演示数据，                  10479
                                               不会删除用户自建的项目与图斑              10480
      python seed_demo.py --reset          连同标准库与指标定义一并重建                  10481
"""                                                                                      10482
import json                                                                              10483
import sys                                                                               10484
sys.stdout.reconfigure(encoding="utf-8")                                                 10485
from app import create_app                                                               10486
from app.models import (db, User, Project, MinePatch,                                    10487
    ReferenceEcosystem,                                                                  10488
                            MicroUnit, IndicatorData, Standard,                          10489
    StandardRule,                                                                        10490
                            ConstraintCheck, EvaluationResult,                           10491
    ScoreDetail)                                                                         10492
app = create_app()                                                                       10493
#  内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」                        10494
DEMO_CODE = "DEMO-001"                                                                   10495
#  论文表 1：11 项地方标准与政策文件                                                    10496
STANDARDS = [                                                                            10497
    dict(category="地方标准", province="辽宁",                                           10498
    name="矿山废弃地植被自然恢复技术规程",                                               10499
         doc_number="DB21/T 3702—2023",                                                  10500
```

```
        publish_date="2023-03-30",                                              10501
        implement_date="2023-04-30",                                            10502
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",                    10503
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),                      10504
               ("无地质灾害", "==", "是", "必要条件", None),                      10505
               ("无水土污染", "==", "是", "必要条件", None),                      10506
               ("与参照系差距", "<", "20", "充分条件", "%")]),                    10507
    dict(category="地方标准", province="湖北",                                   10508
    name="废弃露天矿山生态环境自然恢复标准",                                        10509
        doc_number="DB42/T 2421—2025",                                          10510
    publish_date="2025-08-20",                                                  10511
        implement_date="2025-10-20",                                            10512
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",                          10513
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),             10514
    dict(category="地方标准", province="内蒙古",                                 10515
    name="历史遗留废弃矿山自然恢复判定规范",                                        10516
        doc_number="DB15/T 4280—2026",                                          10517
    publish_date="2026-01-30",                                                  10518
        implement_date="2026-03-01",                                            10519
        summary="调查内容包含参照生态系统调查",                                    10520
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),             10521
               ("地形地貌与周边协调", "==", "是", "必要条件", None),              10522
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),        10523
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),            10524
    dict(category="地方标准", province="江西",                                   10525
    name="废弃矿山自然恢复技术规范",                                              10526
        doc_number="DB36/T 2221—2026",                                          10527
    publish_date="2026-02-02",                                                  10528
        implement_date="2026-08-01",                                            10529
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",                    10530
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),                 10531
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),          10532
    dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",        10533
        doc_number=None, publish_date="2026-06-18",                             10534
    implement_date=None,                                                        10535
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",                  10536
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),                  10537
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),        10538
               ("与参照系差距", "<", "20", "充分条件", "%")]),                    10539
    dict(category="政策文件", province="安徽",                                   10540
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",      10541
        doc_number="皖自然资修函[2020]34号",                                     10542
    publish_date="2020-03-03", implement_date=None,                             10543
        summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",10544
        rules=[("责任主体", "==", "政府", "必要条件", None),                      10545
               ("无地质灾害", "==", "是", "必要条件", None),                      10546
               ("无水土污染", "==", "是", "必要条件", None)]),                    10547
    dict(category="政策文件", province="山东",                                   10548
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",10549
        doc_number="鲁自然资字[2022]104号",                                      10550
```

```
        publish_date="2022-08-02", implement_date=None,
            summary="条件与安徽省相同，另增加自然恢复认定标准",
            rules=[("无地质灾害", "==", "是", "必要条件", None),
                   ("无水土污染", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="湖南",
            name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",
            doc_number="湘自资办发[２０２３]５８号",
        publish_date="2023-07-20", implement_date=None,
            summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度
        >７５%、连续３年正向增长",
            rules=[("植被覆盖度", ">", "７５", "必要条件", "%"),
                   ("植被覆盖率趋势", "==", "连续３年正向增长", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None),
                   ("符合国土空间规划", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="新疆",
            name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",
            doc_number="新自然资源办函〔２０２４〕２１９号",
        publish_date="2024-12-20", implement_date=None,
            summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
            rules=[("植被覆盖度", ">=", "周边３０%", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None),
                   ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="广西",
            name="广西壮族自治区历史遗留矿山自然恢复技术指南",
            doc_number="桂自然资办〔２０２４〕２５０号",
        publish_date="2024-12-30", implement_date=None,
            summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
            rules=[("无地质灾害", "==", "是", "必要条件", None),
                   ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                   ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="西藏",
            name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
            doc_number="藏自然资发〔２０２５〕１９号",
        publish_date="2025-09-24", implement_date=None,
            summary="含修复成本高、生态修复效益不显著等适用条件",
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
```

```
            {                                                                   10601
                "unit_type": "碎石边坡", "ratio": 0.6295,                       10602
                "data": {"soil_ratio": 0.34, "thickness": 10,                   10603
    "ph": 9.24,                                                                 10604
                         "organic_matter": 4.89,                                10605
                         "heavy_metals": {"Hg": 0.05, "As":                     10606
    5.17, "Ni": 12.60,                                                          10607
                                              "Cu": 11.70, "Cd":                10608
    0.04, "Pb": 6.50},                                                          10609
                         "coverage": 87.92, "shannon": 0.60,                    10610
                         "slope": 28.75, "aspect": "阳坡"},                     10611
            },                                                                  10612
            {                                                                   10613
                "unit_type": "平台", "ratio": 0.2319,                           10614
                "data": {"soil_ratio": 0.56, "thickness": 30,                   10615
    "ph": 8.72,                                                                 10616
                         "organic_matter": 10.92,                               10617
                         "heavy_metals": {"Hg": 0.02, "As":                     10618
    3.13, "Ni": 12.33,                                                          10619
                                              "Cu": 11.14, "Cd":                10620
    0.09, "Pb": 9.98},                                                          10621
                         "coverage": 58.42, "shannon": 0.72},                   10622
            },                                                                  10623
        ],                                                                      10624
}                                                                               10625
def clean_case(reset=False):                                                    10626
    """清理数据。                                                               10627
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰               10628
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，       10629
    以便把演示环境恢复到初始状态。                                             10630
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目           10631
    会在每次启动时被删除。                                                     10632
    """                                                                         10633
    from app.models import EvaluationResult, ScoreDetail,                       10634
    ConstraintCheck                                                             10635
    demo_projects = Project.query.filter_by(code=DEMO_CODE)                     10636
    all()                                                                       10637
    demo_ids = [p.id for p in demo_projects]                                    10638
    patch_ids = []                                                              10639
    if demo_ids:                                                                10640
        patch_ids = [p.id for p in MinePatch.query.filter(                      10641
            MinePatch.project_id.in_(demo_ids)).all()]                          10642
    if patch_ids:                                                               10643
        unit_ids = [u.id for u in MicroUnit.query.filter(                       10644
            MicroUnit.patch_id.in_(patch_ids)).all()]                           10645
        if unit_ids:                                                            10646
            IndicatorData.query.filter(                                         10647
                IndicatorData.unit_id.in_(unit_ids)).delete(                    10648
                synchronize_session=False)                                      10649
        result_ids = [r.id for r in EvaluationResult.query.f                    10650
```

```
    ilter(                                                              10651
            EvaluationResult.patch_id.in_(patch_ids)).all()]            10652
        if result_ids:                                                  10653
            ScoreDetail.query.filter(                                   10654
                ScoreDetail.result_id.in_(result_ids)).delet            10655
    e(                                                                  10656
                synchronize_session=False)                              10657
        EvaluationResult.query.filter(                                  10658
            EvaluationResult.patch_id.in_(patch_ids)).delete            10659
    (                                                                   10660
            synchronize_session=False)                                  10661
        ConstraintCheck.query.filter(                                   10662
            ConstraintCheck.patch_id.in_(patch_ids)).delete(            10663
            synchronize_session=False)                                  10664
        ReferenceEcosystem.query.filter(                                10665
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele            10666
    te(                                                                 10667
            synchronize_session=False)                                  10668
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_            10669
    ids)).delete(                                                       10670
            synchronize_session=False)                                  10671
        MinePatch.query.filter(MinePatch.id.in_(patch_ids))             10672
    delete(                                                             10673
            synchronize_session=False)                                  10674
    if demo_ids:                                                        10675
        Project.query.filter(Project.id.in_(demo_ids)).delet            10676
    e(                                                                  10677
            synchronize_session=False)                                  10678
    if reset:                                                           10679
        StandardRule.query.delete()                                     10680
        Standard.query.delete()                                         10681
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，                  10682
    #  避免长期运行后 score_details 表堆积无效数据。                   10683
    _purge_orphan_details()                                             10684
    db.session.commit()                                                 10685
    return len(demo_ids)                                                10686
def demo_patch():                                                       10687
    """取内置演示案例的图斑。                                           10688
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），               10689
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。       10690
    没有演示案例时才退回第一个图斑。                                   10691
    """                                                                 10692
    proj = Project.query.filter_by(code=DEMO_CODE).first()              10693
    if proj is not None:                                                10694
        m = MinePatch.query.filter_by(project_id=proj.id).fi            10695
    rst()                                                               10696
        if m is not None:                                               10697
            return m                                                    10698
    return MinePatch.query.first()                                      10699
def _purge_orphan_details():                                            10700
```

```
    """清理指向已不存在评价记录的孤儿得分明细。
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。
    """
    live_ids = [r.id for r in EvaluationResult.query.all()]
    if live_ids:
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(
    live_ids)).delete(
            synchronize_session=False)
    else:
        ScoreDetail.query.delete()
    db.session.commit()
def seed(reset=False):
    with app.app_context():
        db.create_all()          # 全新环境下先建表，保证本脚本可独立运行
        _purge_orphan_details()
        admin = User.query.filter_by(username="admin").first
    ()
        if not admin:
            from werkzeug.security import
    generate_password_hash
            admin = User(username="admin",
                         password_hash=generate_password_has
    h("admin123"),
                         role="admin")
            db.session.add(admin)
            db.session.commit()
        # 幂等：演示案例已存在就不重复写入，更不清空用户自己的项目
        existing = Project.query.filter_by(code=DEMO_CODE).f
    irst()
        if existing and not reset:
            print("演示案例已存在，跳过写入（用户自建项目不受影响）")
            _report()
            _seed_reference_data(reset=False)
            return
        clean_case(reset=reset)
        project = Project(name=CASE["name"], code=DEMO_CODE,
                          description="论文第 5 节典型案例",
    created_by=admin.id)
        db.session.add(project)
        db.session.flush()
        patch = MinePatch(project_id=project.id,
    name=CASE["patch_name"],
                          admin_region="河北省唐山市玉田县",
                          area_ha=CASE["area_ha"],
                          mineral_type=CASE["mineral_type"],
                          mining_method="露天开采",
    stop_years=None,
                          responsible_entity_missing=True)
        db.session.add(patch)
        db.session.flush()
```

```
        ref = ReferenceEcosystem(patch_id=patch.id,
                                 coverage_benchmark=75.0,
    shannon_benchmark=0.7)
        db.session.add(ref)
        for u in CASE["units"]:
            unit = MicroUnit(patch_id=patch.id,
    unit_type=u["unit_type"],
                             ratio=u["ratio"], area=None)
            db.session.add(unit)
            db.session.flush()
            d = u["data"]
            ind = IndicatorData(
                unit_id=unit.id,
                soil_ratio=d.get("soil_ratio"),
                thickness=d.get("thickness"),
                ph=d.get("ph"),
                organic_matter=d.get("organic_matter"),
                heavy_metals=json.dumps(d["heavy_metals"],
    ensure_ascii=False)
                              if d.get("heavy_metals") else
    None,
                coverage=d.get("coverage"),
                shannon=d.get("shannon"),
                slope=d.get("slope"),
                aspect=d.get("aspect"),
            )
            db.session.add(ind)
        #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、
        #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。
        db.session.add(ConstraintCheck(
            patch_id=patch.id,
            policy_pass=True, location_pass=True,
    pollution_pass=True,
            geo_pass=True, farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        print("已写入案例：", CASE["name"])
        print("  图斑：", CASE["patch_name"])
        print("  微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台
    23.19%")
        print("  管理员账号：admin / admin123")
        _seed_reference_data(reset=reset)
        _report()
def _seed_reference_data(reset=False):
    """写入标准库与指标定义。
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护
    的标准条目与指标定义。
    """
    if reset or Standard.query.count() == 0:
        seed_standards()
```

```
    from app.services.indicator_service import
    seed_indicators
    n = seed_indicators()
    print("指标定义：%d 项" % n)
def _report():
    total_p = Project.query.count()
    total_patch = MinePatch.query.count()
    print()
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))
    if total_p > total_patch:
        print("  提示：项目是归档单位，评价针对的是图斑。")
        print("        新项目需先建立图斑，才会出现在后续各模块中。")
def seed_standards():
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""
    StandardRule.query.delete()
    Standard.query.delete()
    db.session.commit()
    for s in STANDARDS:
        std = Standard(category=s["category"],
    province=s["province"],
                       name=s["name"],
    doc_number=s["doc_number"],
                       publish_date=s["publish_date"],
                       implement_date=s["implement_date"],
                       summary=s["summary"])
        db.session.add(std)
        db.session.flush()
        for item, op, thr, ctype, note in s["rules"]:
            db.session.add(StandardRule(standard_id=std.id,
    item=item,
                                        operator=op,
    threshold=thr,

    condition_type=ctype, note=note))
    db.session.commit()
    print("已写入标准/政策文件：%d 项，规则 %d 条"
          % (len(STANDARDS), sum(len(s["rules"]) for s in
    STANDARDS)))
if __name__ == "__main__":
    #  --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。
    #  无论是否 --reset，都不会删除用户自建的项目与图斑。
    seed(reset="--reset" in sys.argv[1:])
#  ==== 测试/test_ahp.py ====
#  -*- coding: utf-8 -*-
"""地基自检：验证 AHP 引擎与应用可启动。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import numpy as np
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
```

```
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import ahp_service as ahp
LINE = "=" * 66
# ------------------------------------------------------------------
# 1） 完全一致矩阵 → 权重应精确还原，CR=0
# ------------------------------------------------------------------
print(LINE)
print("测试 1　完全一致判断矩阵的权重还原与一致性")
print(LINE)
# 由权重构造完全一致矩阵：a_ij = w_i / w_j
w_true = np.array([0.44, 0.31, 0.25])
A = np.outer(w_true, 1.0 / w_true)
r = ahp.weights_and_consistency(A)
print("  判断矩阵 A：")
for row in A:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print()
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in
    r["weights"]))
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))
print("  λmax = %.6f    CI = %.3e    CR = %.3e    一致: %s"
      % (r["lambda_max"], r["ci"], r["cr"],
    r["consistent"]))
assert np.allclose(r["weights"], w_true, atol=1e-6),
    "权重未精确还原"
assert r["consistent"], "完全一致矩阵应通过检验"
# ------------------------------------------------------------------
# 2） 上三角输入 → 完整矩阵 → 一致性定位
# ------------------------------------------------------------------
print()
print(LINE)
print("测试 2　上三角录入与最不一致项定位")
print(LINE)
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）
M = ahp.matrix_from_upper([2, 5, 3])
r2 = ahp.weights_and_consistency(M)
bad = ahp.find_most_inconsistent(M)
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")
for row in M:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print("  权重:", "  ".join(f"{v:.4f}" for v in r2["weights"]),
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))
```

```
print("   最不一致比较项： 第(%d,%d) 项，当前 %.3f，理想 %.3f"                    10901
        % (bad["i"] + 1, bad["j"] + 1, bad["value"],                          10902
     bad["expected"]))                                                        10903
# ------------------------------------------------------------------------10904
     ----------------                                                         10905
# 3） 多专家聚合（几何平均）                                                    10906
# ------------------------------------------------------------------------10907
     ----------------                                                         10908
print()                                                                       10909
print(LINE)                                                                   10910
print("测试 3  多专家判断矩阵聚合（几何平均）")                                  10911
print(LINE)                                                                   10912
m1 = ahp.matrix_from_upper([2, 4, 3])                                         10913
m2 = ahp.matrix_from_upper([3, 5, 2])                                         10914
agg = ahp.aggregate_matrices([m1, m2])                                        10915
print("   专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")                          10916
print("   聚合后上三角： a12=%.4f a13=%.4f a23=%.4f"                          10917
        % (agg[0, 1], agg[0, 2], agg[1, 2]))                                  10918
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"                  10919
# ------------------------------------------------------------------------10920
     ----------------                                                         10921
# 4） 论文表 3 权重可还原性                                                     10922
# ------------------------------------------------------------------------10923
     ----------------                                                         10924
print()                                                                       10925
print(LINE)                                                                   10926
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")                   10927
print(LINE)                                                                   10928
j = ahp.judge_from_weights([0.44, 0.31, 0.25])                                10929
print("   λmax = %.6f    CR = %.3e    一致： %s" %                             10930
     (j["lambda_max"], j["cr"], j["consistent"]))                             10931
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"               10932
# ------------------------------------------------------------------------10933
     ----------------                                                         10934
# 5） 应用可启动 + 建表                                                        10935
# ------------------------------------------------------------------------10936
     ----------------                                                         10937
print()                                                                       10938
print(LINE)                                                                   10939
print("测试 5  Flask 应用启动与建表")                                          10940
print(LINE)                                                                   10941
from app import create_app                                                    10942
from app.models import db                                                     10943
app = create_app()                                                            10944
with app.app_context():                                                       10945
     db.create_all()                                                          10946
     tables = sorted(db.metadata.tables.keys())                               10947
print("   已创建 %d 张表：" % len(tables))                                     10948
for t in tables:                                                              10949
     print("     -", t)                                                      10950
```

```
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
#  ==== 测试/test_evaluate.py ====
#  -*- coding: utf-8 -*-
"""评价引擎自检：用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
#  ------------------------------------------------------------------
#  论文第 5 节案例数据（表 5，逐字抄录）
#  ------------------------------------------------------------------
#  三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
#  注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
```

```
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
# ------------------------------------------------------------
    ----------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------
    ----------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据  ：%s" % res["grade_info"]["meaning"])
print("  推荐措施  ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
```

```
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
# ==== 测试/test_web_ahp.py ====
# -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_
    hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码:", r.status_code, "(应 200)")
```

```
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」：", "计算结果" in body)
print("  含权重 0.44：", "0.44" in body, " 含 0.31：", "0.31" in
    body, " 含 0.25：", "0.25" in body)
print("  含一致性检验通过：", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」：", "不通过" in body)
print("  含「建议优先调整」：", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
```

```
print("网页界面全部测试通过 √")
print(LINE)
#  ==== 测试/test_web_modules.py ====
#  -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
#  工作台：页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含账户 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
```

```
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码：", r.status_code)
print("  含案例项目：", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
# 新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」：", "测试项目A" in r.get_data(as_text=True
    ))
assert "测试项目A" in r.get_data(as_text=True)
# 清理测试数据，避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
    irst()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元：", all(t in body for t in ("岩质边坡",
    "碎石边坡", "平台")))
print("  含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5  标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南",
    "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("  含多省份并列：", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
```

```
print(LINE)
print("测试 6　限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("　含五项条件：", all(x in body for x in ("政策约束", "区位约束",
    "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
    "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("　全通过后显示「通过，可进入定量评价」：", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("　一项不通过后显示「不通过」：", "不通过" in body)
print("　含失败项「区位约束」：", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("　已恢复为全部通过：", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
```

```
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
```

```
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.i
    d.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
```

```
    WTF_CSRF_ENABLED = False                                          11401
    TESTING = True                                                    11402
app = create_app(TestConfig)                                          11403
client = app.test_client()                                            11404
# 登录（案例种子已创建 admin 用户）                                    11405
client.post("/login", data={"username": "admin",                      11406
    "password": "admin123"})                                          11407
from app.models import MinePatch, EvaluationResult                    11408
with app.app_context():                                               11409
    import seed_demo                                                  11410
    patch = seed_demo.demo_patch()                                    11411
    pid = patch.id                                                    11412
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价          11413
from app.views.evaluate import load_patch_input,                      11414
    save_evaluation                                                   11415
with app.app_context():                                               11416
    units = load_patch_input(pid)                                     11417
    rec = save_evaluation(pid, units)                                 11418
    rid = rec.id                                                      11419
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"            11420
          % (rid, rec.final_score, rec.grade))                        11421
    from app.models import ScoreDetail                                11422
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co11423
    unt()                                                             11424
    print("        逐层得分明细已落库：%d 条" % n_detail)             11425
print()                                                               11426
print(LINE)                                                           11427
print("测试 1  报告列表")                                            11428
print(LINE)                                                           11429
r = client.get("/report/")                                            11430
print("  状态码：", r.status_code)                                   11431
body = r.get_data(as_text=True)                                       11432
print("  含「所属项目」列：", "所属项目" in body)                    11433
assert r.status_code == 200                                           11434
print()                                                               11435
print(LINE)                                                           11436
print("测试 2  得分明细页")                                          11437
print(LINE)                                                           11438
r = client.get("/report/%d" % pid)                                    11439
body = r.get_data(as_text=True)                                       11440
print("  状态码：", r.status_code)                                   11441
print("  含「得分明细」：", "得分明细" in body)                      11442
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)          11443
print("  含二级指标明细（植被覆盖度/坡度/坡向）：", all(n in body for n in   11444
    ("植被覆盖度", "坡度", "坡向")))                                  11445
print("  含权重/标准化值列：", "标准化值" in body and "贡献" in body)  11446
assert r.status_code == 200                                           11447
assert "植被覆盖度" in body and "标准化值" in body                    11448
print()                                                               11449
print(LINE)                                                           11450
```

```
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码:", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
    "")[:60])
data = r.data
print("  文件大小:", len(data), "字节")
print("  是否为有效 docx(PK 头):", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题:", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节:", "评价结论" in text)
print("  含得分明细章节:", "得分明细" in text)
print("  段落数:", len(d.paragraphs), " 表格数:", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检:「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录,先把仓库根目录加入模块搜索路径,
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------
```

```
    ------------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
#  1）复选框必须有可点击关联的 label（这是原先失效的根因）
#     只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#     否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
    "[不符合]"))
#  2）勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
#  3）不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
             follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
    h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
    else "[不符合]"))
assert not rt2
#  4）携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1],
    domain="localhost")
r3 = c3.get("/", follow_redirects=False)
```

```
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]",
     r3.status_code))
assert r3.status_code == 200
# ------------------------------------------------------------------
     ----------------
print()
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
     "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
     "[不符合]"))
assert "confirm(" in html
# 图斑列表：地址用的是项目主键，不是图斑主键
with app.app_context():
     from app.models import MinePatch, Project
     import seed_demo
     patch = seed_demo.demo_patch()
     patch_id = patch.id
     project_id = patch.project_id
html = client.get("/project/%d/patches" %
     project_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
     html and "btn--danger" in html else "[不符合]"))
assert "/project/patch/" in html and "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
     ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
     html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
     else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
     "code": "DEL-1"})
with app.app_context():
     from app.models import db, Project
```

```
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
```

```
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------
    ------------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/profile?tab=logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 账号编辑 / 删除（职务只有管理员与操作员两种）
with app.app_context():
    u = User(username="tmp_user", role="operator",
             password_hash=User.query.first().password_hash)
```

```
        db.session.add(u); db.session.commit()
        uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "账号编辑页可访问")
#  保护校验要趁系统里只有一名管理员时做：
#  先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
with app.app_context():
        admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id,
            data={"role": "operator", "active": "1"})
with app.app_context():
        ok(db.session.get(User, admin_id).role == "admin",
        "拒绝把最后一个管理员降级")
c.post("/users/%d/edit" % uid, data={"role": "admin",
        "active": "1"})
with app.app_context():
        ok(db.session.get(User, uid).role == "admin",
        "职务已修改并保存")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
        ok(db.session.get(User, uid) is None, "账号删除生效")
#  清理日志核对用的项目
with app.app_context():
        for p in Project.query.filter_by(name="日志核对项目").all():
            db.session.delete(p)
        db.session.commit()
# ------------------------------------------------------------------
        ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
        fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
                data={"image": (io.BytesIO(png), "影像.png"),
                      "image_note": "2025年6月无人机正射影像"},
                content_type="multipart/form-data",
        follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
        rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
        rel.replace("uploads/", "", 1))
```

```
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
```

```
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
    f-8")), "d.csv")},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(
    )
```

```
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
    "0.4000")])
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")
# 报告是评价结论的呈现，需先真实执行一次评价才能打开
with app.app_context():
    from app.views.evaluate import load_patch_input,
    save_evaluation
    save_evaluation(patch_id, load_patch_input(patch_id))
r = c.get("/report/%d" % patch_id)
body = r.get_data(as_text=True)
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %
    body.count("<svg"))
exe = browser.browser_path()
if exe is None:
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
    r = c.get("/report/%d/pdf" % patch_id)
    data = r.data
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
    len(data))
    d = fitz.open(stream=data, filetype="pdf")
    txt = "".join(d[i].get_text() for i in
    range(d.page_count))
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
    d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
# ==== 测试/test_workflow.py ====
```

```
#  -*- coding: utf-8 -*-
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
```

```
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
        from app.models import ScoreDetail
        # 先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(patc
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        pr = Project.query.filter_by(name="门禁-" +
    tag).first()
```

```
            if pr:
                db.session.delete(pr)
            db.session.commit()
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ----------------------------------------------------------------
    ------------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
```

```
    n = results_of(pid)                                                    12051
    ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",          12052
       "结果数 %d" % n)                                                    12053
    with app.app_context():                                                12054
        pr = wf.patch_progress(pid)                                        12055
    ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")              12056
    blocked = [b for b in pr["blocking"] if b["step"] ==                   12057
        "constraint"]                                                      12058
    # 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项   12059
    ok(bool(blocked) and "未通过" in blocked[0]["label"],                   12060
       "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else   12061
        "")                                                                12062
    ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),        12063
       "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")  12064
    cleanup(pid, "3")                                                      12065
    print()                                                                12066
    print("场景 4：判定通过，但指标数据全空")                                12067
    pid = new_patch("4"); add_units(pid); add_constraint(pid,              12068
        passed=True)                                                       12069
    r = post_eval(pid)                                                     12070
    n = results_of(pid)                                                    12071
    ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",        12072
       "结果数 %d" % n)                                                    12073
    with app.app_context():                                                12074
        pr = wf.patch_progress(pid)                                        12075
    ok(any(b["step"] == "data" for b in pr["blocking"]),                   12076
       "拦原因指向「录入指标数据」")                                        12077
    ok(pr["data"]["missing"] == ["碎石边坡"],                               12078
       "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))        12079
    cleanup(pid, "4")                                                      12080
    print()                                                                12081
    print("场景 5：单元面积占比合计不为 100%")                               12082
    pid = new_patch("5"); add_constraint(pid, passed=True)                 12083
    with app.app_context():                                                12084
        u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,        12085
        ratio=0.5)                                                         12086
        db.session.add(u); db.session.commit()                             12087
        db.session.add(IndicatorData(unit_id=u.id,                         12088
        soil_ratio=0.3, thickness=10,                                      12089
                                     ph=8.0,                               12090
        organic_matter=5.0, coverage=80.0,                                 12091
                                     shannon=0.6, slope=20.0,              12092
        aspect="阳坡"))                                                    12093
        db.session.commit()                                                12094
    r = post_eval(pid)                                                     12095
    ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")              12096
    cleanup(pid, "5")                                                      12097
    print()                                                                12098
    print(LINE)                                                            12099
    print("二、前置条件齐备时才放行")                                        12100
```

```
print(LINE)
pid = new_patch("OK"); add_units(pid); add_constraint(pid,
    passed=True); add_data(pid)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")
r = post_eval(pid)
n = results_of(pid)
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)
print("  重定向到:", r.headers.get("Location"))
print()
print(LINE)
print("三、报告不得先于评价产生")
print(LINE)
pid2 = new_patch("R")    # 什么都不做
r = get_report(pid2)
ok(r.status_code == 302 and "/report/" ==
    (r.headers.get("Location") or ""),
   "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
    rr = c.get("/report/%d%s" % (pid2, suffix),
    follow_redirects=False)
    ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
    %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
# 报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
# 也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
    add_data(pid3)
with app.app_context():
    pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
   "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
   "直接访问 /report/<id> 被退回，且不产生任何评价结果",
   "HTTP %d" % r.status_code)
# 真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
```

```
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
   "报告数值取自评价记录，而非重新计算",
   "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
# 回归保护：seed_demo.clean_case() 曾执行 Project.query.delete()，
# 导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
# 演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
   m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
```

```
print(LINE)
#  论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
#  允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
      "area": "1.0"},
          follow_redirects=True)
with app.app_context():
      n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
      "area": "2.0"},
                follow_redirects=True)
with app.app_context():
      n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
      "area": "1.0"},
                follow_redirects=True)
with app.app_context():
      n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
      "area": "3.0"},
          follow_redirects=True)
with app.app_context():
      uid = MicroUnit.query.filter_by(patch_id=pid4,
                                      unit_type="碎石边坡").first()
      ).id
r = c.post("/unit/edit/%d" % uid,
                data={"unit_type": "平台", "area": "3.0"},
      follow_redirects=True)
with app.app_context():
      types = sorted(u.unit_type for u in
                     MicroUnit.query.filter_by(patch_id=pid4)
      all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
      print("存在未通过项：")
      for f in _fails:
           print("  -", f)
      sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
```

```
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
```

```
        db.session.add(p); db.session.flush()                    12301
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"  12302
    % TAG)                                                       12303
        db.session.add(patch); db.session.flush()                12304
        db.session.add(MicroUnit(patch_id=patch.id,              12305
    unit_type=UNIT_TYPE,                                         12306
                                 area=1.0, ratio=1.0))           12307
        db.session.flush()                                       12308
        u = MicroUnit.query.filter_by(patch_id=patch.id).fir     12309
    st()                                                         12310
        db.session.add(IndicatorData(                            12311
            unit_id=u.id, soil_ratio=0.34, thickness=10,         12312
    ph=9.24,                                                     12313
            organic_matter=4.89, coverage=87.92,                 12314
    shannon=0.60,                                                12315
            slope=28.75, aspect="阳坡"))                         12316
        db.session.add(ConstraintCheck(                          12317
            patch_id=patch.id, policy_pass=True,                 12318
    location_pass=True,                                          12319
            pollution_pass=True, geo_pass=True,                  12320
    farmland_pass=True,                                          12321
            passed=True, reason="全部通过"))                     12322
        db.session.commit()                                      12323
        return patch.id                                          12324
def cleanup(pid):                                                12325
    with app.app_context():                                      12326
        from app.models import ScoreDetail                       12327
        rids = [r.id for r in EvaluationResult.query.filter_     12328
    by(patch_id=pid)]                                            12329
        if rids:                                                 12330
            ScoreDetail.query.filter(ScoreDetail.result_id.i     12331
    n_(rids)).delete(                                            12332
                synchronize_session=False)                       12333
        EvaluationResult.query.filter_by(patch_id=pid).delet     12334
    e()                                                          12335
        ConstraintCheck.query.filter_by(patch_id=pid).delete     12336
    ()                                                           12337
        uids = [u.id for u in MicroUnit.query.filter_by(pat      12338
    h_id=pid)]                                                   12339
        if uids:                                                 12340
            IndicatorData.query.filter(IndicatorData.unit_id     12341
    .in_(uids)).delete(                                          12342
                synchronize_session=False)                       12343
        MicroUnit.query.filter_by(patch_id=pid).delete()         12344
        m = db.session.get(MinePatch, pid)                       12345
        if m:                                                    12346
            db.session.delete(m)                                 12347
        p = Project.query.filter_by(code="WEIGHT-%s" %           12348
    TAG).first()                                                 12349
        if p:                                                    12350
```

```
                db.session.delete(p)                                    12351
            db.session.commit()                                         12352
def run_eval(pid):                                                      12353
    c.post("/evaluate/%d" % pid, data={},                               12354
    follow_redirects=False)                                             12355
    with app.app_context():                                             12356
        return EvaluationResult.query.filter_by(patch_id=pid12357
    ) \                                                                 12358
            .order_by(EvaluationResult.id.desc()).first().fi12359
    nal_score                                                           12360
print(LINE)                                                             12361
print("一、默认应使用论文表 3 的权重")                                  12362
print(LINE)                                                             12363
with app.app_context():                                                 12364
    ws.reset_weights(UNIT_TYPE)                                         12365
    w, label, scheme = ws.effective_weights(UNIT_TYPE)                  12366
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",  12367
    label)                                                              12368
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",      12369
   "土壤指标 %.2f" % w["l1"]["土壤指标"])                               12370
pid = build_case()                                                      12371
base_score = run_eval(pid)                                              12372
print("  默认权重下的评价值 F = %.4f" % base_score)                    12373
print()                                                                 12374
print(LINE)                                                             12375
print("二、保存 AHP 算出的权重后，评价应改用新权重")                    12376
print(LINE)                                                             12377
with app.app_context():                                                 12378
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）  12379
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",                 12380
    0.20),                                                              12381
                                ("terrain", 0.10)])                     12382
    n_items = WeightItem.query.count()                                  12383
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)    12384
with app.app_context():                                                 12385
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)               12386
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案", 12387
    label2)                                                             12388
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",             12389
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])                              12390
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")  12391
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,                   12392
   "未覆盖的二级权重仍保留论文取值",                                    12393
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])                      12394
new_score = run_eval(pid)                                               12395
print("  新权重下的评价值 F = %.4f" % new_score)                       12396
ok(abs(new_score - base_score) > 1e-4,                                  12397
   "权重变化确实改变了评价结果",                                        12398
   "%.4f → %.4f" % (base_score, new_score))                             12399
print()                                                                 12400
```

```
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
        .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
   "评价记录中标注了权重来源",
   stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到论文取值")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",
   label3)
back_score = run_eval(pid)
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
   base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
```

```
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
#  ==== app/services/__init__.py ====
#  -*- coding: utf-8 -*-
#  ==== app/services/cleanup.py ====
#  -*- coding: utf-8 -*-
"""级联清理：删图斑、删项目时，把挂在它下面的数据一并清掉。
为什么单独抽出来：`Project.patches` 与 `MinePatch.units` 上有 ORM 级联，
但图斑下面的**指标数据、限制性判定、评价结果、得分明细、参照生态系统**
都没有级联关系。原先只有「删除图斑」那条路径里手写了清理，
于是「删除项目」会留下一批孤儿行——自检实测：删一个项目后，
上面五张表各残留 1 行（项目与图斑本身都删掉了，数据却还在库里）。
两条删除路径都调用这里，才不会漏。
"""
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail,
    ConstraintCheck,
                        ReferenceEcosystem)
def delete_patch_children(patch_id):
    """清掉挂在某个图斑下的全部附属数据（不含图斑本身）。
    返回各表清理条数，便于调用方打印或断言。
    """
    counts = {}
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patch_id=patch_id).all()]
    if unit_ids:
        counts["indicator_data"] = IndicatorData.query.filter(
            IndicatorData.unit_id.in_(unit_ids)).delete(synchronize_session=False)
    result_ids = [r.id for r in
                  EvaluationResult.query.filter_by(patch_id=patch_id).all()]
    if result_ids:
        counts["score_details"] = ScoreDetail.query.filter(
            ScoreDetail.result_id.in_(result_ids)).delete(synchronize_session=False)
    counts["evaluation_results"] = EvaluationResult.query.fi
```

```
    lter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["constraint_checks"] = ConstraintCheck.query.filt
    er_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["micro_units"] = MicroUnit.query.filter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["reference_ecosystems"] = ReferenceEcosystem.quer
    y.filter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    return counts
def delete_patch(patch_id):
    """删除一个图斑及其全部附属数据。图斑不存在时返回 (None, {})。"""
    patch = db.session.get(MinePatch, patch_id)
    if patch is None:
        return None, {}
    counts = delete_patch_children(patch_id)
    db.session.delete(patch)
    db.session.commit()
    return patch, counts
def delete_project(project):
    """删除一个项目：先清掉每个图斑的附属数据，再删项目本身。
    项目下的图斑由 `Project.patches` 的 ORM 级联删除，但图斑下面那五张表
    必须先手工清，否则会留孤儿。
    """
    cleaned = 0
    for patch in project.patches.all():
        counts = delete_patch_children(patch.id)
        cleaned += sum(counts.values())
    name = project.name
    db.session.delete(project)
    db.session.commit()
    return name, cleaned
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/parsing.py ====
# -*- coding: utf-8 -*-
"""表单数值的安全解析。
背景：页面上的数字输入框原来写成 `float(request.form.get("area"))`，
用户填「约 3.5」或全角数字就会撞上 ValueError —— 页面上是默认英文 500 页。
真人演示时这是最容易踩到的一类。统一走这里：
没填返回 None，填了但不是数字抛 FieldError，让视图 flash 一句中文再回退。
"""
import math
class FieldError(ValueError):
    """表单里某个字段填得不对，携带给用户看的中文提示。"""
def parse_float(raw, field, *, minimum=None, maximum=None):
    """把表单值解析成 float。
    raw      表单原始值（可能是 None / "" / "abc" / "1e400"）
    field    字段名，用于拼提示
```

```
    minimum / maximum  可选的取值范围，越界同样抛 FieldError
    """
    if raw is None:
        return None
    text = str(raw).strip()
    if text == "":
        return None
    try:
        value = float(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填数字" % field)
    # nan / inf 能通过 float()，但存进库里会把标准化算成满分，必须挡住
    if not math.isfinite(value):
        raise FieldError("「%s」请填正常的数字" % field)
    if minimum is not None and value < minimum:
        raise FieldError("「%s」不能小于 %g" % (field, minimum))
    if maximum is not None and value > maximum:
        raise FieldError("「%s」不能大于 %g" % (field, maximum))
    return value
def parse_int(raw, field, *, default=None, minimum=None,
    maximum=None):
    """解析整数，用于查询参数与计数类字段。"""
    if raw is None:
        return default
    text = str(raw).strip()
    if text == "":
        return default
    try:
        value = int(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填整数" % field)
    if minimum is not None and value < minimum:
        value = minimum
    if maximum is not None and value > maximum:
        value = maximum
    return value
# ==== app/utils/timefmt.py ====
# -*- coding: utf-8 -*-
"""时间显示：库里一律存 UTC，展示时换成服务器本地时间。
为什么不在写库时直接用本地时间：一个系统里时间基准只能有一个。
`app/models.py` 里所有 `created_at` / `checked_at` 都用
    `datetime.utcnow`，
存的是 UTC；如果写库改成本地时间，已经存进库的几千条记录会新旧混着差 8 小时。
所以统一在「展示」这一层换算，老数据不用动。
用法（Jinja）：
    {{ row.created_at | localdt }}                    →
    2026-09-26 23:33
    {{ row.created_at | localdt("%Y-%m-%d %H:%M:%S") }} →
    2026-09-26 23:33:28
"""
```

```
from datetime import datetime, timezone
DEFAULT_FORMAT = "%Y-%m-%d %H:%M"
def to_local(value, fmt=DEFAULT_FORMAT, empty="—"):
    """把 UTC 时间换算成服务器本地时间并格式化。
    传进来的值按「无时区即 UTC」处理；带时区的原样换算，便于以后改用
    带时区的时间戳。空值返回占位符，免得页面上出现 None。
    """
    if value is None:
        return empty
    if value.tzinfo is None:
        value = value.replace(tzinfo=timezone.utc)
    return value.astimezone().strftime(fmt)
def now_utc():
    """取当前 UTC 时间（带时区），用于只作展示、不落库的时间戳。"""
    return datetime.now(timezone.utc)
# ==== app/views/__init__.py ====
# -*- coding: utf-8 -*-
```